



*The Asian Necropolis
Inadvertent*

AFTER ALL THE CONVOYS, PIRACY, AND PLUNDER HAD BROUGHT THE GOLD AND SILVER FROM THE NEW WORLD TO EUROPE, ALMOST NONE OF THAT MASSIVE MOVEMENT OF PRECIOUS METALS ENDED UP WHERE THE EUROPEANS EXPECTED IT TO END UP.

ELEVEN

and Hien Tsung's Innovation

The entire flow lingered only briefly in Europe and then continued eastward to Asia. There is even some evidence that the outflow of gold and silver to the Far East may have exceeded the total imports from America between 1600 and 1730. During the first 25 years after the establishment of the East India Company in 1600, bullion accounted for 75 per cent of all the cargo shipped eastward.

Asia turned out to be a sponge for gold and silver. Only a tiny quantity ever came back to Europe. The reasons for this one-way movement are not obvious, and one is tempted to agree with Kipling that east is east and west is west and let it go at that. But what happened is not that trivial.

The Europeans may have grumbled about the loss of their beloved gold and silver to the East, but their desire for the spices, tea, silk, and other luxuries of Asia was so insatiable that they had no alternative. The people in Asia clearly considered gold and silver to be more desirable than the tin, lead, mercury, woolens and furs that the Europeans offered for sale. That the one-way trade continued for so long is perhaps evidence enough of how satisfactory it was to the Asians, but another revealing piece of evidence is available: the prices of goods in China, Japan and India varied very little relative to the precious metals. If the process had been an unstable one, the Asians would have refused to continue selling their products to Europe or would have demanded a far larger quantity of gold and silver relative to the physical volume of tea, silks and spices shipped out.

If you believe that the Asians had a peculiar set of priorities, you will have to concede that the priorities of the Europeans were just as strange. Europeans also put the highest value on the precious metals. They would have considered themselves much wealthier if

Tea harvest in China

they could have retained the gold and silver and shipped out the useful stuff like iron, tin and furs. Even as wise an observer as the Scots philosopher and historian David Hume fell into this trap when he wrote in 1752 that 'The skill and ingenuity of EUROPE in general surpasses perhaps that of CHINA, with regard to the manual arts and manufactures; yet we are never able to trade thither without grave disadvantage.'

More than 100 years before Hume, Thomas Mun, son of a moneyer, master merchant, and key executive of the East India Company, had a keener sense of the reality of this trade: 'If those nations which send out their monies do it because they have few wares of their own, how come they have so much Treasure. . . . I answer, Even by trading with their money; for by what other means can they get it, having no mines of Gold or Silver?' Mun was responding to criticism of the East India Company for shipping so much in precious metals eastward, but in fact the transactions proved to be highly favorable to the Europeans. We have already noted that, in contrast to Asia, the price of gold in Europe fell relative to the prices of other commodities. Hence, it was an illusion that the Europeans were getting the worst of the bargain. With the Asians so accommodating by accepting gold and silver, the discovery of America now enabled Europe to satisfy, to a much greater extent than in the past, the longstanding hunger for the products of Asia. Spices cannot be grown anywhere but in Asia, because their development

depends on the monsoon rains, while silkworms and tea plants also require a particular kind of climate. The Europeans did learn to convert their earth into fine white china and to execute beautiful designs on it, but the products of the Far East were still esteemed for the mark of status they provided and for the adventure implicit in having obtained them.

Even the Europeans far away in the American colonies were infected with the import virus. The Peruvian silver city of Potosí and the port city of Lima were famous for their displays of silk, porcelain, lacquer ware, precious stones and pearls from China, while Mexicans paraded around in cottons from the Philippines, silks from China and calicos from India.

With all that gold and silver pouring in from the New World, the Europeans were behaving precisely as classical economic theory would have predicted. It was the Asian sponge that was in defiance of theory. The patterns of predictable responses to fresh inflows of money were first set forth in systematic fashion by early economists such as Thomas Mun and William Petty in the first part of the 17th century. In 1752, however, David Hume developed the theory more fully in an essay titled 'Of the Balance of Trade'. According to Hume, the movement of money from one area to another is inherently unsustainable and destined to reverse itself. The money piling up in the pockets of the citizens of the gaining country will encourage them to go out and buy things, while the loss of purchasing power in the losing country will lead its citizens to tighten their belts and buy less; prices will rise in the gaining country and fall in the losing country. This shift in demand will in time reverse the flow of money back to the country that first suffered the outflow. As a result, Hume argued, 'It is impossible to heap up money, more than any other fluid, beyond its proper level.'

The sequence of events related to the ransoming of the sons of Francis I was a perfect illustration of Hume's theory – the gold moved from France to Spain in payment of the ransom, squeezing French citizens in the process but simultaneously encouraging the Spanish to spend more money abroad. No wonder, then, that Spain's newly acquired treasure found its way back to France.

Hume used his theory to explain why Spain and Portugal failed to hold on to the gold and silver they imported from their colonies overseas. 'Can one imagine,' he asked, with his own peculiar use of uppercase lettering, 'that it had ever been possible by any laws, or even any art of industry, to have kept all the money in SPAIN, which the galleons had brought from the INDIES? Or that all the commodities would be sold in FRANCE for a tenth of the price they would yield on the other side of the PYRENEES, without finding their way thither and draining from that immense treasure?' Hume's theory would also explain why the countries that accumulated gold by exporting to Spain would in turn display a voracious appetite for imports from the East (they were under no obligation to import from Spain), especially as Europeans were able to keep replenishing their supply of precious metals after 1500 from both New World sources and increased domestic production. Thus, the Europeans played according to Hume's rules.

Not so in Asia. Contrary to Hume's hypothesis, the precious metals did 'heap up' in China, India, and Japan. Hume may have been correct that fluid cannot accumulate beyond its proper level when the fluid is contained, but what about when the fluid flows into a vessel so enormous that the inflow is meaningless? A 17th-century English merchant, echoing a metaphor in Ecclesiastes, made an appropriate observation when he lamented that 'Many streams run thither (India), as all rivers to the sea, and there stay.' The economic historian Earl Hamilton has noted that 'The East [was] a necropolis of European treasure even in Roman days.'

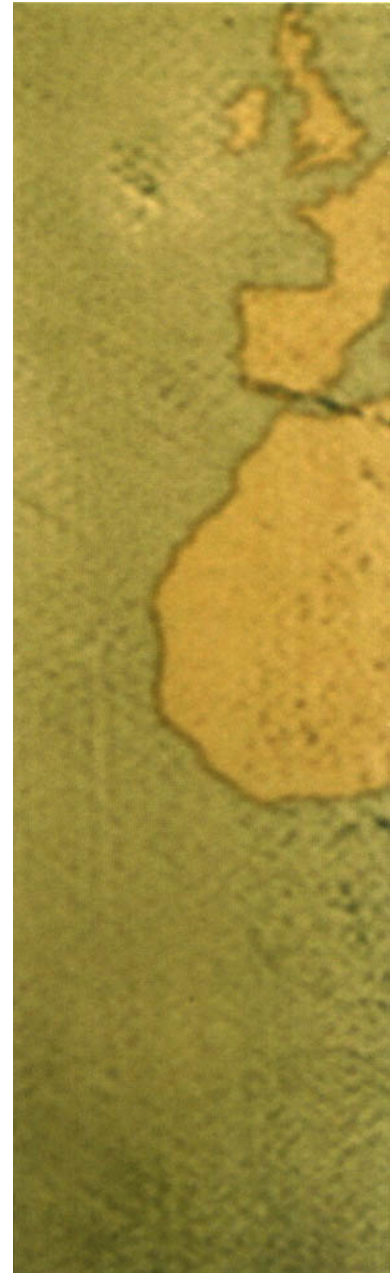
Why did the immense quantities of gold and silver that were shipped to China, Japan, and India accumulate instead of provoking a reverse flow of demand for goods from Europe? Did the Asians of the 1500s and 1600s have a natural predilection for the Protestant Ethic, with its focus on the virtue of saving, a set of beliefs that most of them back then had probably never heard of? Were Asians so innocent, or so neurotic, about gold and silver that the sheer joy of ownership was enough to keep them importing useless precious metals in exchange for valuable commodities that they could themselves have eaten and worn? Or was a different set of forces acting upon the Asian scene?

One thing is certain: Asians derived much pleasure from their ownership of gold. Gold's natural attributes of malleability, indestructibility, and dazzling beauty appeal to people in any part of the world. The Asian rulers were just as convinced as Hatshepsut, Croesus, Justinian, Abbé Suger, Atahualpa and Francis I that gold conveyed both a sense of power and a sense of magical beauty.

Marco Polo is a revealing authority for this view. In 1271, he traveled from Venice to China, remaining in the area for 20 years. Although he did not reach Japan, he obtained descriptions of it from his Chinese contacts. Even allowing for the obvious exaggeration in his many colorful tales, the frequency with which Marco Polo refers to gold is notable. The amount of space he gives to gold and silver together is much greater than the attention he gives to any of the other Asian products in greatest demand in the West, such as spices and silk.

Most of the time, Marco Polo was in the service of the great Mongol leader Kublai Khan, a man who sent emissaries every two years to the Tartars to select between 400 and 500 of the most beautiful girls to join his already overflowing inventory of concubines; the Great Khan made do with 30 or 40 girls whom he selected from each biennial group. These served the Khan in rotating groups of six 'in his chamber and his bed, ministering to all his needs'. Those women were in addition to his four official wives, each of whom had 300 ladies in waiting, and each one of those – according to Marco Polo – has in her court 10,000 persons'. This may be a huge embellishment of the truth, but it contains an important hint about puzzling patterns of Asian foreign trade.

A man supporting a household like Kublai Khan's is clearly not a penny-pinching believer in the principles of the Protestant Ethic. On the contrary, Marco Polo's





Marco Polo's travels to China



A 5th Century example of a Chinese 'Cash Coin'

descriptions of the Khan's palaces are breathtaking. He contends that the palace in the capital city of Cathay was the largest palace ever built, where 'there is nothing to be seen anywhere but gold and pictures'. He also describes the magnificence of another ruler, who built a tower in his palace grounds that was covered with gold a full finger's breadth in thickness, and was so completely covered that the structure appeared to be made entirely of gold.

Marco Polo also reported on Japan, 'a very big island [where] the people are fair-complexioned, good-looking, and well-mannered'. Gold is found there 'in measureless quantities', and one reason they possess so much of it is that 'no trader, nor indeed anyone else, goes there from the mainland.' He then describes a palace of the island's ruler, 'that is in truth a veritable marvel. Just as we roof our houses and churches with lead, so this palace is roofed with fine gold. And the value of it is almost beyond computation. Moreover, all the chambers, of which there are many, are likewise paved with fine gold to a depth of more than two fingers' breadth. And the halls and the windows and every other part of the palace are likewise adorned with gold.' This is the passage from Marco Polo's *Travels* that spurred Columbus on when he reached Cuba and thought he was in Cipangu (Japan).

Gold seems to have been almost everywhere Marco Polo visited. In Vochan gold was so plentiful that one ounce of it exchanged for only five ounces of silver at a time when one ounce of gold in Europe fetched upward of ten ounces of silver and the gold/silver ratio would approach 1:14 after the big American silver mines brought about a radical increase in the European supply of silver relative to gold. In Tibet, where the natives used salt for money and were 'very poorly clad, in skins, canvas and buckram . . . there are rivers and lakes and mountains in which gold-dust is found in great quantity. . . . The province produces plenty of . . . cloth of gold.' Marco Polo also mentions gold in India, where he claims it was so plentiful that it exchanged on a ratio of 1:6 for silver. In addition to modest indigenous supplies of gold, India imported large amounts of both gold and silver in the 1500s and 1600s in exchange for cotton, most of it coming in via the major trading post of Malacca in Malaysia.

Everything that we have seen so far reveals that Asians did not perceive gold as money in the same way that Westerners viewed it. Even before Croesus, people in Europe and the Near East were using gold as money, first in bars and then in coins. For at least 2000 years before Columbus discovered America, Europeans had viewed gold coins as the ultimate expression of financial might and sophistication. Coinage, however, democratized gold, because it circulated among members of the public. Asian rulers held no such notions. They shared the Western delight in gold's beauty and what it signified in terms of power, but they considered gold too important to be used as money that would be passed around

from one dirty and ignoble hand to another. Releasing gold to public circulation would dilute the power of the state.

As a result, most Chinese money over the centuries has been fabricated out of materials of little value. Small amounts of gold coins were minted after the 16th century, but they were used primarily for ceremonial purposes, such as being thrown on occasion at professors when the students approved of their lecture. China did not issue any coins of precious metals in meaningful amounts until 1890, and then in silver, although Mexican silver pesos – of all things! – circulated in some quantity in China between 1700 and 1826.

This conception of a monetary system had a long tradition in China. In 255 BC, when the many feudal Chinese states were united by a great general named Qin Shu Huangdi into a single political entity, Qin promptly declared himself the emperor and built the Great Wall to protect his domain (Qin is pronounced 'Chin' and is the derivation of the word China). It was his tomb that was guarded by the famous terracotta army discovered by archeologists at Xian in 1974. At that time, the longstanding Chinese coinage system – which probably predated the use of coins in the West – consisted of awkward cast-bronze pieces that looked like hoes, knives, and cowrie-shaped shells. Qin replaced these forms with cast-bronze round coins punctured in the middle with a square hole. These small base-metal coins came to be known as cash, the Tamil word for money that is now the common expression reserved for ready, liquid money. Although the denominations and weights of the coins changed over the centuries, the familiar characteristic design of round Chinese coins with the square holes remained unchanged for over 800 years.

The holes in the coins made it possible to string together a large quantity for carrying or for trading: the coins were made out of such low-value material that a trader or his customer had to handle many coins even for transactions of modest size. If some of the coins had been minted from metals of more value – at least in the eyes of the beholders – to provide for larger transactions, fewer coins would have been needed and carrying them around would have been less burdensome. As in the West, the coins could have been carried in pockets or purses.

About 1000 years after Qin, during the reign of Hien Tsung (806–21), a severe shortage of copper induced the emperor to use sheets of paper for money in place of bronze coins. If there was no point in making payments with useful stuff, the emperor reasoned, why not go all the way and adopt paper? This newfangled idea appears to have been more of a historical accident than a stroke of financial genius, but the long perspective of history suggests that Hien Tsung's inadvertent innovation should join printing, gunpowder, and the compass among China's most enduring contributions to the civilization of the world. Hien Tsung not only passed on his invention to posterity; his successors also paved the way for the inevitable route that most paper money systems have followed: over-issue and uncontrollable inflation.

The lessons were learned in China early on. In a book called *A Treatise on Coinage*, published in 1149, the historian Ma Twan-lin warned in strikingly modern terms that 'Paper should never be money [but] only employed as a representative sign of value existing in metals or produce. . . . The government . . . wished to make a real money of paper, and thus the original contrivance was perverted.' We shall see that this argument reappears almost verbatim in Britain in the course of the Napoleonic wars, dressed in different clothes but containing the same substance. About 600 years had to pass, until 1455, before the Chinese decided that they could control their money supply more effectively with metallic coins than with paper money, but they were still so far ahead of the West that another 300 years would pass before printed banknotes became common in Europe.

Marco Polo was so impressed with the paper money of China that he considered it a kind of magic. As he characterized it, Kublai Khan's 'mint . . . is so organized that you might well say that he has mastered the art of alchemy.' The paper for the money was manufactured out of mulberry tree bark and cut up into various sizes to reflect the differing denominations; a note representing 1000 coins measured 9 by 13 inches – an awkward size but light as a feather, while 100 coins weighed 8 pounds. The proper

River Li, China



officials then wrote their names on the papers and stamped them with the seal of the Great Khan. At that point, according to Marco Polo, 'The procedure of issue is as formal and authoritative as if they were made of pure gold or silver. . . . The money is authentic. . . . Of this money the Khan has such a quantity made that with it he could buy all the treasure of the world.'

Marco Polo reported that the Khan ordered every payment, everywhere in his empire, to be made in paper currency. Then Marco Polo delivered the punchline that exposed why this money worked as well as it did and why he thought that the Khan could buy all the treasure of the world with it: *'And no one dares refuse it on pain of losing his life.'* There's legal tender for you! But no force was involved, according to Marco Polo. Everyone was 'perfectly willing' to accept these papers in payment, 'since wherever they go they pay in the same currency.' The system provided a neat system for the government to finance itself.

The Khan did not restrict his expenditure of the paper money merely to current operating expenses of his government. He deployed his power to greater advantage. Whenever traders arrived in his domain with pearls, precious stones, gold, silver or other valuables, or when any town possessed gems or precious metals, they were required to surrender all their treasure to the Great Khan. These people accepted the paper money in exchange for their valuables 'willingly', because they could use it to pay for the goods they bought throughout the Great Khan's dominions. By this means, Marco Polo concluded, 'The Great Khan acquires all the gold and silver and pearls and precious stones of his territory,' and this is how the Khan 'has more treasure than anyone else in the world'. This 'willingness' may have had to do with more than the Great Khan's raw power, although surely that was a necessary condition for the success of the paper money. China's huge landmass may have made its economy more self-sufficient than any individual country or city-state in Europe. Thus, concerns about acceptability of the paper money abroad, if any, were at least secondary.

What Marco Polo failed to point out was that the Khan thereby accumulated assets that were accepted as assets anywhere in the world, while the people who held paper money in exchange held assets that were accepted only in the Great Khan's dominions. Seen from that perspective, Marco Polo was correct that the Khan had mastered the art of alchemy. His paper money turned into gold – at least at his palaces.

Was the exchange unfair? In 1933, the US government prohibited the ownership of monetary gold by any individual, company, or political entity except the federal government itself (jewelry and works of art were excluded from this prohibition). All monetary gold within the borders of the United States or imported into the United States had to be turned over to the government and converted into dollar-denominated bank deposits or everyday coin and currency. This law was still less restrictive than Kublai Khan's, because foreign governments and central banks such as the Bank of France or the Bank of England were free to convert their dollar balances back into gold – until 15



*A tapestry of Marco Polo on
his travels*

August 1971. At that moment, President Nixon joined the Great Khan. As the expression of the time put it, the gold window was shut down. Even foreign governments and central banks could no longer exchange dollars for gold. In his own fashion, Richard Nixon had mastered the art of alchemy.

Meanwhile, the story in Japan was much the same as in China. The Japanese used gold for adornment, not money. Unlike many other forms of art, the work of the goldsmith in Japan developed late and was never as notable as Japanese pottery, sculpture, or painting. When it came to money, the Japanese just copied the Chinese practice, as they had already copied art and the alphabet from China, except that the Japanese clung to metallic money rather than converting to paper.

Copper was the first metallic money, followed by bronze, which economized on the copper. When the Chinese changed over to paper money in the 9th century, they had no more use for their metal coins, so the Japanese obliged them by importing the superfluous coins. For the next few centuries, the Chinese would continue to supply the Japanese with all the bronze coins they needed. As trade was considered demeaning in Japan, and the economy was in any case less developed than in China, the demand for money was correspondingly smaller. Marco Polo himself had observed that no traders ventured from Japan to the mainland, which meant that foreign trade played no role in the Japanese economy until much later on.

Around 1600, a rising price for gold provoked a minor mining rush in Japan, to satisfy increased demand in China as well as the Japanese emperor's desire to advertise his power. Portuguese shipping was then just beginning to reach the Japanese seas, and so a modest amount of foreign trade was developing. The Portuguese exchanged Chinese silks for Japanese silver, which they then sold in Macao, Malacca and India in exchange for spices until the Japanese spoiled a profitable triangular line of trade by cutting off the Portuguese. Why? The Portuguese had tried to convert the Japanese to Christianity.

The Japanese finally began issuing their own coins around this time, but these home-grown specimens were still copies of the Chinese money and were even inscribed with the appropriate Chinese characters. Japan also started issuing a limited amount of gold and silver coins in oval or rectangular shapes to suggest the ingots that had circulated for a short time as money in the Middle Ages, but these had such high value – one was worth about a ton of the copper coins in daily circulation – that they were used for the most part in ceremonial transactions among the nobility. The stones of Yap would have functioned more efficiently! Even in the latter half of the 19th century, after Admiral Perry's arrival, the Japanese coinage continued to copy foreign coins, this time of American and British money; yen, which means 'round coin' in Japanese, was also the word they used for dollar, with sen, the word for a copper coin, standing for cent.

To return to the main question: Why did the gold and silver of the West move in only one direction toward the East? What was going on in Asia that led people to value precious metals more than the food, clothing and home decoration that they shipped in such volume to Europeans?

One answer has already been suggested. From the viewpoint of the Asians, the supplies of gold and silver they were importing were money only in a very indistinct sense. They did not perceive the precious metals as something that people use to exchange for something else, either now or at some point in the future. Rather, the Chinese, Japanese and Indians considered the precious metals to be commodities – that is, goods with a genuine use value that rendered these goods worth keeping for their own sake rather than as a means of payment. Gold was for decorating a bride, for baubles, for ornaments, and, most important, for hoarding. Indeed, burying treasure in the East was akin to conspicuous consumption, similar to the insatiable demand for fancy gold watches in our own time. This view of gold explains why Asia was a sponge for precious metals rather than one of Hume's nations where 'it is impossible to heap up money, more than any other fluid, beyond its proper level.' The Asians were not playing the money game.

What, then, was the utility of gold and silver in the eyes of the Asians? As with everyone else, beauty and resistance to rust were important elements. Adam Smith charmingly observes, 'A silver boiler is more cleanly than a lead, copper or tin one; and the same quality would render a gold boiler still better than a silver one. The principal merit [of the precious metals], however, arises from their beauty, which renders them peculiarly fit for the ornaments of dress and furniture. No paint or dye can give so splendid a color as gilding.'

That is not all: scarcity matters, too. Smith refers to rich people who never appear to themselves so rich 'as when they appear to possess those decisive marks of opulence which nobody can possess but themselves. . . . Such objects they are willing to purchase at a higher price than things much more beautiful and useful, but more common.' Wealthy Asians were no exception to this general rule.

The strangest feature of gold is that there has never been a moment when it has ceased to be scarce. There have been moments when the supply of gold increased faster than the supply of silver, and gold's price fell relative to the price of silver (to say nothing of the dramatic drop in the price of gold after disinflation took hold in the world economy of the 1980s), but the price nevertheless has always remained high enough to indicate that gold was far from a glut on the market. Steel sells for 2¢ an ounce; gold sells for more than 10,000 times as much.

Gold remains a definitive mark of opulence. Thus, the mystique of gold as a store of wealth, which is critically dependent on its scarcity, has added lustre to gold as a symbol of power, whether it accents the halo of a saint or covers the roof of a palace, whether it dangles from the ear or surrounds the neck, and whether it fills the room of a captive

emperor or the storage cells of the Federal Reserve Bank of New York. Darius must have had great fun in his bathtub.

In the long run, the cupidity and stupidity of human beings are what motivate the drama, although nature contributes by distributing gold arbitrarily and unevenly around the globe. There was never a time when gold was not in constant demand for ostentation or for hoarding.

Hoarding is similar to buying an insurance policy. Like an insurance policy, hoarding gold has a cost, for the idle metal earns nothing. However, you sleep better knowing that you hold some kind of a hedge against the chance that the catastrophes you fear may actually occur. This motive is as powerful among poor peasants and laborers as among kings and princes. But the insurance policy can work only as long as everyone else continues to consider gold an item of highest value. For that essential condition to be met, people must continue to agree with one another that gold is scarce.

The distribution of income in Asia provides a second answer to our question about the strange equilibrium that kept gold moving in the eastern direction and useful commodities such as spices and silk moving to the west. Evidence developed by modern economists indicates that average per capita wealth in Asia in the 16th and 17th centuries was close to average per capita wealth in Europe. Adam Smith himself pointed out that China was one of the richest and most fertile countries in the world.

Averages can be dangerously deceptive when the dispersion around them is wide. The per capita figure for Europe is of some value as at least suggestive, but the per capita measure for China, or any part of Asia at that time, obscures the reality that a tiny number of people lived in an environment of luxury and indulgence while the masses existed at levels that were horrible even by the standards of the poorest people in Europe.

Under conditions like this, the gold-consuming nobility of China were aware of no loss to themselves in shipping the products of their land out to Europe, because the privileged class was so small and wielded so much power that their consumption could be, and usually was, as conspicuous as they wanted it to be without any sense of prodigality on their part. Gold and silver, on the other hand, had a limitless market in Asia, for both ostentation and insurance against uprisings and war. What looked like an irrational exchange with the Europeans at first glance thus becomes an understandable set of transactions when viewed from the perspective of class structure and tastes. Similar observations apply, with only minor variations, to Japan, India and the island kingdoms of the Pacific.

As Marco Polo did not have a graduate degree in economics, we can pardon his hyperbole when he says that stamping the paper money of the Great Khan made the process 'as formal and authoritative as if [the notes] were made of pure gold or silver. . . . The money is authentic.' Nevertheless, his evaluation of the paper currency of 13th-century China

was keen enough to touch on matters that will concern us repeatedly in later pages, topics such as the interaction between specie – money in metallic form – and money in its more ephemeral forms such as paper, book-keeping entries and computer blips. Marco Polo's respect for Kublai Khan's stamp on the mulberry currency also raises important questions about money issued by the state and its relationship to the private money that enters today into the huge volume of transferring funds, either by check or electronically, from one private bank account to another.

Kublai Khan's stamp bestowed authenticity and the attributes of gold and silver on his paper money only because Kublai Khan's state was all-powerful – in China. Marco Polo missed that essential point. No one in England would have been able to use the Chinese notes to pay for a pint of beer at the local tavern, and no French king could have paid ransoms with it. Hence, this money did not have the true attributes of gold and silver. In fact, no one outside China would even have known what to do with the mulberry notes unless they were planning to go to China to buy some silk or vases. Otherwise, the Khan's currency was valueless outside China.

People demand that their money must have value. In fact, valueless money is not even money, because it would not serve as a means of payment and would be nothing that anybody would want to accumulate or consider as wealth. Metallic money, or paper money convertible into metal, is usually considered to have more value than a system that uses paper only. As Ma T'wan-lin reminded us, 'Paper should never be money [but] only employed as a representative sign of value existing in metals or produce.' The presumption here is that metals are more limited in supply than paper, which means that metallic systems should prevent money from becoming valueless.

Yet there is a strong superficial resemblance between Croesus turning out great supplies of gold coins by mining the Pactolus and a Chinese emperor who turns on the printing press. Croesus and the emperor both produce an expansion in the supply of money. The difference is in the international consequences. A gold coin would buy a pint of beer anywhere when foreign paper notes would not. Nevertheless, from a domestic viewpoint, the similarities between Croesus and the emperor are much closer than they appear at first glance. The world just thinks about them differently.



Chinese vase



The Great

IN 1661, CHARLES II OF ENGLAND ISSUED AN ORDER-IN-COUNCIL MANDATING THE ADOPTION OF A REVOLUTIONARY INNOVATION IN THE MANUFACTURE OF COINS – THE USE OF MACHINERY IN PLACE OF HAMMERING OUT THE COINS BY HAND.

TWELVE

Recoinage and the Last of the Magicians

The innovation in question was not a gadget that someone invented one day and that went into effect the next. Technological change is often disruptive to established habits. The workers and officials in the national mints who earned their living by doing things the old-fashioned way fiercely resisted the introduction of any modification in the manufacture of coins. Even with a king's order-in-council to force its introduction, a full installation of the new method had to wait more than 30 years, and then came about only because a financial crisis had forced the issue.

It was in the aftermath of that crisis that, quite without warning, the markets for the precious metals in Britain suddenly decided that silver was out and gold was in as the standard for the value of the pound sterling. Without anyone having planned this sequence of events, we shall see that they led, step-by-step, from Charles's order of 1661 to the establishment of a set of hallowed institutions based on a central role for gold. These institutions would rule the world economy throughout most of the 19th and 20th centuries. Thus, although the story here begins by focusing on silver, the golden threads are laced through it from start to finish.

The Lydians and ancient Greeks who first introduced coins into general use hammered their coins out by hand, one at a time. In view of the millions of coins produced over the next 1000 years all over Europe and in the East, it is astonishing that nobody succeeded in developing a faster method. But no one did. Indeed, more than speed of production should have motivated a change. From the very beginning of coinage, the smooth edges of hammered coins had encouraged people to clip or file off tiny pieces of metal that could be accumulated until the quantity was sufficient to be melted down into bullion, which

the clippers then resold to the mint for a fresh supply of coins. The process was too profitable to be stifled by the severe punishments dealt to the clippers who were caught at their work. In the 13th century, Jews were often accused of clipping even when they were innocent. In 1270 alone, 280 Jews were beheaded for the crime.

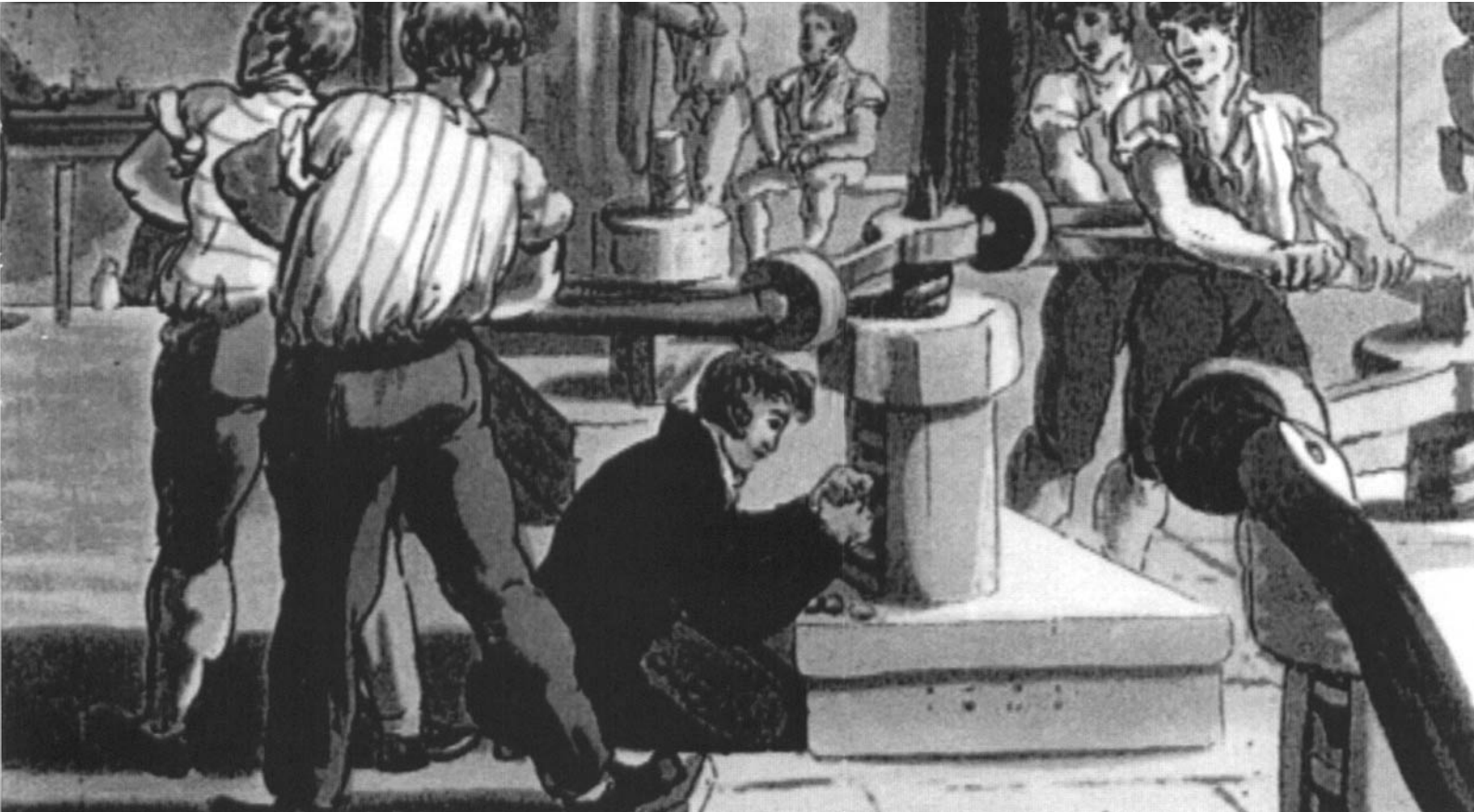
Despite the impact of clipping on the currency, the traditional methods of minting coins continued without any semblance of change until early in the reign of Elizabeth I, when a man named Eloy Mestrell experimented with using horses to power the coin-stamping machines and using this machinery to redesign the edge of the coins so that clipping would be immediately visible. Ingenious as he may have been, Mestrell generated little enthusiasm for his efforts and was fired in 1572. That was not the last to be heard of Mestrell, for he was hanged in 1578 for counterfeiting!

Nevertheless, Mestrell's efforts encouraged others to keep trying. In the 1620s, the chief engraver at the Paris Mint, Nicholas Briot, succeeded in introducing a workable technique for frustrating the coin-clipper. Briot's approach was to edge the coin with either a kind of graining or with inscriptions that would make the clipper's efforts immediately visible, no matter how tiny the piece he clipped. No luck: the traditionalists in the Paris Mint refused to go along, leaving Briot instead of the coin-clippers in a state of frustration.

Briot was not ready to give up. In 1625 he went to England, where he started producing what came to be known as milled coins at the Tower Mint in London. Again there was resistance from the hierarchy at the Mint. As the moneyers were paid by the number of pounds of coins they struck, they were not lightly going to relinquish much of their metal to a Frenchman and his new-fangled devices. Briot's total output in 1631–2 amounted to around 26 pounds of gold coins and 211 pounds of silver coins, compared with more than 4000 pounds of gold and 50,000 pounds of silver turned out by conventional methods; by 1638–9, Briot's silver coin production was approaching 1000 pounds, but this was still a minor effort. The evidence also suggests that Briot's coins were turned out by hand rather than by machine. Briot temporarily faded from view from about 1640 to the 1660s.

In 1645, the French, thinking better of the matter, finally eliminated the hammer at the Paris Mint and replaced it with horse-drawn machines that performed the entire job from rolling out the metal to stamping out the designs on the faces and milling the edges to foil the clippers. That success led the English Commonwealth, headed by Oliver Cromwell, to invite Pierre Blondeau, the chief engineer at Paris, to follow in Briot's path to London. The authorities at the Mint permitted Blondeau to produce milled coins, but from just a small portion of a huge silver treasure that the English had plundered from a captured Spanish galleon; the rest of the silver was transformed into coinage by way of the hammer.

This story has a happy ending. With the restoration of Charles II to the throne of England in 1660, the process of change accelerated. Briot was recalled to London,



Mechanical coin press

Blondeau was given a 21-year contract, and Charles's order-in-council of 1661 declared that 'All coin [was] to be struck as soon as possible by machinery, with grained or lettered edges.' Blondeau died in 1672, before he could complete his contract, but during his employment at the Mint he had the moneyers fully subordinate to him and sworn to secrecy about what they learned from him.

In addition, Charles II issued a warrant on Christmas Eve 1663 to create a new gold coin that would be produced entirely by the mechanized methods. This coin soon became identified as the guinea, because it was fabricated from gold imported from West Africa by the newly established Africa Company. Charles's warrant defined the new coin as equal to £1 (20 shillings of silver). This was a substantial piece of gold – at about eight grams, or a quarter of an ounce, it weighed more than twice as much as the genoin or florin of the 13th century. Appropriately enough, the guinea was stamped with a little elephant, the sign of the Africa Company. The edges were inscribed with a motto that read *Decus et Tutamen*, or Ornament and Safeguard, which is believed to have come from a clasp on the purse of Cardinal Richelieu, Blondeau's former patron. New issues of silver coins manufactured by these methods soon followed after the appearance of the guinea.

The handsome new coins, with their elegant designs and milled edges, were a vivid contrast to the coins that had been circulating in Britain for 60 years since the previous recoinage in 1601. Those were a tattered lot, repeatedly clipped and sadly worn from constant movement from hand to hand. Nevertheless, people continued to use the worn coins because anyone who brought them back to the mint to exchange for new full-weight coins would have received much less in value than the face value that was stamped on them.

Although clippers were hanged 'by the half-dozen', according to one contemporary authority, the gallows seem to have been an inadequate deterrent for such a simple and



*The Royal Exchange and
the Bank of England*

profitable means of getting rich. The clipping continued at a merry pace, especially on half-crowns (two shillings and sixpence) and shilling coins that were larger and thicker than the smaller denominations. In 1652, Blondeau estimated that the weight of the average old coin was 20 to 30 per cent below its original weight. Hopton Haynes, who served as assay master of the Mint, calculated that a bag of coins with a total face value of £100 in 1695 had half the weight that the same bag would have had in 1686.

In a way, the clippers performed a public service, for the coins over time had become unequal in weight and thickness as well as less than perfectly round. Haynes observed that the clippers would file with such skill that the coins became 'as flat and as smooth as the blanks at the Mint are before they have been in the press'. Samuel Pepys recounts an anecdote about a workman from the Mint who made a profit by stamping out counterfeit small-change groats that were as good or better than the true groats then circulating. Groats were small coins equal to four pennies; three groats made one shilling. The workman was caught, but 'He was neither hanged nor burned [because] the cheat was so ingenious . . . and so little hurt to any man in it, the money being as good as commonly goes.'

In addition to the difficulties over the quality of the coinage, problems had also developed in the ratio of gold prices to silver prices. The great silver mines of Mexico and Peru poured their output into Europe in such volume during the first half of the 1600s that the price of silver in Europe began to fall. Thus, a man who wanted to exchange silver for gold would have to offer increasing amounts of silver to obtain an unchanged quantity of gold – or, to put it the other way, an ounce of gold commanded increasing quantities of silver.

Meanwhile, the price of silver in India during the first half of the 1600s was so high that only 9 or 10 ounces of it were needed to buy an ounce of gold, compared with 15 ounces in Britain. The price of an ounce of silver in India, in fact, was far higher than the amount of coinage that an Englishman could obtain by bringing an ounce of silver to the Mint.

The economics of the business was irresistible. There was a great rise in exports of silver, with most of it shipped out by the East India Company. British merchants and manufacturers complained bitterly that the company was shortchanging British goods, especially woolen cloth, which had been called – by contemporary observer Edward Misselden – ‘the flower of the king’s crown, the dowry of the kingdom, the chief revenue of the king . . . the gold of our Ophir, the milk and honey of our Canaan, the Indies of England’. The Indians, unfortunately, had no use for the woolens, but they certainly had a steady demand for silver.

Pressure built up to raise the price of silver at the Mint in order to keep more of it at home and to provide for a larger supply of silver coins. From January 1690 to December 1695, the coinage of silver had amounted to only £19,383. Money was so scarce that the government by proclamation raised the face values of foreign silver coins in order to discourage people from exporting them. Meanwhile, the demand for gold was surging, because it was so profitable to exchange gold for silver that would be exported to Asia. This was one of the forces that would push the price of guinea coins in the market above the officially declared value of 20 shillings.

The result was that Charles II’s impressive revolution in minting methods and associated innovations in currency administration ended up by serving little purpose. Most of the new silver coins disappeared into hoards or were shipped to Asia for fancy prices instead of functioning as British money. Meanwhile, the currency was further reduced by the continued prevalence of clipping, so an increasing number of coins was becoming unacceptable in trade or in payment of debts. If Britain were to have a proper currency for daily use as well as for accumulated wealth, a major reform would be unavoidable.

The big obstacle to a total recoinage was the uncertainty as to who would bear the cost of the difference between the face value on the coins and their true value based on their shrunken weight. Blondeau had warned that the longer the authorities waited, the more the coins would be clipped and the more costly the ultimate outcome would be. Nevertheless, the government dithered with the problem for so long that Charles II had died, James II had been overthrown and William and Mary were occupying the throne by the time the monarchy finally got around to the task in 1696.

Important events immediately preceded the decision to inaugurate a recoinage. As usual, those events revolved around a war, this time a mighty effort to defeat or at least contain Louis XIV of France, the most aggressive European leader since the Romans and up to the time of Napoleon. Hostilities had broken out in 1689 and immediately took

their toll on the Exchequer. By 1697, William III was over £20 million in debt. Taxation, personal loans and lotteries helped to raise revenue, but not enough. The result of the shortfall was the establishment of the Bank of England, an unusual deal between the government and the men of 'quality' who were shareholders of the Bank (that uppercase 'B' forever after identified that bank as the Bank). Under this arrangement, the Bank would lend the government £1.2 million at the moderate interest rate of 8 per cent, in return for which the institution would be established as the first private company to do business as a limited-liability corporation, or so-called joint stock company – in the rapidly growing field of banking.

The founding of the Bank would turn out to be a momentous step in the history of Britain, as the institution over time would steadily increase its influence – even its power – over the banking system and the general economy, the gold stock, and Britain's financial relations with the rest of the world. In later years, the Bank came to be known familiarly as the Old Lady of Threadneedle Street, an expression whose meaning varied from a friendly nickname to a bitter expression of disdain, depending upon the circumstances.

This was, however, just one step in a broad advance in economic growth and increasing financial sophistication in Britain, unleashed after the Glorious Revolution of 1688 had resolved, once and for all, the religious uncertainties surrounding the monarchy and permitting the country finally to get down to business. As credit throughout the British economy expanded at a rapid rate, the inevitable price inflation soon took over, affecting all commodities, then the precious metals, and finally a wave of speculation in the youthful stock market. As always happens in such environments, countless fraudulent issues were uncritically gobbled up by a greedy public in a market where losing money appeared to be an impossible outcome. The economic historian Charles Kindleberger cites 'a proposal by several ladies . . . to make, print and paint and stain calicoes'. (Subscribers must be women dressed in calico.) Daniel Defoe, the author of *Robinson Crusoe*, issued a tract that described the 'Scandalous Trade' as one in which 'There is not a man but will own 'tis a compleat System of Knavery . . . founded in Fraud, born of Deceit, and nourished by Trick, Cheat, Wheedle, Forgeris, Falsehoods and Delusions . . . preying on the Weakness of those whose Imaginations they have ever elevated or depres'd.'

Circumstances were ripe for the speculative fever to spread to the guinea, originally coined with gold that was worth 20 silver shillings. With the silver currency continuing to deteriorate, the rumor mills were busy churning out the news that the recoinage was finally about to happen – but under terms that were still uncertain. Just as in our own times, the uncertainty led to a 'flight to quality'; people began to shift from silver coins into guineas, even if they had to pay a premium to protect the value of their assets.

In March 1694, guineas were trading at 22 shillings, but a year later they were trading at over 25. The guinea hit 30 shillings in June 1695, creating a rush to bring gold to the Mint for coining into guineas, which in turn drove the price of gold from 80 to 109

shillings. At that point, the swollen supply of guineas was tempering the rise in their price – £750,000 of gold was coined in 1695, compared with only £65,000 the year before – at the same time that the leap in the price of gold made the other side of the transaction too expensive to justify its continuance. The government had another weapon to throw at the speculators: the tax collectors announced that they would not accept guineas as payment of taxes at a value as high as 30 shillings.



A clipped coin

The speculation in the guinea finally persuaded the authorities that they could no longer postpone the recoinage. Matters had progressed to a point where no old silver coins were coming into the Mint for coining, because they weighed so little that the quantity of new silver coins given out in exchange would be too far below the face value of the old coins to make the exchange practicable. As Sir Dudley North, a contemporary expert, saw the matter, 'There is a great fear that if clipped money be not taken there will be no money at all.'

Although recoinage occupied the attention of the politicians for most of the 1690s, a long series of parliamentary reports and committees had managed to produce a large volume of words but no legislation. The shift from words to deeds finally began in September 1695 following the publication of *An Essay for the Amendment of the Silver Coins* by William Lowndes, a veteran civil servant and secretary of the Treasury. The Essay was a remarkable document in which Lowndes traced in great detail the entire history of English coinage over the 629 years since the Norman Conquest.

On the basis of this analysis, Lowndes recommended replacing the clipped money with new milled silver coins. These new coins would reflect the diminished value of the silver in the clipped money handed in: the new silver shilling coins would have only 80 per cent as much silver as the old shilling coins they were replacing. This step was the equivalent of raising the price of silver at the Mint, because anyone bringing a given amount of silver to the Mint for coinage would now receive 25 per cent more shilling coins than formerly.

The harm was already done, Lowndes argued, and this step would simply confirm what everyone recognized. Why not acknowledge the reality of the situation? Indeed, without this change, no silver would be brought to the mint for coining. If there was to be a shortage of coins to pay for merchandise or to repay debts, business would be depressed and production would be curtailed. There was no dishonor in this process: the great Queen Elizabeth herself had taken essentially the same step in the 43rd year of her reign. Lowndes was also emphatic that the recoinage should not wait until the end of the war, because such a step 'does but postpone the Cure of a Disease which may destroy us before such remedy can take effect'.

Lowndes's recommendations immediately ran into the opposition of Charles Montagu, the chancellor of the Exchequer, the cabinet member under whom Lowndes worked. Montagu enjoyed the vigorous support of the distinguished philosopher John Locke. Locke had been deeply involved in political activities for many years and was one of the original subscribers to the Bank of England, but he also became one of the most articulate proponents of the Age of Reason; his reputation in this area was launched when he was 58 years old, by his essay of 1690, *Essay Concerning Human Understanding*. His position in the controversy over the recoinage, however, though cloaked in what appeared to be cold logic, was heavily colored by emotion.

Locke presented himself as a so-called hard-money man who would tolerate no tinkering with the traditional official weights and standards of the British coinage, regardless of the physical damage that the coinage might have suffered in the interim. To Locke, a coin that read 'one shilling' on it was a one-shilling coin; that the coin had been clipped down to a shadow of its former self was irrelevant. One shilling stood for a specific weight of silver and should continue to stand for that same specific weight to eternity. To require the holder to exchange old shilling coins for new coins equal to less than one shilling was the equivalent of government appropriation of private property.

Locke argued further that, if Britain was losing silver to foreign countries, the proper solution should be to reduce the demand for imports, even though the country was at war. He personified the issue with a 'Country Farmer who lives within Compass, increases his Stock by diligence and frugality, is never in debt at the year's end but has a balance always to receive at the foot of his accounts'.

Locke's logic for his case was weaker than his zeal, but the zeal was eloquent and persuasive. Montagu and Locke won the day. The battered shillings would be exchanged at their original face value for the new shillings, with the loss to be borne by the Treasury - and ultimately by the taxpayer. The stage was now set for the recoinage to begin.

The British of the 16th and 17th centuries were inordinately fond of labelling major events 'Great'. Henry VIII's attack on the integrity of the currency came to be known as the Great Debasement. When the members of Parliament in 1641 decided to tell Charles I what they thought of his reign, they called their resolution the Grand Remonstrance; the battle to overthrow him that began the following year was known at the time as the Great Rebellion (today as the Civil War). The plague that hit England in 1665 and killed 100,000 people was dubbed the Great Plague. The terrible fire that destroyed most of London in the following year is referred to as the Great Fire. And the recoinage that began at the end of 1695 has come to be known as the Great Recoinage.

Carrying out a great recoinage of the currency is an extremely complicated process, but in this instance it turned out to be extremely messy as well. The king's first proclamation appeared on 19 December 1695, citing that 'The Lords Spiritual and

*Silver and gold globe*

Temporal, and the Knights, Citizens and Burgesses in Parliament assembled, having taken into consideration the great Mischiefs which this our Kingdom lies under, by reason that the Coin, which Passes in Payment is generally clipped. . . . the most Effectual Way to put a stop to this Evil, is to prevent the currency thereof.' The proclamation then proceeds to specify a series of dates after which no clipped coins could be offered in payment to anybody except in payment of taxes or loans to the king. By 2 April 1696, 'No such Money Clipped . . . shall Pass in any Payment whatsoever.'

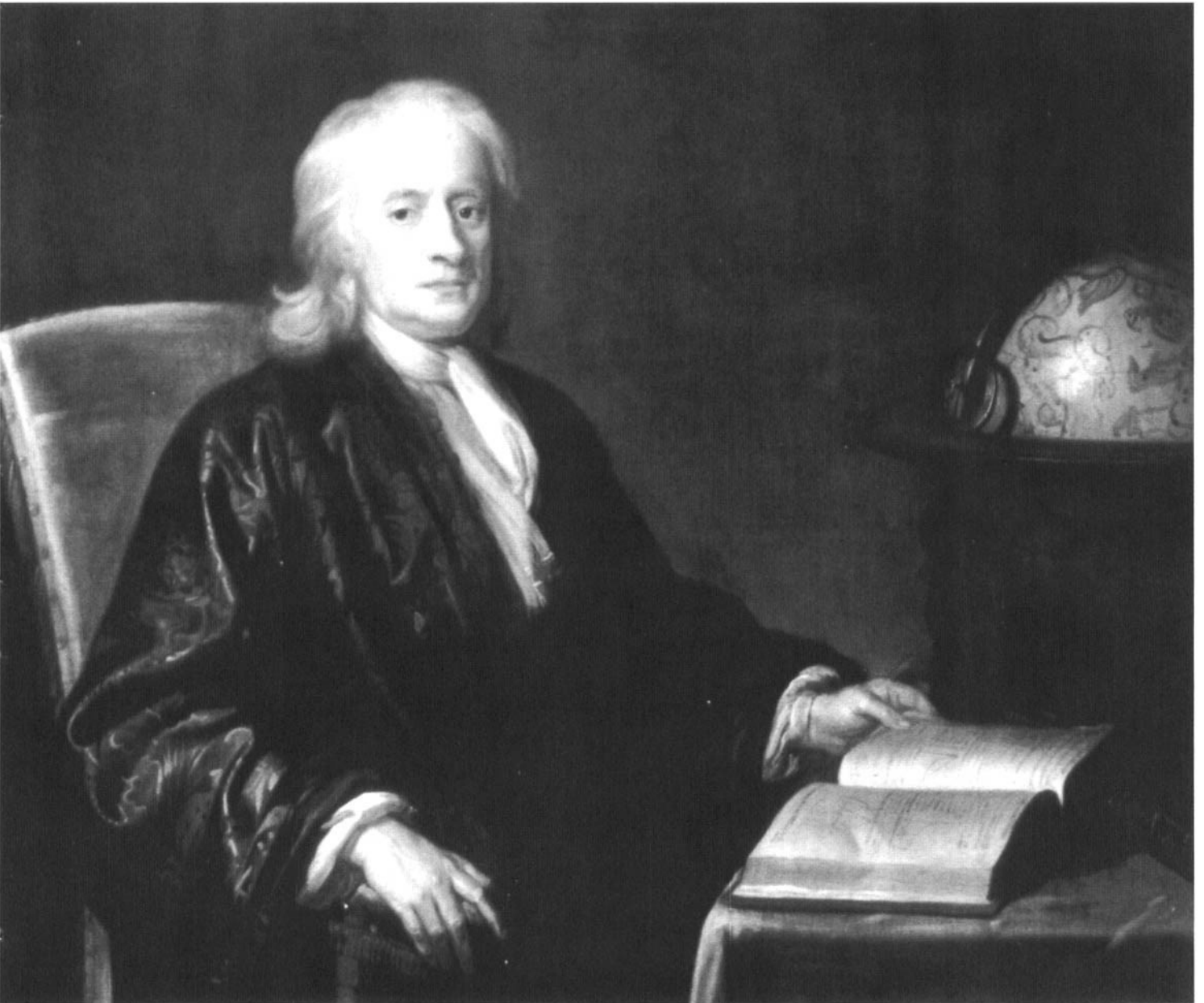
The immediate result was panic. Nobody wanted to take clipped coins in payment, so business ground to a halt, and in any case most people did not pay all their taxes during the short span of time before the stated dates. On 21 January 1696, a month after the original proclamation, Parliament relented by passing 'An Act for Remedying the Ill State of the Coin,' which extended the process into the latter part of June and restored a semblance of order.

Nevertheless, turbulence bubbled up from time to time over the details of the arrangements. Although £4.7 million (containing no more than £2.5 million in silver by weight) had been received at the Exchequer by the final cutoff date of 24 June, well over £2 million more remained in the hands of the public, especially among the little people who had been unable to get to the Exchequer in time. The government losses between what they received and what they had committed to pay out were to be covered by a tax on windows, but the losses had been underestimated by a wide margin, forcing the government to borrow to cover the deficit. The Mint, meanwhile, was so overwhelmed by receipts of clipped money that people who brought coins in for exchange had to depart empty-handed with only a promissory document. The resulting shortage of metallic money disrupted retail trade. At the same time, the market price for silver remained above the face value of the silver in the coins, so many of the new coins disappeared from circulation. It was November before the job was complete, by which time the horses that powered the presses of the Mint had been so busy that £700 had to be paid out just to haul away their manure.

There were riots, petitions, and instructions to justices to 'administer the poor law, but above all to keep the peace'. Edward Bohun wrote from Ipswich on 31 July 1696, that 'Our tenants can pay no rent. Our corn factors can pay nothing for what they have had and will trade no more. . . . Many self murders happen in small families for want, and all things look very black, and should the least accident put the mob in motion no man can tell where it would end.'

In November, 'An Act for *Further* Remedyng the Ill State of the Coin' (my italics) established 1 July 1697 as the final cutoff, after which no more old coins could be brought in for exchange. By the time the long process had come to an end, three years after its commencement, £6.8 million pounds of new milled silver coins had been issued, almost all of them in exchange for clipped money and relatively little in exchange for bullion or plate. The experts estimate that another million of clipped coins owned by poor people never came in for exchange.

When it was all over, the Great Recoinage had restored the weight of British money to what it had been before the Great Debasement, some 150 years in the past. The achievement is impressive despite all the blunders and botches, because the British carried it out while the country was involved in a major war against a powerful enemy, the kind of environment in which preserving the sanctity of the currency is usually among the lowest of national priorities. In the more distant past, sound money had been almost a religion with the English: 'the ancient right standard of England' had been respected with few alterations from the Conquest to the onset of the Hundred Years' War. From Edward III to Henry VIII, however, most people believed it was the king's currency to do with as he preferred.



Isaac Newton



Cambridge

Even so, the depreciation in the pound over the centuries was much more limited than in all other countries – indeed, the uninterrupted history of the pound sterling from the end of the 8th century, when 240 of Offa's pennies were called a pound, to modern times is unique among the currencies of the world. The position that Locke took in 1695 was an effort to restore the older tradition of the sanctity of the weight of the currency. He caught the spirit of the occasion when he took issue with Lowndes's proposed method of recoinage, declaring, 'It will weaken, if not totally destroy, the public faith when all that have trusted the public and assisted our present necessities upon Acts of Parliament . . . shall be defrauded of 20 per cent of what those Acts of Parliament were security for.'

Lowndes was hardly out to defraud the public or use the government to rob their pocketbooks. Rather, he was reluctant to accept the notion that any particular metallic weight was sacred. Greater flexibility in managing the currency, he maintained, might cause less damage in the long run than clinging to some arbitrary number established in the distant past. The Great Recoinage ultimately cost the taxpayers a lot of money, because the new coinage required so much more silver than the silver the Exchequer received from the old coins that were turned in for exchange. Furthermore, the immediate impact of the recoinage on the economy was deflationary, which may have helped the bankers and the rich who had applauded Locke's position but was painful indeed to anyone who owed money.

At this point, an unexpected character appears on center stage: the most distinguished scientist of his age, and surely among the most influential scientists who ever lived, Sir Isaac Newton. In March 1696, just a few months into the turbulence of the Great Recoinage, Newton took on the post of warden of the Mint at the invitation of his good friend Charles Montagu, chancellor of the Exchequer.

What could possibly have been the chancellor's motivation in choosing Newton for such a task? Newton had spent most of his life as a total nerd, uncommunicative, introverted, unapproachable, and far removed from the chaotic realities of politics and finance. Yet he was also a passionate believer in the pseudo-science of alchemy, to which he ascribed profound religious as well as chemical significance, and was described by English economist John Maynard Keynes in 1942 – exactly 300 years after Newton's birth – as 'the last of the magicians'.

An only child, Newton was born on Christmas Day 1642 to the wife of a farmer who had died soon after Mrs Newton had become pregnant. The baby was so small that neighbors commented he would have fitted into a quart pot. Three years later, Newton's mother remarried and left him in the care of her parents for most of his boyhood – a rift that scarred his personality for life.

His talents, however, were visible early. The headmaster of the local school was convinced that Newton should go to Cambridge University immediately after graduating, but his mother kept him working on the family farm for two years before she relented. Consequently, when he entered Cambridge in 1661, he was older than most of his classmates, which only added to his sense of loneliness and isolation. He also entered on the lowest social rung as a subsizar, which meant that he paid his way by cleaning the rooms and emptying the bedpans of his wealthier classmates.

When Newton graduated from Cambridge, he had already determined that his life's work would be to unravel the laws that governed God's universe. He saw no conflict between his immediate scientific ambitions and his intense concentration on the arts of alchemy. According to the fascinating intellectual biography of Newton, *The Janus Faces of Genius: The role of alchemy in Newton's thought*, by the late Newtonian scholar Betty Jo Dobbs Teeter, Newton believed that the whole truth is made up of many parts. The parts can be found everywhere, not just in mathematics and physics but in alchemy, light and even in ancient theology and prophecy. He was an ardent explorer of many such areas, but always with the discipline and rigor of the theoretical scientist. He never did produce gold from his alchemical experiments, but he learned a great deal of value about chemistry in the process.

Newton progressed rapidly up the academic ladder at Cambridge. He received a full professorship in 1669 at the young age of 27 in recognition of achievements that had already earned him the reputation as the most advanced mathematician of his age.

The circumstances that changed Newton's career were totally unexpected. He had moved out into the real world to some degree by becoming involved with the Royal Society, a group organized for the exchange of scientific ideas and research, but his spiritual residence was in

the ivory tower until 1685. In that year, James II, the new king and a Catholic, determined that he would attack the rigidly Protestant Cambridge establishment by forcing them to admit Father Alban Francis, a Benedictine monk, as a student of Magdelene College 'without requiring him to perform the exercises requisite thereunto . . . and without administering unto him any oath or oaths whatsoever'. The master of Magdelene, John Peachell, was a weak man, an alcoholic, and ill-equipped to deal with a situation that was distasteful to the entire university, especially when the king warned him, 'Disobey at your peril.'

This was just the moment when Newton's *Principia* was about to be published and he was approaching the peak of his scientific achievements. But when he heard of the affair of Father Alban, Newton's anti-Catholic sentiments boiled over and he became deeply involved in the struggle. Despite his efforts, he was little help: the king's intimidation of the scholars at Cambridge was merciless, Peachell was dismissed from his post, Newton's own position hung by a thread and Father Alban took up residence at Cambridge. The victory was short-lived, however. Before the monk had received his degree, James had been replaced by William and Mary, who were Protestants. The traditional anti-Catholic religious barriers at Cambridge remained unchanged, while Father Alban departed the premises.

Newton was never quite the same again. The episode suddenly whetted his appetite for public life at a time when he was already world-famous for his scientific discoveries. He ran for Parliament and won. He began to function as a man about town, for the first

*The Tower of London,
the original home of the
Royal Mint*



time including a lot of female companionship. He renewed his acquaintance with Montagu, whom he had met when the latter was a fellow at Trinity College in Cambridge. Then he met John Locke, who refers to Newton as 'the incomparable Mr Newton'. Newton instructed Locke in mathematics and physics, while Locke exposed Newton to political theories and practice. Newton was an apt pupil, for Locke even consulted with him prior to presenting his original report on the recoinage to Montagu in 1695.

Although he brushed off the rumors that were beginning to circulate around Cambridge about his imminent departure, by the end of the 1680s Newton was eager to obtain a post in government. The opportunity finally came along in March 1696, when Montagu informed him that the position of warden of the Mint, at £500 or £600 pounds per annum, 'has not too much business to require more attendance than you can spare'. In addition to the salary, the post received a royalty on every ounce of gold and silver issued by the Mint. Most wardens of the Mint before Newton looked upon the job much as Montagu described it.

Four days later, Newton broke abruptly with his past studies and experiments, packed up his belongings in Cambridge, and moved to London. On 2 May, he started work at the Tower, the home of the Mint since 1300. In a single moment, he ceased his career as an introverted, secretive, mysterious scientist and transformed himself into the first of the policy wonks. The break was astonishing in itself, but the choice of new career appears even stranger: imagine Albert Einstein leaving Princeton to become second-in-command at the Bureau of Printing and Engraving in Washington DC.

When Newton took up his responsibilities at the Mint, its head, the master, was Thomas Neale, a lazy man with a strong taste for drink. Neale and the Mint staff hardly knew what hit them upon Newton's arrival. Even Montagu himself had no idea that this theoretical academic would turn out to be a motivated, skillful, energetic and demanding administrator who would devote himself not just full-time, but overtime, to the task at hand.

For the first few weeks on the job, Newton took up residence in a tiny dank room that was right next door to the clanking presses being worked by 300 men and scores of horses (remember the £700-worth of manure produced there). He was on the scene when work began at 4 am and when the night shift took over, six days a week. He studied the entire process in great detail and continuously discovered methods to accelerate the output of coins. Later on, he bought himself a nice house in London and began to live like a gentleman, but his fiendish attachment to his work at the Mint persisted.

Despite 16-hour days, Newton was also educating himself into an economist. He spent as much time as he could with such people as Locke, Montagu and Lowndes and read everything he could find on the subject. Then he started writing – voluminously – on the history of economics, commerce and currency systems. Lacking a photocopying machine, he even employed young men to make duplicate copies of everything he wrote. Through it all, he was maneuvering to displace Neale and become the master of the Mint. He made



*A George III Guinea
dated 1787*

himself as visible as possible, clashing with government contractors over the prices they charged the Mint and then lustily entering into conflict with the governor of the Tower, where the Mint was located. He was tireless in overcoming bureaucratic inertia (a physical principle that was integral to his scientific work) and went so far as to use secret agents around the countryside to root out the villains who continued to clip the coinage. This once puritanical introvert began to frequent the lowest public houses in the city to arrange secret meetings with informants from the brothels and gin houses. He carried out interrogations and attended hangings, always keeping detailed accounts of everything.

In December 1699, Thomas Neale died and Newton at long last achieved the promotion he had desired for so long. He became master of the Mint.

We must now briefly retrace our steps. During the Great Recoinage of 1695–6, the government attempted to bring down the bloated price of the guinea by refusing to accept guineas in payment for taxes at a price higher than 22 shillings. Nevertheless, at 22 shillings it was still profitable to import gold to be coined into guineas, exchange them for silver coins, and melt the silver into bullion for export to the East. As the basic day-to-day coinage of British was silver, and as silver was the standard that defined the pound sterling, this process could not be allowed to continue indefinitely. The difference between the two metals as coins and as bullion was unsustainable.

Something had to give. There was no doubt that it was the price of gold that was going to have to back down. As a special Report by the Council of Trade put it on 22 September 1698, 'For it be impossible, that more than one Metal should be the true Measure of Commerce; and the world by common Consent and Convenience [has] settled that Measure in Silver; Gold, as well as other Metals, is to be looked upon as a Commodity . . . its value will always be changeable.'

In February 1699, the Treasury reduced the acceptable price of guineas to 21s 6d, hoping thereby to halt the process. The gold imports fell off slightly, but then a record import of £1.5 million came into England in 1701, and silver continued sailing off toward Asia. Newton in his position of master of the Mint issued reports on the problem in both 1701 and 1702, pointing out that, at current rates of exchange, a guinea's weight of gold was worth from 9 pence to a full shilling (12 pence) higher than in the other countries of Europe. His strong recommendation was to reduce the guinea further to 21 shillings. Renewed fighting with France cut off the imports of gold for a while and made any further changes in the coinage unnecessary until the Treaty of Utrecht was signed in 1713. At that point, the flood of gold imports gathered renewed strength. Over £4 million came in over the next three years. When the East India Company exported three million

ounces of silver in 1717, the authorities once again turned hopefully to the wisdom of Sir Isaac Newton.

Newton's 'Representation to the Right Honourable the *Lords Commissioners* of His Majesty's Revenue' has become a famous document in the history of money. The reading of it is a tedious business, and the essence of the content is no more than simple arithmetic reciting the values of various weights of gold and silver in different countries. A great scientist's mind was hardly necessary for this particular task. Nevertheless, his words have earned their immortality from the timing of their appearance and the recommendations with which he finishes his essay. Among the latter was that 'There seems nothing more requisite, than to take off about 10d. or 12d. from the Guinea, so that Gold may bear the same Proportion to the Silver Money in England which it ought to do by the Course of Trade and Exchange in Europe.' On the basis of Newton's advice, the Treasury issued a proclamation on 22 December 1717, prohibiting anyone to pay or receive the gold guinea coins at a value different from precisely 21 shillings. The outcome was not the expected outcome. Newton had it wrong on two scores.

First, as matters turned out, 21 shillings was still too high a value for the guinea: defying Newton's predictions, the imports of gold and exports of silver continued, even though at a reduced rate. The process persisted, in fact, for about 30 years, beyond 1717, by which time full-weight silver coins had disappeared from circulation.

Second, Newton was confident that the laws of supply and demand would solve the matter so that the problem would simply go away with the passage of time. He was confident that the continued increase in the supply of gold would bring down the price of guineas as denominated by silver shillings. 'If things be let alone,' he wrote, 'till silver money be a little scarcer, the gold will fall of it self . . . And so the Question is, Whether Gold shall be lowered by the Government, or let alone 'till it falls of it self, by the Want of Silver Money?'

That is not what occurred at all. Newton's forecast turned out to be wrong on a more fundamental level than merely expecting the laws of supply and demand to bring everything to rights. He was correct that, in the end, gold would have to decline in value relative to silver. But, like many economists since then, he went astray in assuming that the future would look like the past. Economics is evidently a lot more difficult than physics, even for a genius like Newton.

The unexpected happened: the price of gold did not 'fall of itself'. In fact, it did not fall at all. Instead, the guinea held steady at 21 shillings, while silver coins began to exchange at more than face value. Gold still lost value relative to silver, but the price that moved to accomplish that shift was the price of silver, not the price of gold. Although the ultimate outcome was the same either way, *the markets themselves, without any acts or orders-in-council or representations, had silently but decisively established gold in place of silver as the standard for the pound.*

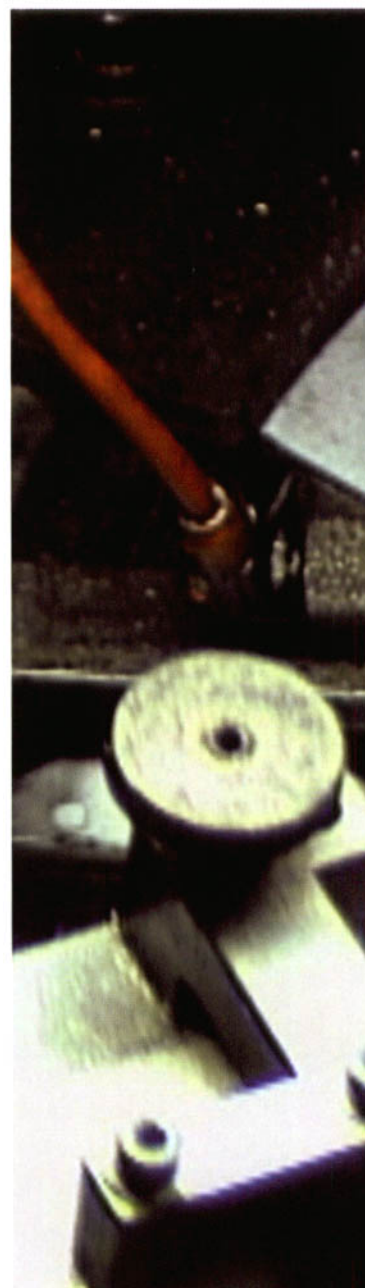
As so often happens, the markets were way ahead of the officials. As late as 1730, John Conduitt, Newton's successor at the Mint, was still reciting the old story that 'Gold is only looked on as a commodity, and so should rise or fall as occasion requires. An ounce of fine silver is, and always has been, and ought to be, the standing and invariable measure between nation and nation' (*italics added*). Reality, however, had moved strongly and decisively in the opposite direction since 1717. For more than 200 years, the price of gold in Britain would remain set at £3 17s 10 ½d (derived by translating the 129.4 grains of gold in the guinea into its money price of 21 shillings), while the price of silver would succumb to violent fluctuations. Until the official devaluation of the pound in the terrible crisis of 1931, £3 17s 10 ½d became a kind of magic and worshipful combination of numbers that governed British monetary policy.

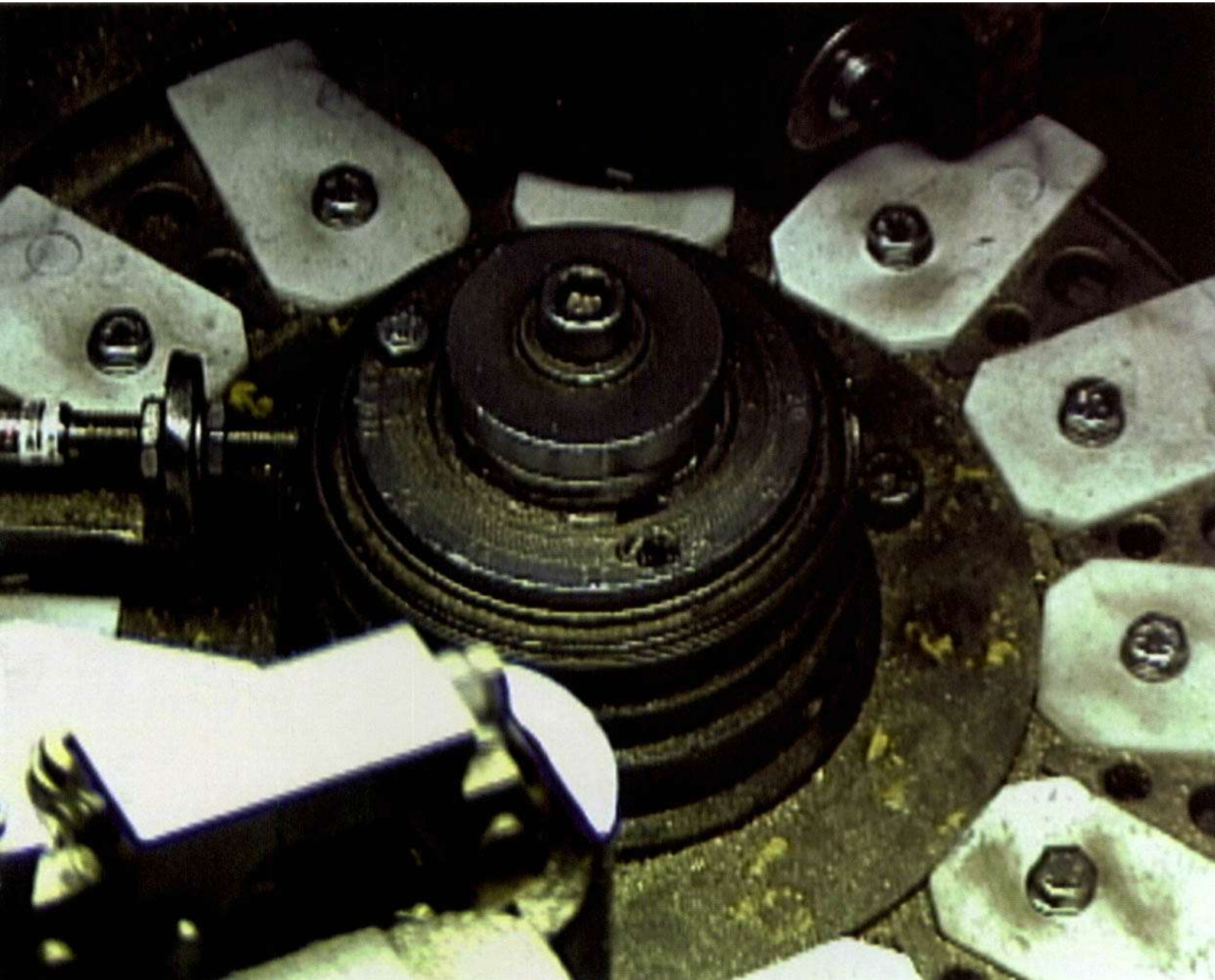
This unanticipated outcome was a direct reflection of the increasing popularity of the guinea. The guinea's consistent weight and fineness made such a vivid contrast with the rotten state of the silver coinage up to the Great Recoinage that people preferred to accept the guinea wherever possible. Bankers held it as reserves, tax collectors welcomed it in order to avoid the arguments about what a worn silver piece might be worth, and economic activity in Britain at that time had developed to a point where a large-denomination coin such as the guinea was no longer just an inconvenient curiosity.

From the moment when Elizabeth I ascended to the throne in 1558 to the foundation of the Bank of England in 1694, a period of 136 years, the Mint had issued no more than £15 million in gold coinage, of which half was in guineas that appeared after 1663. During the 45 years from 1695 to 1740, the Mint produced £17 million gold coins. The story on silver is precisely the opposite: £20 million in the earlier period versus £1 million in the latter.

The guinea came to a strange end. A direct descendant of Croesus' stater, Constantine's bezant, the ducat, the genoin and the florin, the guinea remained the basic gold coin of Britain for another 100 years after the events just described. In 1821, under George IV, the guinea came to an official demise when it was replaced by the sovereign, which was set equal to precisely £1 instead of the awkward 21 shillings of the guinea.

The guinea lingered on not as a coin but as a denomination or offbeat unit of account. Prices quoted in guineas had snob appeal and served as reminders of a great past. Doctors on Harley Street quoted their fees in guineas, and fine jewelry and clothing were priced in the same fashion. But 21 shillings no longer made sense as a denomination when Britain joined the rest of the world and abandoned their historic shillings and pence for the metric system in 1969. The guinea finally vanished, remaining as a romantic memory or, on occasion, as a precious Christmas gift to children from doting grandparents who still owned a few of the beautiful coins stamped with the little elephant.





A modern mint



ON 22 FEBRUARY 1797, THREE FRIGATES FROM THE FRENCH NAVY SAILED INTO THE HARBOR OF THE TINY FISHING VILLAGE OF FISHGUARD ON THE SOUTHWEST COAST OF WALES AND PROCEEDED TO LAND ABOUT 1200 ARMED SOLDIERS. THIS LITTLE FORAY WAS CONFRONTED ALMOST IMMEDIATELY BY THE LOCAL MILITIA UNDER THE COMMAND OF LORD CAWDOR.

THIRTEEN

The True Doctrine and the Great Evil

The French also caught sight of an approaching troop whose red cloaks and tall black hats convinced them that they were facing a contingent of the crack British unit, the Grenadier Guards – an even more serious danger.

This fearsome band was nothing of the sort. The French soon discovered that it was nothing more than a gathering of Welsh women in traditional festival costumes. The costumes did not inhibit the women from beating up the bewildered French soldiers. One of these women, Jemima Nicholas, was so adept at wielding her pitchfork that she was credited with the capture of 14 French soldiers. The French commander, an Irish-American general, was compelled to submit to an ignominious surrender.

Rumors of an impending French invasion had been circulating for several months, with a series of reports from Paris referring to that danger. French warships had gathered at Brest at the end of 1796 and had headed off toward northern Ireland around Christmas, evidently hoping that the Irish would join in their invasion. On 21 February 1797, there was a false rumor of French ships off Beachy Head on the southeast coast of England, which provoked English warships to head out to sea from Portsmouth. The next day, the battle of Fishguard took place.

Just what the French had in mind with this expedition remains obscure. Napoleon was off invading Italy, and Fishguard was about as far away from British power centers as anyone could get. The most reasonable explanation is that the French were making a low-cost but deliberate effort to create a panic in Britain. Seen from this viewpoint, their project was a smashing success.

*The French leaving the
frigate Vengeance to row to
Fishguard*



Jemima Nicholas. Modern portrait.

The excitement provoked by the French invasion launched a rush to withdraw gold from the banking system as frightened citizens stormed banking offices to cash in their paper money, almost all of which consisted of banknotes issued by the Bank of England as well as notes issued by smaller banks. While the Bank of England notes were the most widely accepted form of paper money in circulation – and traditionally referred to as Bank notes – the greatest fear was over what would happen to those small banks and the acceptability of their banknotes as means of payment in the event of an invasion.

In contrast to the paper currency, gold was perceived as the ultimate money, indestructible in value and forever acceptable no matter who would be running the government, even if it were the French. The very idea that the Bank's gold reserve was rapidly diminishing therefore served as a self-fulfilling prophecy as people dashed to the banks to convert the paper notes into gold while there was still some gold left to withdraw.

Jemima Nicholas. Tapestry.



On Saturday, 18 February, even before the incident at Fishguard, fears of a French invasion had provoked a run on the banks in Newcastle (some 350 miles from Fishguard), forcing the local banks to shut their windows on the 20th. The panic soon spread to London and other major centers. The withdrawals were draining the Bank of England's gold stock at the rate of £100,000 a day, out of a gold reserve that had already shrunk under wartime conditions from £7 million at the end of 1794 to £5 million at the end of 1795 and then to only about £2 million at the end of 1796.

Word of the 'invasion' at Fishguard reached London on the morning of Saturday the 25th, three days after the fact, forcing the management of the Bank to face the unprecedented and unpleasant prospect of refusing to redeem Bank notes in gold. The directors informed Prime Minister Pitt, who sent an urgent message to George III at Windsor on Saturday evening to join a council to be held at the Bank on Sunday. In addition to the king, the Sunday meeting included Pitt, the lord chancellor, the governor and deputy governor of the Bank, and two of the directors. At the conclusion of the meeting, the government issued an order-in-council whose most important passage read as follows:

It is indispensably necessary for the public service that the directors of the Bank of England should forbear issuing any cash [i.e. gold] in payment until the sense of the Parliament can be taken on that subject and the proper measures adopted thereupon for maintaining the means of circulation.

On 9 March, the House of Commons transformed the order-in-council into a full-fledged act, the so-called Restriction Bill, which indemnified the Bank against the legal consequences of refusing to pay out gold in exchange for the Bank notes. The bill also made it official that all payments in Bank notes were to be 'deemed payments in cash'.

A public relations campaign swung into action at once to reassure the public and to prevent the gold crisis from turning into an even more intense panic. On Monday, 27

Tapestry showing the French attack on Fishguard.





The Royal Oak. The pub in Fishguard which was the site of the French surrender

February, *The Times* of London published a leader invoking the spirit of the embattled English at the time of the Spanish Armada, when the danger to life and property was much greater. At noon that day, the leading bankers and merchants of London met at the Mansion House, home of the lord mayor, where they passed a unanimous resolution declaring that it was their intention to receive without hesitation notes of the Bank up to any sum of money that would be owed to them and that they would make their own payments in Bank notes as well. This resolution, which received 4000 additional signatures, was published in newspapers throughout the country. The directors of the Bank then issued a notice declaring that the Bank was in a 'most affluent and prosperous situation, and such as to preclude every doubt as to the security of its notes', with assets (primarily claims on borrowers) some £4 million in excess of liabilities, not even counting the debt owed by the government of close to £12 million. This reassurance about the financial condition of the Bank was critically important, as many smaller banks had either failed or suffered a dangerous deterioration in their condition during the excitement provoked by the original declaration of war against the French in 1793.

Despite the soothing words about the Bank, the order-in-council of 26 February and the Restriction Bill that followed were shockers. The crisis was unique. Governments had often devalued and debased their coinages in other wars, but then only coins were involved. The Chinese had made a mess with their paper currency, but then no coins were involved. This was the first time that the markets had attacked a paper currency freely convertible into gold coins or bullion, a right that had existed for over a century. The Bank of England itself had been established as the paradigm of financial soundness and responsibility. The Bank's notes had been 'as good as gold'. Indeed, Bank of England currency, decorated with the corporation's seal and engraved with Britannia seated on a bank of money, was deemed more acceptable than obligations of the government itself.

Suddenly, all that changed. With the vaults almost bare of gold, the Bank notes were now no more than pieces of paper. The claims on borrowers may have exceeded the Bank's



liabilities, but those claims were far from the same thing as gold; even if the borrowers repaid all their loans, the Bank would still have only Bank notes on hand to pay out.

The impact of this unexpected rupture can be gathered from the memoirs of a Scots banker, Sir William Forbes, who recalled the experience in his *Memoirs of a Banking House*, published in 1803. Forbes describes what happened on the morning of 1 March when the first word of 'this interesting event' reached Edinburgh: 'Our counting house . . . was instantly crowded to the door with people clamorously demanding payment in gold of their interest-receipts . . . they were mostly of the most ignorant classes . . . all bawling out at once for change. . . . Both gold and silver specie was hoarded up and instantly disappeared.'

Nevertheless, the way the British dealt with these matters, once the crisis had passed, was as extraordinary as the decisiveness of the authorities at the moment of a crisis that none of them had ever faced before. As Forbes continued his account, 'It was a matter of agreeable surprise to see in how short a time after the suspension of paying in specie, the run on us ceased [and] how quietly the country submitted . . . to transact all business by bank notes for which issuers give no specie as formerly.'

There was no argument against the principle that gold must continue to be the central foundation of the monetary system, and that some day redemption in specie would be restored. The language of the long sequence of government orders that was to follow gives every indication that most people believed that the arrangements were temporary, with normal conditions soon to be restored. The entire objective was to reach a point where notes could once again be converted into gold on demand. The embargo would endure for 24 years, until 1821, a length of time that even the most pessimistic observers never anticipated. But widespread confidence in an ultimately happy ending to the tragedy meant that acceptability of the Bank notes was at no point brought into question anywhere. The notes continued to function as though nothing had happened to the gold reserves.

Let us step back for a moment, in order to gain better perspective on these dramatic events. Some brief history and overview of the mechanics of the British monetary system might be helpful.

It is important to recognize that the British monetary system – like the British political system – had evolved by trial and error, without the rigidities of legal stipulations and regulations that characterized the French system of government and that Napoleon and his successors did nothing to diminish. Public and private money circulated together and reinforced each other. The government's money was the coinage, most of which consisted of the golden guinea, although silver coins and token coins of smaller denominations also circulated. (In 1816, Parliament ruled that silver coins henceforth were tokens that would be acceptable based only on their face value, not their metallic content.) More than 100 years before the crisis of 1797, however, private paper money had begun to substitute for coins in large transactions. Bills of exchange, as we have already seen, were often endorsed over from one holder to another and thereby became a kind of paper money. In addition, many people who owned gold coins would deposit them with a goldsmith for safekeeping, accepting in exchange a receipt for the gold that could then be used as a means of payment because it was redeemable in gold on demand. Indeed, the total output of gold during the century as a whole averaged only 20 tons a year.

More important, banking grew at a rapid rate throughout the 18th century, and it was customary for banks to pay out the proceeds of their loans in the form of promissory notes to customers; these notes, in a variety of denominations and engraved and watermarked to defy forgers, would then circulate from hand to hand as money. When the Bank of England was established in 1694 and made the original loan of £1.2 million to the government, part of the proceeds paid out to the government was in the form of Bank notes, which the government used to purchase supplies for its campaign against Louis XIV. Those notes circulated among businesses and the public as money but were also held by other banks as a reserve to cover withdrawals of deposits.

The consequences of this unplanned structure were profound. As the volume of banknotes replaced the coinage of government in daily circulation, the supply of money in daily use was now joined directly to the volume of credit provided by the privately owned banking system. In essence, the private sector overtook the public sector as the primary engine of money creation. Although most money in use today is in the form of checking accounts rather than paper currency, the basic 18th-century relationship between bank credit and money supply remains intact. By the end of the 18th century, in fact, a rising share of money was already being held in the form of checks and deposits much as we know them today.

These developments do not mean that gold coins or bullion ceased to be important because their hand-to-hand circulation declined. English money was clearly defined by its weight in gold. Even though guineas resided for the most part in the safes of goldsmiths or the faithfully guarded vaults of the Bank, the assurance that they were there was what



The Bank of England

appeared to make the system work. The various forms of private paper – bills of exchange, goldsmiths' receipts, and banknotes issued by commercial banks throughout the country – could always be exchanged for the notes of the Bank of England, and Bank notes could always be exchanged for gold, or specie as it was often called in those days. For example, before 26 February 1797, anyone with £210 in notes could go to the Bank at any time and receive 200 golden guineas in exchange.

Most of the time, nobody bothered. Yet when business was booming and the price of gold in the marketplace was rising, or when sterling was losing value in the foreign exchange markets, 200 golden guineas could command more than £210 in the City markets or the equivalent of more than £210 in foreign financial markets. At that point, it would be profitable to convert £210 in notes into 200 guineas and then exchange the guineas for some larger amount in the financial markets.

In 1783, after the end of the American Revolution, the Bank had made an energetic effort to halt such a process before it had much opportunity to begin. Private business activity was expanding rapidly, with prices about 10 per cent higher than they had been four years earlier, and the Bank's stock of gold coins began to shrink. In response, the directors of the Bank of England refused additional credits to merchants and other borrowers, who had to turn elsewhere for credit. The result was a sharp rise in interest rates, a cooling of speculative fevers, an immediate improvement in the exchange rate between sterling and continental currencies, and a return flow of guineas to the vaults of the Bank. Thus, the staunch faith in gold had transformed the shiny metal that most people viewed as the ultimate form of wealth into a powerful vehicle of checks and balances (a role that gold would continue to play, in one form or another, and not just in Britain, for the next 174 years). What everyone considered to be the riskless asset seemed to subdue the risks in the entire economy.

The crisis of 1797 and the Restriction Bill that sundered the linkage between Bank notes and gold tore apart this unofficial but powerful system of controls. With the golden counterweight no longer operative, the economy was now operating completely on paper currency. Hien Tsung's inadvertent innovation, which, with 'continentals' and *assignats*, had caused disasters in financing the American and the French revolutions just a short time back, was now about to make its debut on the more conservative shores of Britain. In the words of Lord Lansdown, contemplating the consequences of the Restriction Bill, 'A fever is as much a fever in London as in Paris or Amsterdam; the fall will be slow perhaps, and gradual for a time; but it will be certain.'

The original issue of the *assignats* in December 1790 had amounted to 800 million livres; on 23 October 1795, over 20 billion were outstanding. The Paris riots that ensued paved the way for Napoleon's rise to power and forced the government to abandon the worthless *assignats* for a metallic-based system based on gold and silver.

Where did the French gold come from? Much of it was a return of capital that had fled the country during the assignat regime. In fact, the events in France were the main reason that the Bank of England's gold reserves were so low when the Fishguard adventure occurred: gold that had earlier been transferred to Britain by frightened Frenchmen now began to return to France, driving the Bank of England's gold reserve from £6 million in early 1795 to only £2 million in early 1797. Napoleon had inadvertently laid the groundwork for the panic that ensued from the Battle of Fishguard.

In addition, however, acting in the tradition of the great conquerors of the past, Napoleon lost no time in launching his career as a ruthless accumulator of monetary treasure (among other things) from vanquished nations. In a note to the Directoire Executif on 1 June 1796, he informed his colleagues that 'Two millions of gold are en route [from Italy]. They leave tomorrow with 100 carriage horses, the most beautiful one can find in Lombardy. They will replace the mediocre horses that pull our carriages.'

Napoleon remained steadfast as a 'hard money' man throughout his reign. He had no choice. If he had even whispered the possibility of issuing an inconvertible paper currency, the nightmare of the assignat experience would have provoked an immediate flight of all the gold in France to safer havens abroad. Thus, Napoleon's keen monetary management skills enabled him to succeed where every other leader had failed. This was probably the only major war in history to be conducted without currency depreciation in one form or another.

Lord Lansdown was right: the fall was slow and gradual, but inflation ultimately took hold in Britain. Prices in 1802 were lower than they had been in 1800, but they climbed 30 per cent between 1802 and 1807 and by another 15 per cent over the next three years. Prices had doubled from their 1797 level when Napoleon succumbed at Waterloo in 1815.

The inflation was associated with a substantial increase in money and credit. No longer constrained by the gold supply, the Bank's loans to business – so-called commercial

bills under discount – more than quadrupled between 1797 and 1810. The Bank has never supplied the financial markets with such an explosion of credit except during the extraordinary conditions of the two world wars of the 20th century. By no coincidence, the Bank's note issue expanded from approximately £10 million to £25 million over the same period of time, with half the increase having occurred just since 1807, while deposits at the Bank were increasing at about the same pace. The Bank's holdings of coin and bullion fluctuated with the fortunes of war but were at no point equal to as much as 50 per cent of the Bank notes outstanding; in 1794, before all these troubles began, the gold reserve had been equal to 70 per cent of the outstanding notes.

Beginning in 1808, the price of gold began a rapid ascent. By 1809, the gold in a guinea was fetching £4 10s an ounce in the marketplace, well above the price of £3 17s 10 ½d that had defined the value of gold when Isaac Newton was pondering the matter back in 1717. Sterling was also losing value relative to the currencies of other nations – toward the end of 1809, the pound was exchanging in Hamburg, Amsterdam and Paris from 16 per cent to more than 20 per cent below its official par values. The result was a 50 per cent decline in the Bank's holdings of gold coins and bullion between February 1808 and August 1809, even though these holdings had been replenished since the low point at the enactment of the Bank Restriction Act in 1797. The restoration of convertibility appeared further off than ever.

The financial community was outraged. On 29 August 1809, a 38-year-old stockbroker, as spokesman for this community, submitted the first of three letters on this matter to the *Morning Chronicle*, complaining that the public 'do not seem to be sufficiently impressed with the importance of the subject, nor of the disastrous consequences which may attend the further depreciation of the paper.' His name was David Ricardo, and this was the first time his name had appeared in print.

Ricardo was born in 1772, when Adam Smith was 50 years old and Thomas Malthus, Ricardo's beloved friend and unremitting intellectual opponent, was six. (Ricardo first met Malthus in 1809 at the very moment he was sending in his letters to the *Morning Chronicle*.) Ricardo's father was a Jewish merchant banker and stockbroker – stock jobber, as the British call it – who took his son in as an employee when the boy was only 14. The firm prospered, even though it had to limit its trading activity to a section of the Royal Exchange known as Jews Walk. Ricardo remained in business with his father for seven years, until he fell in love with a Quaker girl. At the age of 21, he broke with his family, married Miss Wilkinson, adopted the Quaker religion, went into business at the Exchange on his own and lived happily until he died suddenly at the age of 51.

As the British government debt climbed ever higher during the course of the Napoleonic conflict, Ricardo became one of the major underwriters of those government securities each time they were issued to the public. Like investment bankers today who like to take care of their friends, Ricardo would occasionally allot a small cut of these deals to his chum Malthus, a parson and academic of modest means whose fame as an



economist would in time rival Ricardo's. In 1815 – at the moment when the battle of Waterloo was approaching – Malthus could not stand the strain of being exposed to what might happen to his little nest egg if Wellington were to lose. He pleaded with Ricardo 'to take an early opportunity of realizing a small profit on the share you have been good enough to promise me'. Ricardo obliged but held on to his own much more substantial position. For Malthus, Napoleon's defeat at Waterloo was good news and bad news: good news as it was for all Britons but bad news for the enormous opportunity missed. For Ricardo, it was 100 per cent stupendous news. Just two years later, he published his magisterial work, *The Principles of Political Economy and Taxation*. The remarkable friendship with Malthus survived these events unscathed. After Ricardo's death, Malthus declared that 'I never loved anybody out of my own family so much.'

About six months after the publication of Ricardo's letters in the *Morning Chronicle*, and after extended debate in Parliament and the press, a little-known member of Parliament named Francis Horner moved for the establishment of a parliamentary committee to look into the whole matter in detail, to examine expert witnesses, and to prepare a report and recommendations

The Houses of Parliament, London

for the House of Commons upon the completion of their task. What came to be known as the Bullion Committee numbered 22 members. The most active of these, and the primary authors of the final report, were Francis Horner, Henry Thornton and William Huskisson.

The report was a bestseller for its time; it had sold out three months after its appearance in August. For someone who encounters it nearly 200 years after its publication, it is an extraordinary document. A reader could disagree with the hard line taken by the authors and still be startled by the sophistication of the economic analysis. At its best, the report accurately reflects the most important economic ideas developed by David Hume and Adam Smith even while anticipating concepts that would appear in theory books only years in the future, especially in relation to the theory and practice of the gold standard system that would develop over the course of the 19th century. Indeed, the clarity and thoroughness of the presentation of monetary theory matches Milton Friedman at his best. A document of equivalent quality written by today's members of Parliament or the US Congress would be inconceivable.

This Bullion Committee's most important conclusion echoed the principles about the role of gold originally set forth by David Hume and Adam Smith. Their report provides the first and perhaps the most authoritative justification for the establishment of the gold standard as the superior system for managing an economy's money supply. The entire thrust of the statement is to throw up a sharp contrast to what can happen when money is not convertible into metal and such decisions are left to just plain human judgments.

After pointing out the power over the economy that the suspension of cash payments transferred to the directors of the Bank, the committee goes on to contend:

When the currency consists entirely of the precious metals, or of paper convertible at will into the precious metals, the natural process of commerce, by establishing Exchanges among all the different countries of the world, adjusts, in every particular country, the proportion of circulating medium to its actual occasions. . . . If the natural system of currency and circulation be abandoned, and a discretionary issue of paper money substituted in its stead, it is vain to think that any rules can be advised for the exact exercise of such a discretion.

In short, the markets know best; their signals must determine policy. The committee's logic leads them to a clear and unqualified recommendation: 'That the system of the circulating medium of the Country ought to be brought back, with as much speed as is compatible with a wise and necessary caution, to the original principle of Cash payments at the option of the holder of Bank paper.' Nothing else would provide 'sufficient remedy for the present, or security for the future'. These words would continue to echo through the endless debates over money that lay ahead.



A modern sovereign

The House of Commons did not provide an opportunity to debate the findings of the Bullion Committee until the spring of 1811. In the meantime, pamphlets on the subject appeared in great quantity, instructing the public on the fine points of the issues involved. This was also the moment for the appearance of Ricardo's *The High Price of Bullion and the Depreciation of Bank Notes*. The topic became so hot that one newspaper bribed a clerk at the Bank to steal a confidential copy of the names of the Bank's borrowers, which appeared the next day on the paper's pages.

On 6 May 1811, by which time the note issue had risen by another £2 million and the price of gold had extended its climb, Francis Horner finally transformed words into action by submitting to the House 16 resolutions designed to implement the proposals contained in the Bullion Committee's report. Horner began by tracing the sequence of events in the crisis that led to the formation of the Committee, after which he carefully defined the legal definition of a pound sterling in terms of its weight in gold and declared that Bank notes were promises to pay in such money.

His resolutions ended with two specific recommendations. First, that for as long as the suspension of convertibility continues, 'it is the duty of the Directors of the Bank of England to advert to the state of the Foreign Exchanges, as well as to the price of Bullion, with a view to regulate the amount of their Issues.' The second proposal urged all possible haste in returning to convertibility by moving the end of the suspension of cash payments from 'Six Months after the Ratification of a Definite Treaty of Peace', as originally stipulated in the Restriction Bill, to 'Two Years from the present Time'.

Horner's resolutions were debated in the House for four days. The first 15 resolutions were defeated by a vote of 150 to 75; the final resolution to change the cut-off date went down by 181 to 47.

What happened? Despite all its eloquent arguments, the Bullion Committee had simply ignored the brutal reality that Britain was engaged in the greatest war in history up to that time – a true precursor to the total wars of the first half of the 20th century. The most important financial objective for the government was to encourage the highest possible level of production of food, coal, ships, guns and ammunition, and uniforms. Consequently, the leadership of the government was reluctant to put any brakes on the money supply as long as the war continued and the government was spending more money than it was receiving in tax revenues. Although the Bank had increased its holdings of government debt by very little up to 1810, its position in government securities would more than double between 1810 and the end of the war in 1815. The chancellor of the Exchequer was vehement on the subject, announcing that the committee's recommendations were equivalent to 'a declaration that we must submit to any terms of peace rather than continue the war'.

Other members of the House, concerned about precisely the problem that agitated Thornton, were reluctant to return to the old relationship between the pound and gold when prices were already so much higher than they had been when the rupture took place –

the very issue that would nearly tear Britain apart in the 1920s. A third group resented the attack on the management of the Bank and rose to their defense. It is amazing that the most vocal part of the opposition came from sheer incredulity that the Bullion Committee could have figured out matters properly and that the Bank's notes were actually 'depreciating'.

The most dramatic response to the extended debate, and to the defeat of Horner's 16 resolutions, was provided by a young nobleman named Peter King. 'For the defence of his property', Lord King informed his tenants that he would no longer accept Bank notes at face value in payment of their rents. He maintained that there was no reason why he should suffer just because prices had risen so much since they had signed their various leases. Consequently, he proposed to give all of his tenants a choice: they could pay him their rent in an amount in gold equal to the amount of gold that could have been purchased with Bank notes at the time their lease was signed, or they could pay him a sufficient amount of Bank notes to purchase that amount of gold at its present price. He even went so far as to declare his preparedness to pay his own creditors in similar fashion and to reduce the rent payments from his tenants if prices were to decline and the purchasing power of Bank notes improved. He then published his speech in a pamphlet that incorporated tables to help his tenants and any other interested parties to figure out what the adjusted payments would be. Lord King's announcement caused such a furor that Parliament passed special legislation declaring that the face value of contracts was inviolate and could not be so revised.

There were more crises, more hearings, more public debates, more inflation and even deflation before Parliament finally restored the convertibility of Bank notes into gold in 1821. By that time, the usual period of deflation that has followed major wars had driven the price level all the way back down to its level in 1797. The distortion that bothered Lord King in 1810 had been washed out of the system. Full cash payments in gold were restored and a new coin was issued, the sovereign, equal to 20 shillings and $\frac{20}{21}$ of the amount of gold in a guinea; in preparation for this moment, the Mint produced a total of 35 million sovereigns in 1821. The gold standard was now an acknowledged reality, enshrined in official legislation. The British arrangements became the model for the rest of the world to follow for nearly 100 years.

A full-page photograph of a powerful volcanic eruption. A towering, bright yellow and orange column of fire and smoke rises vertically from a dark, silhouetted landscape. The plume is dense and turbulent, with smaller bursts of fire visible at its base and along its sides. The sky is dark, making the intense light of the eruption stand out. The overall mood is one of awe and natural power.

The New

THE YARNS OF THE 19TH-CENTURY GOLD RUSHES IN CALIFORNIA, AUSTRALIA, THE KLONDIKE AND SOUTH AFRICA HAVE BEEN TOLD OVER AND OVER, IN BOOKS AND MOVIES AND ON TELEVISION. EVERY LIBRARY DEVOTES YARDS OF ITS SHELVES TO THE SUBJECT.

FOURTEEN

Mistress and the Cursed Discovery



Gold prospectors

All these accounts of adventure and personal drama are gripping, but they reduce a great story to too small a scale. The scope of the 19th-century discoveries dwarfed everything that had happened up to that point in the history of gold. Gold production in the United States surpassed both iron ore and petroleum in value until after World War I. Seen from that perspective, the gold of Croesus, Pizarro and the caravans of the Sahara shrinks to tiny pinpoints. Furthermore, although the huge increase in gold supplies made possible the establishment of the international gold standard, the economic impact of the discoveries was radically different from the Price Revolution of the 16th Century and quite the opposite of what most experts confidently expected to happen.

True, the Spanish discoveries of gold in the New World lifted world output of precious metals – gold plus silver – in the course of the 1600s to over seven tons a year, about double what it had been before those discoveries. By 1700, the total world stock of precious metals was five times as large as it had been in 1492. Then, thanks to Portuguese discoveries in Brazil, production doubled again in the 18th century.

By 1859, with California, Australia and Siberia all going strong, total world output of gold alone was 275 tons a year, which was more than ten times the average annual output during the 18th century. At that rate, the amount of gold produced in ten years matched the production from all sources over the entire 356 years from Columbus to 1848. And that was before the Klondike, Colorado and South Africa opened up at the turn of the century. World gold production by 1908 was over 100 times what it had been in 1848 and 4.5 times the levels of just 20 years earlier. By 1908, the total amount of gold in all forms – coinage, hoards, adornment and decoration – could have been fashioned into a cube 10

Molten gold



Prospectors came to California from all over the world



meters in each direction — an enormous expansion from the 2-meter cube of 1500 that represented 3000 years of developing civilization (see page 109).

Quite aside from the contrast in supplies, the character of the entire affair bore little resemblance to the gold strikes of the 1600s. The gold of the New World was found by adventurers who led the king's armies and employed military power to lay claim to entire nations in the name of the newfound wealth and plunder. The gold in California and Australia was discovered by independent prospectors panning in the rivers and working for themselves in a full expression of pioneering spirit, followed by businessmen who replaced the individual prospector with heavy capital equipment such as dredges and drills. The gold then traveled to banks and national treasuries on railroads and steamships whose high speeds and black smoke obliterated the memories of the romance and perils of the Spanish Main or the camels of the Sahara. Little of the 19th-century gold production ended up in Asia, although silver continued to move eastward.

In the 1500s, the gold discoveries were accompanied by even more important new supplies of silver in both Peru and Mexico, so silver continued to be the primary form of money in Europe and America until well into the 19th century. The influence of all the imports of gold from America did not even begin to dislodge silver in Britain until 1717, in Isaac Newton's time, and another 100 years had to pass before Parliament established the official gold standard. But the deluge of new gold supplies in the 1800s finally established gold's dominance over silver in the world's monetary system. By 1900, the gold standard had been adopted almost everywhere; in many countries, silver was demonetized (ceased to be acceptable as money) except for small-coin transactions.

Finally, although there were constant warnings about the glut of gold and the inevitability of higher prices, inflation did not take hold until the very end of the 19th century except for very brief periods of time. This experience was a dramatic contrast to the Price Revolution of the 16th Century. Indeed, the behavior of the price level was in many ways the most startling and interesting of the consequences of the 19th-century discoveries of gold.

Tradition paints the 19th-century gold rushes as historical accidents of some kind, with unsuspecting individuals stumbling on to sudden riches and the word spreading like wildfire. That tradition contains an important element of truth, but it is incomplete. The

gold rushes of the 19th century were something more than bursts from the blue. People had been aware of gold-bearing areas in California, Canada, Australia, Alaska and South Africa before the excitement turned hot in those areas. But the rapidly expanding world economy and financial system made the role of gold as money and as a standard even more important than it had been up to that point. Enormous prizes were to be won by the lucky individuals who beat the odds and managed to emerge with some gold from the turmoil and struggle of daily life in the gold rushes.

Although one could hardly describe the situation in the early 1800s as a bullion famine, the copious flow of gold from Latin America had peaked during the 1790s as Brazilian supplies began to dry up. Then in 1810 – the year the Bullion Committee held its hearings – Mexico rose up in a revolution against Spain that ignited a series of wars of independence in Latin America, disrupting the production of precious metals for nearly 20 years. This turbulence in Latin America was one of the reasons that Britain postponed restoring convertibility of Bank notes for six years after the defeat of Napoleon in 1815.

A dire world gold shortage was postponed, however, even before gold was found in California, as a result of discoveries in the land of the first and least famous of the 19th-century gold rushes – Russia. Gold had first been discovered in the Urals in 1744, but output remained modest until 1823, when, sensing opportunity, Tsar Alexander I mounted a major effort to develop the country's bounteous gold resources. Annual production in the Urals rose from less than two tons in 1823 to more than five tons by 1830 and kept on climbing. The explorers were even more successful when they moved

Panning for gold





LEFT: *Johann Sutter*

ABOVE: *The Sacramento River; the valley where Sutter established his sawmill*

*Panning for gold*

eastward into Siberia; production in that area had reached 11 tons a year by 1842. By 1847, the Russians were supplying over 60 per cent of world gold production. As a result of the discoveries in other countries, the Russian share of the total declined after that date, but the Russians found so much gold over the years that their annual output was all the way up to 60 tons when World War I broke out in 1914. By the time Stalin's prisoners were working the notorious mines in the Urals and in Siberia, Russia had been turning out enormous amounts of gold for over 100 years.

The development of Russian gold production bore no resemblance to the Wild West and the entrepreneurial character of the gold rushes that would stir the imagination of adventurers in North America and Australia. Most of those prospectors went home empty-handed, but at least they had had the opportunity to hit the jackpot and enough of them succeeded to keep the crowds pouring in. Not so in Russia. Digging into the freezing tundra, the tsar's miners were essentially serfs who worked for pitiful wages from 5 am to 8 pm, six days a week, in rough and marshy terrains. None of the gold they dug up would be theirs. The whole affair was either under the direct control of the crown or of a small number of rich landlords whose gold revenues were taxed by the tsars.

In California, the discovery of gold in 1848 took place two years before the territory was admitted to statehood and even before a peace treaty had been signed in the Mexican War. The news about Sutter's Mill spread through the neighboring communities in short order and almost emptied out San Francisco within a few weeks. With no radios, television or Internet, however, word reached the rest of the country more slowly. The first deposit of gold from California at the US Mint did not arrive until 8 December 1848, at which point it was greeted by one periodical as 'the new mistress'.

The big rush did not begin until 1849, after President Polk mentioned it in his State of the Union address to Congress, which is why the prospectors came to be known as the Forty-Niners rather than the Forty-Eighters. The long delay between the discovery and Polk's announcement was the primary impetus for the first revolution in telecom – the establishment of the Western Union Company and the wiring of the entire United States for telegraphy. By 1853, over 100,000 people had swarmed into California, including 25,000 Frenchmen and 20,000 Chinese, and annual gold production approached 80 metric tons; production would peak as early as 1853 at around 95 tons.

The name Sutter's Mill has always been associated with the onset of the California gold rush. Poor Johann Sutter! In essence a good man, not a greedy man, Sutter was grieved rather than thrilled to hear about the golden nuggets in the stream on his property. He was so far out of step that he ultimately landed in deep trouble and came to a sad end.

Sutter was born in western Switzerland in 1803 and fled his homeland in 1834, hounded by creditors and facing debtor's prison. He took himself to the United States, eventually landing up in California. In August 1839, he chose a spot in the Sacramento valley, not far from where the state capital is located today.

Sutter named his domain New Helvetia, and managed it as a small empire. A bell rang each morning like reveille at an army post, and Sutter insisted on elegant manners by all workers, white and Indian alike. By 1846, there were 60 buildings there, including a bakery, a barracks, a tannery and a blanket factory, plus 12,000 head of cattle, more than 10,000 sheep, 2000 horses and mules, and fields producing over 40,000 bushels of wheat. Sutter believed he had every chance of becoming the wealthiest man on the Pacific Coast. 'My best days were just before the discovery of gold,' he would recall.

The need arose for a large sawmill. As the valley had no timber, Sutter decided to look for a site in the mountains and settled on Coloma on the south fork of the American River. He assigned the job of building the mill to his chief mechanic, James Marshall. On 24 January 1848, Marshall appeared at Sutter's office back at headquarters and asked to see Sutter alone, insisting that the door be locked. Marshall pulled a white cotton rag out of his trousers. He opened the cloth and held it before Sutter, who saw about 1½ ounces of gold dust in flakes and grains. 'I believe this is gold,' said Marshall, 'but the people at the mill laughed at me and called me crazy.'

Sutter took a dim view of the consequences: 'During the night the thought burst upon my mind that a curse might rest upon this discovery. . . . From the very beginning I knew what the outcome would be, and it was a very melancholy ride on which I started the next morning.' Sutter went up to the mill and told all the workmen that they must keep the discovery a secret for six weeks until the mill could be finished. 'But this was not to be. Women and whiskey let the secret out.' Nevertheless, Sutter succeeded in keeping the discovery a secret within a limited area around Fort Helvetia for over three months.

On 4 May, everything changed when a neighbor who had visited the site ran through the streets of San Francisco with a bottle of gold dust, shouting, 'Gold! Gold! Gold from the American River!' Within a few weeks, the surrounding area went crazy. Even the recently opened school in San Francisco had to be closed because both teachers and pupils had gone off to the mines.

And so begins Sutter's lament: 'All my plans and projects came to naught. One after another, all my people disappeared in the direction of the gold fields. . . . Only the sick and the crippled remained behind. . . . The damage which I suffered in 1848 is inestimable.' Squatters settled everywhere: 'My property was entirely exposed and at the mercy of the rabble. . . . I was alone and there was no law.' The gristmill was never finished; even the stones were stolen, along with cattle and horses, the bells from the fort, hides and barrels.

Sutter spent years of frustrating efforts in the law courts attempting to reclaim his land from the squatters. After five years in Washington, DC, trying without success to adjudicate his claims, he retired to the little Pennsylvania town of Lititz. In 1880, he was back in Washington, trying for the 16th time to have his claim confirmed by Congress, but Congress adjourned without taking action. Two days later, at the age of 77, Sutter was dead. The memory of his sawmill, however, remains very much alive.

When the news of the discovery of Sutter's Mill reached Australia at the end of 1848, a crowd of Australians took off at once across the Pacific to join in the fun. Among them was an English-born man who had scraped together a living for some years in the Wellington district of New South Wales, about 170 miles west of Sydney. His name was Edward Hammond Hargraves, described by the historian Richard Hughes as 'a corpulent bull-calf of a man.' After two years of panning and scrambling in California, with nothing to show for his efforts, Hargraves spent his last dollars to head back to Australia. He was still hopeful, however: he took his panning equipment back home with him, because he was struck by the geological similarities between the gold area in California and the Wellington area in Australia.

On 12 February 1851, Hargraves and his guide were poking on horseback along a tributary of the Macquarie River when, as Hargraves described it, he felt 'surrounded by gold'. Gold came up in four of the first five pans they dipped into the river. Hargraves exclaimed to the guide, 'This is a memorable day in the history of New South Wales. I shall be a baronet, you will be knighted, and my old horse will be stuffed, put in a glass case and sent to the British Museum!' None of those things happened, but it was indeed a memorable day. As the news spread, Hughes relates, 'It was as though a plug had been pulled and the male population of New South Wales had emptied like a cistern, in a rush toward the diggings.' One of the Sydney newspapers reported, 'A complete mental madness appears to have been seized by almost every member of the community.' Within six months, 50,000 people were digging for gold.

By November, bags of gold were pouring in a great flood to waiting ships. The first shipment to London at the end of 1851 was 253 ounces. Six months later, weekly shipments were averaging half a ton. Gold was soon turning the crudest workmen into pretentious gentlemen. 'It is not what you were, but what you are that is the criterion,' as one contemporary observer wrote. Miners were heard to say, 'We be the aristocracy now and the aristocracy be we.'

As newcomers poured into Australia, they transformed a minor-league penal colony of 45,000 men into what would turn out to be a flourishing and well-diversified nation. Hughes summarizes these turbulent times well: 'Gold disturbed the order of Anglo-Australian society – from pastoral aristocrat to convict – with shudders of democracy.' Indeed, the most remarkable consequence of the Australian gold rush was the end of 'transportation' – the forcible exile of English criminals to the horrors of Van Diemen's Land on the island of Tasmania. No terror clung any longer to the name of Australia when a man could find the riches of his dreams there. With a quarter of Britain's men clamoring for tickets to Australia, the governor-general had to admit that 'Few English criminals . . . would not regard a free passage to the gold-fields . . . as a great boon.' In medieval times, gold had saved lives by serving as ransoms; in 19th-century Australia, gold led to the end of the barbarous conditions on Tasmania in December 1852, less than two years after Hargraves felt 'surrounded by gold'.

*Monument to the Discovery
of Gold in South Africa, by
George Harrison, 1886*



Although the rush to the Klondike was dramatic and colorful because of its location and its hostile terrain, the Klondike was relatively unimportant in the long history of gold. In this instance, a couple of fishermen prospecting for salmon in a tributary of the Yukon called the Thron-diuck River – later transformed to Klondike – spotted gold in the waters one August afternoon in 1897. The advance guard of the rush came from Circle City, which was down the Yukon, a bustling center of gold prospectors with the usual complement of dance halls and saloons. It took until the following spring before the first shipments of Klondike gold sailed south to California. Fifteen hundred people in Seattle, including the mayor, sailed north within ten days of the first news.

Before it was all over, 100,000 people had set out for Dawson City, less than half of whom were able and willing to hang in during the rugged trip and actually made it to the gold-bearing areas. Four thousand found gold and about 400 struck it rich. The most abundant areas were few in number and had been pretty much used up by 1900. For all the hoopla, all the gold mined in Alaska since 1880 has amounted to less than 10 per cent of the gold mined in all the other parts of the United States over the same period of time.

The South African story has a different flavor. Yes, South Africa has its counterpart of James Marshall or Edward Hargraves in the person of George Harrison, who happened upon an outcropping of gold in 1886 while digging up stone to build a house for a widowed neighbor not far from the city of Johannesburg. But South African gold does not appear in nuggets and little of it shows up in surface outcroppings. Instead, the gold lies embedded in an imposing body of ore in what are called reefs that average only about one foot thick and lie as deep as one mile underground. Depth is by no means the least of the miner's problem: the ore is so low-grade that a ton of it contains no more than an ounce of pure gold, which does not willingly separate itself from the raw rock.

Despite the rush that at its peak brought as many as 2000 immigrants a week to South Africa, gold mining there required so much capital that it was big business almost from the very start, with the diamond men from Kimberly leading the way. Even so, like the Klondike, the boom in South Africa seemed to be coming to an end within three years of Harrison's discovery at Widow Oosthuizen's farm: the high hopes lay buried in the hideous heaps of slag that piled up like grotesque mountains from which just a trickle of gold – and an equally pitiful volume of profits – had been dislodged. Despite repeated efforts with different kinds of chemical processes to extract more gold from the ore, the tons of gold-bearing rock just continued to accumulate, refusing to part with their precious contents in sufficient quantity to make the business profitable.

As the ore brought to the surface of the earth appeared thinner and thinner, all the earlier excitement yielded to pessimism. The stocks of the mining companies crashed, some falling 95 per cent from their previous values. One observer predicted, 'Grass will grow in the streets of Johannesburg within a year' – yet another prediction that turned



The cyanidation process is still used today

out to be wrong almost as soon as the words were uttered. As often happens with people who yield to panic, selling gold-mining shares at that moment would turn out to be a colossal error.

In late 1889, Allan James arrived on the scene, representing a Scottish corporation called the African Gold Extracting Company. James announced that his company had come up with a process called cyanidation that would solve all of South Africa's problems. Although scientists in Britain, the United States and New Zealand had experimented with cyanide as far back as the 1840s, without clear success, the process that James offered was one that worked. The cyanide was stirred and decanted after a few hours. These steps tended to separate the gold from the ores, and then zinc was applied to precipitate the gold. The cyanidation process had actually been developed by John Stewart MacArthur, a chemist from Glasgow, who had teamed up with an unlikely pair of physicians with intense curiosity in such matters, Robert and William Forrest.

A pilot plant was erected in May 1890 that performed all the miracles that James had promised. The process made possible the profitable production of gold from newly mined ore, but that was not all: the huge piles of slag could now be attacked and profitably turned into lovely yellow metal. The African Gold Extracting Company soon negotiated a royalty with the mining firms that was the equivalent of \$1.36 per ounce of pure gold, to be earned on every ton of ore that was treated with the cyanidation process; with the world price of gold at that time around \$21 an ounce, this was a handsome royalty indeed.

The cyanidation process was an immediate and smashing success. The sophistication of the MacArthur–Forrest achievements turned South African gold production into one of the high-tech industries of the late 19th century. Cecil Rhodes' consulting engineer was on the right track in 1890 when he predicted that, thanks to cyanidation, the value of the annual gold output in South Africa would exceed £20 million before the turn of the century, at a time when the total world output of gold was not much more than that. Gold production rose from less than a ton in 1886, when the first discovery was made at Widow Oosthuizen's farm, to 14 tons in 1889 and then to nearly 120 in 1898 before the outbreak of the Boer War. The 120 tons had a market value in 1889 of about £16 million.

After four years of paying those fat royalties to MacArthur and his associates, the gold-mining companies came to the conclusion that the African Gold Recovery Company (as it subsequently came to be known) was getting too much of a good thing. The Chamber of Mines entered into negotiations to reduce the royalty rate, but MacArthur and the others stood fast. Their greed would do them in. After four months, the Chamber gave up on negotiating and brought suit in the High Court to contest the validity of the MacArthur patents. Their argument claimed that the patentees were not the true inventors, that the process was not new when it was patented, that the final specification was faulty, and that others had used the process in South Africa before MacArthur's group had done so.

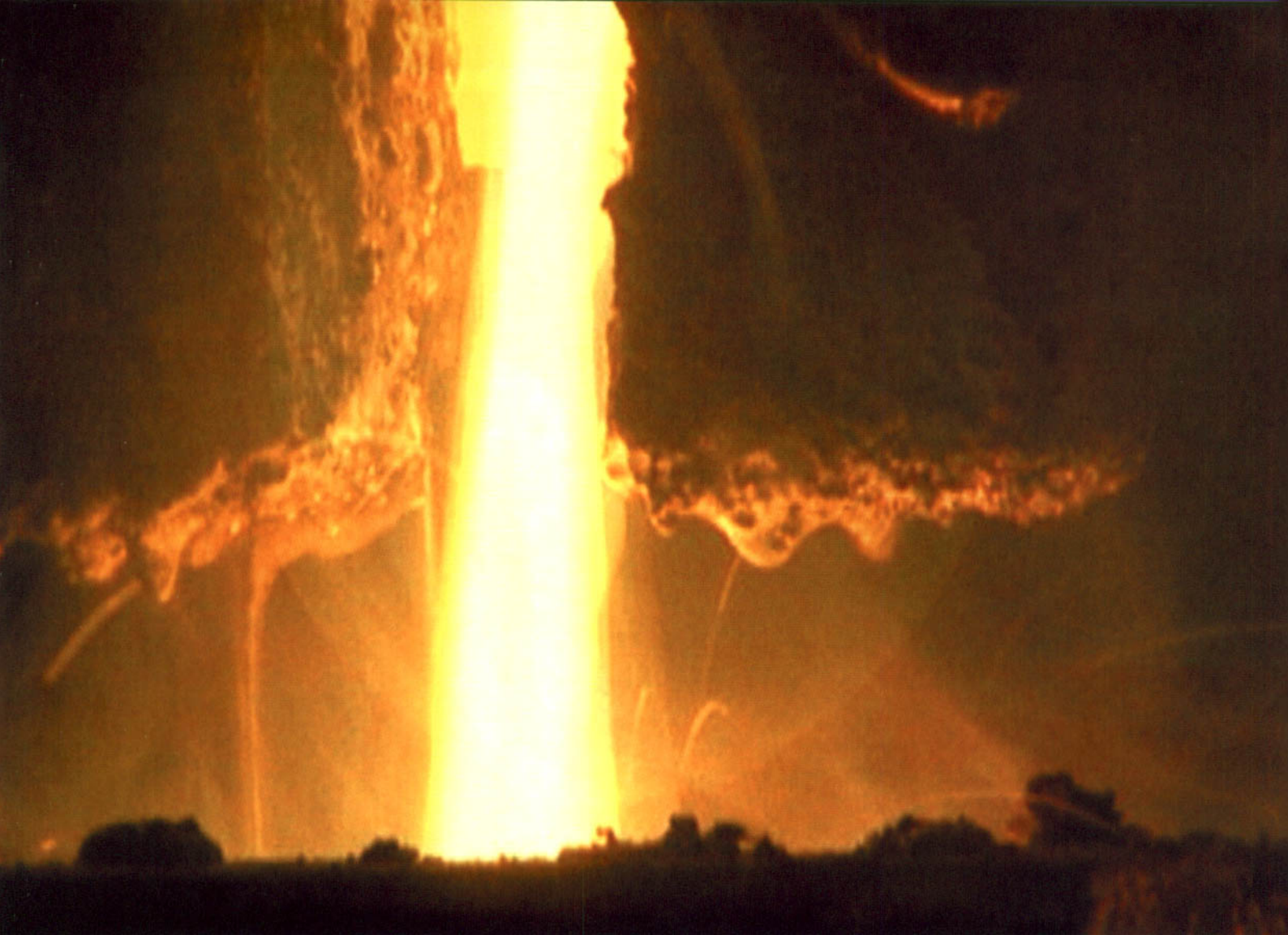
The result was an extraordinarily protracted and expensive lawsuit. Evidence was taken from mining experts in the United States, Canada, Australia, Hungary, Korea, Japan, South America, India and Russia. The president of the Royal Society in London and a covey of physicists and chemists in scores of universities either testified or provided affidavits. The judges did not render their verdict until November 1896, nine months after the hearings had started.

Their decision (two in favor and one against) ran to 70 pages, but the bottom line was loud and clear: 'It was found on the facts by the majority of the Court that the processes above mentioned were not novel and had been anticipated, and consequently the Court declared the said patents void. . . . There is nothing novel in the weakening or diluting of a well-known solvent or re-agent for the purpose of winning gold from quartz.'

MacArthur followed Johann Sutter into oblivion and poverty, dying a poor man in 1920. Sutter was ruined because he had too little greed at the sight of gold. MacArthur was ruined because he had too much.

Newly minted coins





Molten gold

Despite dire predictions, gold prices during the last 40 years of the 19th century never rose by more than 5 per cent above the levels of the late 1850s, and they remained well below those levels from 1875 to the outbreak of World War I. The contrast with the 1500s and 1600s is all the more remarkable because the magnitude of the 19th-century discoveries was so much greater than the new supplies that opened up after Columbus first crossed the Atlantic in 1492.

One explanation for the wide difference in economic performance between the 1500s and the 1800s is in the frequency of warfare in the 1500s. Although living standards did improve during the 16th century, warfare mercilessly consumed much production and mercenary manpower. The 100 years from the fall of Napoleon to the onset of World War I were also marked by wars, including the American Civil War, the Franco-Prussian War in 1870 and the British expedition to the Crimea in the 1850s, but those struggles, though bloody, were relatively brief, and both the United States and Europe enjoyed peace just about all the rest of the time.

The rate of economic growth over the last 50 years of the 19th century was grossly underestimated by contemporary economists. We cannot sit in judgment of this forecasting error, because what happened was so far outside the bounds of what anyone had ever experienced before. The authors of the Bullion Commission report of 1810 would have been astonished at how the pace of economic activity would appear to outrun the growth in the stock of gold.

Consider just the decade of the 1870s in the United States as an example – a decade marked by persistently declining prices and a serious depression that began in 1873 and was still felt to some degree in 1879. Thanks in large part to immigration, population rose by 30 per cent. The miles of railroad track in operation more than doubled (the golden spike that linked the railroad system from the Atlantic to the Pacific had been driven in place in 1869). In New York State, railroad traffic by the late 1870s was for the first time exceeding traffic on canals and rivers. The number of farms rose by 50 per cent and the dollar value per acre rose even as the prices of farm products were dropping – eloquent testimony to how agricultural productivity was cutting production costs. The output of coal, pig iron and copper more than doubled; lead multiplied sixfold. Overall, industrial production grew at an annual rate of better than 5 per cent a year from the end of the Civil War to 1900, which meant that output at the end of the century was 5.5 times as great as it had been in 1865.

There was a modest degree of correlation between gold production and price inflation, especially after 1870, but nobody took any notice. Year-to-year price changes were, in any case, so irregular and erratic that no underlying inflationary trend was able to rise to the surface until the full impact of the South African discoveries made themselves felt in the world economic system after 1900 – and even then the perception of inflation was delayed or muted. The most striking evidence of the public's relaxed view of the probability of inflation is in the behavior of interest rates, which moved up and down without any systematic linkage to what was happening in the price level for commodities and services.

However, there was a demand for hoarding in a civilized state of society that would turn out to be untold magnitudes greater than any amount of gold that individuals might hide to escape spoliation. The growing acceptance of the gold standard, following Britain's first steps in that direction in the early 1800s, created an enormous demand for gold for hoarding – not by individuals but by the central banks of the nations, such as the Bank of England and the Bank of France, or the US Treasury. A hoard of gold was the major line of defense against an unexpected surge in imports or outflow of investment capital to other financial centers. With the huge expansion in economic activity and international trade and investment, those gold hoards were essential to give nations freedom of action and to attract new capital to their shores. 'Sound money' and 'sound banks' were defined by the accessibility of gold to be paid out on demand to anyone who came to the windows and asked for it. Countries gaining gold were held in high repute; countries on the losing side were viewed as in deep trouble.

As a consequence, little of the 'vast production' of gold moved around from hand to hand. The largest part of it was transferred into idle hoards in monumental buildings that looked like Greek temples, where the heaps of gold enjoyed the more polite name of 'reserves'.



GOLD
999.9

1000.9

THE PRIMITIVE NOTIONS SUPPORTING THE GOLD STANDARD
TRACE THEIR ORIGINS BACK TO THE PRESTIGE AND
INTERNATIONAL ACCEPTABILITY OF CROESUS'S STATERS,
WHEN GOLD FIRST STARTED TO PERFORM AS MONEY IN A
SYSTEMATIC FASHION.

FIFTEEN

Strength and Failure

From this root, the religion of the international gold standard had just one absolute commandment: gold, and gold alone, is the ultimate form of money, the standard. A country on the gold standard defined its money – pound, franc, lira, dollar – in terms of a specified and immutable quantity of gold. Its own citizens, and the world around them, expected the authorities to maintain full and free convertibility of its banknotes and bank deposits into gold at that fixed ratio, come what may.

The numbers are fussy, and the convention of measuring weights in grains is annoying, but the ratios are important for they define how many units of one currency such as pounds sterling will exchange for units of another currency such as dollars or francs. For example, Britain's sovereign coin, which was the equivalent of one pound sterling, was defined in 1870 as 113.0016 grains of gold, a setting that reached all the way back to a parliamentary resolution of 1717. This was equal to about a quarter of an ounce, the same weight as a modern 25c piece. The dollar's weight in gold was set at 23.22 grains or \$20.67 per ounce in 1837, after having fluctuated between 24.68 grains and 22.85 grains from 1791 to 1837.

The result of the definitive settings of 113.0016 grains to the pound and 23.22 grains to the dollar was an automatic and invariable rate of exchange between the two currencies, an act of faith that £1 would equal \$4.86 (the result of dividing 113.0016 by 23.22) for ever and ever. Under those conditions, fluctuations over time around this rate would be minimal and determined primarily by the costs incurred in shipping gold across the Atlantic between London and New York.

These publicly announced relationships to gold were inviolate. The international gold standard amounted to nothing more than that. As long as the gold parities were intact, dollars or krone were as welcome as pounds in London and marks or rubles were

as welcome as francs in Paris, in an era in which globalization of capital and trade was as forceful a process as it has been over the past 25 years.

The sophisticated dynamic of the system arose from the robust tension between the good news and the bad news in the gold standard. The good news was in the many advantages of having a currency that was universally acceptable because it was backed by stuff before which humanity had been genuflecting since the beginning of time. This meant that holders of your currency would not desert you at the first sign of trouble. In rare cases such as Britain's sterling, foreigners were willing, or often even more willing, to hold sterling and make their payments in sterling in preference to their own currency, just as people around the world use dollars today.

The bad news was the shattering loss of acceptability that would afflict a nation that mismanaged its affairs or was caught in the whirlwind of a global economic crisis. Then investors, traders, and speculators would rush to redeem their paper and bank deposits for gold, depleting the country's precious stockpile. This cataclysmic outcome was frequent among the smaller but aspiring nations, most of which were dependent on the volatile markets for raw materials. The senior members of the gold standard club went to the brink more often than the nostalgia about the gold standard might suggest, but none of them ever succumbed to apostasy despite frequent narrow escapes.

Until the 1870s, every movement toward a European gold standard was unplanned. Two metals were necessary, because gold coins are so valuable that the system could not function without a widely accepted subsidiary coinage. In most countries, convertibility of paper currency and bank deposits into gold was perceived as nothing more than a variation on the theme of a bimetallic standard that consisted of both gold and silver; elsewhere, silver was the sole standard. The only constant was that specie – metallic-based money – was imperative all the way. The world would not outlive that primitive notion for many decades to come.

Despite gold's glitter, glamour, and prominence in many aspects of life, silver was the primary form of money right up to the middle of the 19th century. The plentiful supply of good silver coins, in contrast to the pathetic clipped coins of the late 17th century, provided for payments in small transactions, while gold lorded over the system of large transactions and liquidity reserves.

Silver was always more plentiful than gold, as Offa's pennies remind us. It is no coincidence that the French word for money, *argent*, translates as silver, or that 'sterling' has defined British money for centuries, even when the pound was defined only by gold. Even during the disputes over gold during the days of the Bullion Committee and its immediate aftermath, silver was a major focus of attention. Both Ricardo and Locke favored one metal over two as the standard, but in 1816 Ricardo opted for silver rather than gold for this purpose. He based his argument on silver's 'greater regularity of supply

and demand', and its use by all foreign countries, arguing that the inconvenience of its bulk could be entirely offset by substituting paper money as the 'general circulation medium'.

Silver might have been number one for ever, but it has two disadvantages. First, silver lacks gold's glamour because it tarnishes so much faster than gold; silver has never driven men and women to the extremes of greed that have been motivated by gold. Second, silver's bulk is much greater than gold's. In the days of Locke and Newton, 1000 guineas at £3 17s 10½d weighed about 18½ lb, while the same amount of money – £1050 – in silver coins would have come to nearly 280 lb. The cost of transporting a given amount of money in gold coin was therefore much lower than the cost of moving the same value in silver. With commerce, industry and international trade and finance growing with unparalleled vigor in the course of the 19th century, this simple physical difference against silver operated as a tax on international movements and in the end may have been important in tipping the scales in favor of gold. Another factor in favor of gold was that a given physical volume of each metal at the Mint could produce a much higher value of gold coins than silver coins.

The bimetallic system did have its advantages, although any arrangement based on metal was destined to be vulnerable to shocks such as new discoveries or new mining technologies, wars, the changing tastes of the Asians, and the fruits of plunder. A convincing justification for bimetallism was made in 1791 by the secretary of the US Treasury, Alexander Hamilton, in a remarkable document in which he invented the US monetary system, including its metallic standards, the decimal system, coin denominations and many smaller details. The coins were to include three gold coins – \$10, \$5 and \$2.50 – then a dollar, half-dollar, quarters, dimes and half-dimes in silver, plus a copper penny and a half-penny. The decimal structure of denominations was a radical break with the long tradition of Offa's pennies, with 240 to the pound, and the use of the multiple of 12 in so many other monetary systems that developed in Europe.

After great deliberation, Hamilton concluded that 'The Secretary is upon the whole strongly inclined to the Opinion that a preference ought to be given to neither of the Metals for the monetary unit.' His document emphasizes the importance of having a money that would be acceptable in the varied systems of all countries and therefore warns against the consequences of converting one of the metals into 'mere merchandize [*sic*]'. He went on to make an even more important point that reveals his extraordinary foresight: 'To annul the use of either of the metals, as money, is to abridge the quantity of circulating medium; and is liable to all the objections, which arise from a comparison of the benefits of a full, with the evils of a scanty circulation.' An insufficient supply of gold – except briefly in the 1850s – would turn out to be a recurring concern for the world economy during the entire 19th century.



Coin bag

The sequence of events leading to the final triumph of gold in Europe was indeed dramatic. As in the 16th and 17th centuries, North America and Asia continued to swing monetary developments in Europe and were the source of most of the severely protracted disturbances.

The story begins in the fledgling United States of America, which in 1791 followed Alexander Hamilton's advice to set the mint price of gold equal to \$19.3939 and the mint price of silver at \$1.2929, which works out to a ratio of 15:1. This happened to be the moment when supplies of gold from Brazil began to decline, to be followed soon afterward by the Latin American revolutions and a precipitous drop in precious metal shipments from the Western Hemisphere. Then Napoleon insisted on a specie-backed money system in France, followed by Britain's attempt to return to convertibility of sterling into gold. With supply falling and demand rising, the price of gold in the markets around the world soon climbed to more than 15 times the mint price of silver. One could now bring 15 ounces of silver to the US Mint and receive one ounce of gold in return, then go into the market and use that ounce to buy more than 15 ounces of silver, and then repeat the process.

The process was so irresistible that the United States found itself with only silver circulating and gold gradually disappearing from the monetary system. As a practical matter, the country was on a silver standard. This disparity between gold and silver persisted for over 40 years. By 1834, the gold-silver price ratio in world markets had changed from 15:1 to about 15.625:1. A growing shortage of gold coin was now interfering with commercial and financial transactions.

In 1834, the US Congress finally recognized that 15:1 was no longer the appropriate ratio of the mint price of gold to the mint price of silver and that an adjustment closer to the reality of world markets could be postponed no longer. But Congress did not set the new ratio of mint prices at 15.625. Instead, it went all the way to 16:1, which worked out to the gold price of \$20.67 an ounce that prevailed for another 99 years.

This step threw the prevailing flows of metal into reverse. Now it was profitable to bring an ounce of gold to the mint, obtain 16 ounces of silver, go into the marketplace and

repurchase the ounce of gold for the equivalent of only 15.625 ounces of silver, and repeat the process. The members of Congress were fully aware that they were stimulating the demand for gold in their choice of 16:1.

The United States was now effectively on a gold standard, even though official legislation establishing the gold standard would not be enacted until 1900. Silver continued to function as subsidiary coinage, but gold was the major holding into which currency and bank deposits could be converted.

These shifts were the inevitable consequence of using a commodity as a standard of value for the monetary system. The demand for gold and silver depends on more than monetary factors, for these metals have additional uses such as adornment or as hoards against uncertain futures. At the same time, nobody knows when new discoveries will occur. Thus, a variety of forces played upon world prices for gold and silver in the 19th century, creating constant disturbances within the monetary system as divergences developed between prices set in the marketplace and prices set at the mints. The experience of 1834 was just the first act in the drama. More violent upheavals were yet to come.

The discoveries at Sutter's Mill in California in 1848 and Hargrove's discovery in Australia in 1852 shook the world. The supply of gold coming into the markets ballooned and pushed the price of gold in the marketplace downward. Now silver appeared relatively expensive. People ceased to bring silver to the mints for coining and hoarded it by cashing in their paper money or bank deposits and even giving up gold in exchange for it. In 1850, about three times as much money circulated in silver coin as in gold coin; by 1860, the amounts were roughly even.

Silver was the primary factor that kept gold from turning into a glut on the market after the discoveries in California and Australia. As it had for a long time, silver traveled to Asia during the 1850s in high volume, especially to India. Shipments of silver into that country quadrupled from the 1840s to the 1850s; French exports alone rose steadily from 82 million francs in 1850 to 458 million francs in 1857. Then came the American Civil War, which shut down American exports of cotton and led to an abrupt surge in the demand for Indian cotton; India obliged by importing even more silver in exchange.

Just as suddenly, everything went into reverse after the Civil War came to an end. The demand for gold showed no signs of diminishing while the Indian demand for silver had a precipitous drop. During 1870–5, total Indian imports of silver were smaller than in the year 1865 alone. When a huge silver deposit known as the Comstock Lode was discovered in Nevada in 1859, the glut had clearly switched over to silver.

For silver, the timing could not have been worse. In view of Britain's economic leadership in foreign trade, other countries began to give serious consideration to shifting to a pure gold standard like Britain's – but the difficulty of disposing of their stocks of silver in an oversupplied market was a major deterrent. Germany was especially eager to

Iguassu Falls, Brazil



make the changeover to gold, for the Germans wanted to be perceived by the world as a great power. Germany also wanted to be on the same standard as Britain in order to meet the growing need for sterling to pay for raw material imports from the outposts of the British Empire.

Germany seized on the opportunity provided by its victory over France in 1871. The indemnity paid by the French relieved the Germans of the necessity of liquidating silver to finance its purchases of gold. They waited until 1873 to begin their silver sales, even expecting some to be bought by the French. The latter not only refused to cooperate but took a more drastic step. On 5 September 1873, the day after the last payment on the indemnity, France limited its silver coinage to 280,000 francs a day and cut it again in November to 150,000. The result was another sharp drop in the demand for silver.

The French ended up outdoing themselves. Too many other countries joined in the selling wave in an effort to avoid being caught with a stock of silver money that was worth less almost by the hour. The game became a self-fulfilling prophecy. A decision that the French perceived as a tactical rather than a strategic step provoked a steady fall in the price of silver from over 60 pence per ounce in the 1860s to only $52\frac{3}{4}$ pence in 1876 and 51 pence by the end of the 1870s. By that time, an ounce of gold in the marketplace was bringing 18 times as much as an ounce of silver; by the end of the century, the amount was up to 30 times as much as an ounce of silver. As the glut of silver drove the ratio of market prices further and further away from the ratio of prices at the mints, anyone who used gold to buy silver at such depressed prices in the market and then brought the silver into the mint for coinage enjoyed a profitable, riskless, and irresistible activity. The process rocked the gold standard to its very foundations and threatened to bring down the whole structure. The only defense was to eliminate silver as a monetary metal except for small coins.

That is precisely what happened during the course of the 1870s, beginning in 1873 in France, and also in the United States. That sequence of events explains why so many countries hastened to follow Germany into a pure gold-standard system.

Thus, the fabled international gold standard was built from the dust of the disaster to silver – by 1893, even the mints in India were closed to the coinage of silver. The gold discoveries in California and Australia and the Comstock Lode in Nevada, the appetite of Indians for silver, the American Civil War and Germany's overwhelming ambition to be a great power in effect backed the world into a system that no one had anticipated and that many people were reluctant to accept. Once in place, however, the system displayed remarkable durability for the next half-century.

One of the most spectacular – but by no means atypical – crises in the age of the gold standard occurred in 1890, largely as a result of ill-advised and mismanaged investments in Argentina by the respected London house of Baring Brothers. Barings came perilously close to bringing the Bank of England itself to its knees.



Along with their devotion to the gold standard, the British had long led the way among the European nations in the development of financial markets and institutions. As the resources of the City expanded, the British capital market became the leading source of funds for economic growth all around the world. Britain was the largest supplier of capital for the development of the railroad system in the United States, but British capital also poured into many developing economies in Latin America and in Asia. Argentina was a special favorite.

The enthusiasm for Argentina from 1870 to 1890 seemed to be well justified by the impressive growth achieved by this nation. Immigration to Argentina in those years doubled the population, with Europeans who were very different from the 'flotsam and jetsam' that headed to the United States. Many brought capital with them to acquire and develop the vast territories of land available for ranching and for agriculture. The result was a surging demand for roads, railways, waterworks, and industrial enterprises. Just during the brief period of 1882–9, European investors poured \$1 billion into Argentina.

This drama played itself out in classic fashion. By the end of the 1880s, credit was pouring in so rapidly that service on the outstanding debts began to outrun capacity to pay, the flood of money was clinging to the sticky fingers of the politicians, and inflation began to raise its ugly head. In just one year, money in circulation increased by \$270 million.

Meanwhile, back in Europe, business activity was accelerating, stimulated in part by an enormous borrowing of £11 million by the British government to build iron-clad battleships, followed by similar efforts in France, Russia and Germany. The stock markets across Europe were roaring ahead and building speculative froth of their own; on Wall Street from 1884 to 1889, stock prices appreciated by over 50 per cent. At the same time, capital was also moving to the newly discovered nitrate mines of Chile and the gold and diamond mines in South Africa.

In 1889, concerned about the swindling and greed that were sweeping the speculative manias, the Bank of England and the German Reichsbank sharply raised their key interest rates. The impact was immediate and fevers cooled – which meant that security prices fell and took an excruciating toll on those poor investors who had let themselves be lured in at the very top of the bubbles.

The timing was unfortunate. The Argentinian wheat crop failed, followed by a bloody political revolution in the summer of 1890. Financial panic broke out in Buenos Aires. The repercussions on the leading banking houses in London were immediate, most notably on Baring Brothers. The latter had underwritten £42 million of Argentine securities during the 1880s – £28 million in 1888 and 1889 alone – a significant portion of which remained unsold, leaving Barings the unwilling owners. The message in the family motto that had become the company logo, *Virtus in arduis* (Fortitude under difficulty), now seemed very garbled indeed.

*Gold Piggy bank*

As the Barings were among the most respected families in Britain, the shock was devastating far beyond even the huge amounts involved. The diary of one contemporary reported, subsequent to the resolution of the crisis, that "The West End social life of Lord Revelstoke (the partner blamed for the crisis) so enraged the Stock Exchange that it was "ready to lynch him". Barings had been a leader in international financial markets 'when the Rothschilds still bought and sold old clothes'. In 1792, they had played a primary role in financing the British war effort against the French Revolution and Napoleon. In 1803, they played the lead in the financing of the American purchase of the Louisiana Territory from the French. They led the European financing of US foreign trade and the US government up to the outbreak of the Civil War. In the early 1840s, the managing partner, Alexander Baring, later Lord Ashburton, was ambassador to the United States and negotiated the Maine–Canada border with Secretary of State Daniel Webster.

The Barings crisis could not have hit at a worse moment. The Bank of England's gold reserve had been as low as £9 million in 1889 and was less than £11 million at the height of the crisis. Baring Brothers alone was going to require at least £4 million immediately in order to avoid closing its doors. The Bank of England now anticipated withdrawals of gold by the Bank of Spain and the State Bank of Russia. Although the Bank's interest rate had just been raised to 6 per cent, William Lidderdale, governor of the Bank and no fool,

was convinced that further increases in the Bank's interest rate would merely signal the intensity of the difficulties they were facing. 'It would have taken a very high rate indeed to bring gold over in quantity,' Lidderdale later explained to the chancellor of the Exchequer. Nevertheless, *The Economist* reported on 15 November that the Bank's gold reserve was 'just about sufficient for ordinary home requirements, but it was too small to meet exceptional demands.'

In desperation, the Bank turned to the Rothschilds, a powerful competitor of Barings. Rothschilds and the Bank together put up £15 million as a start to the complex and painful process of paying off the loans that Barings had been forced to make to finance their unwanted inventory of unsold Argentine issues. The Rothschilds were then asked to serve as intermediary for a loan of £2 million of gold from the Bank of France. The French agreed to the loan, at only 3 per cent interest, and even shipped the gold across the Channel to British soil. Another £1.5 million in German gold coin was obtained from Russia, while the Russians also agreed to refrain from withdrawing their substantial deposits at Barings. At that point, the Bank of France offered an additional £1 million in gold. Always eager to be in good graces with the Bank of England, a consortium of domestic banks in London now stepped forward and agreed to put up most of the remaining funds required to satisfy the creditors of Baring Brothers.

Perhaps the most remarkable aspect of the whole operation was the discretion and secrecy with which it was carried out. The governor of the Bank of England had excellent sources of information on what was going on in the City, and on Friday, 7 November, the first rumors reached him about some big houses that might be in danger. Three days later, Lidderdale had the chancellor approach Rothschild; before the week was out, the Russians had come along. By the 14th, word 'began to ooze out that something was up', but by that afternoon the entire deed was done.

It was indeed an elegant performance. Charles Kindleberger, a leading historian of financial crises, has observed that the rescue operations in the Barings crisis were seen 'as a measure of the strength of the London financial system more than the Baring failure was taken as a sign of weakness.' But without the cooperation of the French and the Russians, the story would have had a horrible ending. However, the tradition of cooperation was well established by that time.

This assistance was predicated on the assumption that preservation of convertibility into gold at fixed rates was the bedrock of economic policy, before which all other considerations had to give way. Without that assumption, the credits that gold-standard countries provided to one another would have been far less generous and would have come with much more onerous conditions.

The classic sequence of events when trouble developed is the opposite of what tends to happen in our own time. Who but the naive today would put such firm confidence in either credibility or cooperation? Now speculators flee a currency in trouble, putting it

into ever greater trouble, destabilizing its relationships with the outside world until they become completely untenable. In the nineteenth century, a pledge to maintain gold parity was accepted at face value. At the end of the twentieth century, such a pledge was usually taken as a sign that whatever parity existed would soon go down the tube. In the contemporary scene, only grudging cooperation is forthcoming, and then with strings attached so lacerating that domestic political considerations may even necessitate rejecting the proffered aid.

Nevertheless, the magnitude of the emergency of 1890 brought home to all the central bankers the necessity of holding much larger gold reserves than they had considered adequate up to that point. Faith in the power of interest-rate policy to hold disaster at bay was not abandoned, but emphasis began to shift toward accumulating a hoard of gold big enough to build credibility to a degree beyond question. This step would reduce the heavy reliance on managing the gold reserves solely by varying interest rates. The German war reserve accumulated after 1900 and, held separate from the Reichsbank's reserve, became an object of envy. Shades of the emperors of Byzantium!

All the experts agree that the gold standard operated in a congenial economic and political environment. Despite recurrent financial turbulence, the period between the end of the American Civil War and the outbreak of World War I was marked by a mighty force of economic growth and industrialization in both America and Europe. Peace helped, too: no major wars were fought within Europe from 1870 to 1914, while the only military activity to engage the United States was the Spanish-American War of 1898.

There was an additional element of great importance, more political than economic. An economy with a fixed rate of exchange between its currency and the currency of other nations must be prepared to see its domestic economy dominated by the requirements of the gold standard. If the gold stock was flowing outward, interest rates had to rise to attract foreign funds and the domestic economy had to be suppressed to curtail imports. No safety nets of any kind were allowed. That was bad news for the workers who lost their jobs and saw their wages cut as well as for business firms whose profits were slashed. But those were the rules of the game. The system simply could not have survived if the objectives of domestic economic stability and high employment had dominated the objective of defending the gold stock. No matter how intense the pain, the political arms of the governments – parliaments or executives – never dreamed of intervening, because the pain was for the Greater Good.

Remember that this was the age of Queen Victoria and King Edward VII. The type of political outcry that we would expect today was in large part absent in the nineteenth-century environment of Europe, or, to the extent that voices were raised, men in positions of power paid little heed to them. Even among economists, macroeconomic considerations and business cycle analysis did not occupy the attention of mainstream

theorists; it was only in the underworld of such people as Karl Marx that those concerns were expressed. The doctrine of laissez-faire and minimal government interference in business and financial affairs was the ruling philosophy for most of the period.

This viewpoint suggests that the success of the international gold standard might have been more symptom than cause. It did operate in an environment in which growth continuously bailed out policy errors, the burdens of international debts within Europe never expanded beyond manageable levels, and international cooperation could be taken for granted rather than depending on painful and conditional support.

A famous contemporary got the point at the time, without any benefit of hindsight: Benjamin Disraeli told a group of Glasgow merchants that 'It is the greatest delusion in the world to attribute the commercial preponderance and prosperity of England to our having a gold standard. Our gold standard is not the cause, but the consequence of our commercial prosperity.' If ever there was a time in human history when circumstances combined to make the world economic system function as people expected it to function, this was it.

The widespread failure to recognize the profound truth in Disraeli's observation is eloquent testimony to the power of the mystique that the gold standard acquired in the course of the nineteenth century, despite the choppy history from which it originated. The unquenchable conviction that the gold standard explained its benign environment is the dominant reason why the Europeans insisted on putting themselves through all the trials and tribulations associated with efforts to restore and then to maintain the gold standard from 1921 until its final demise in 1971.



A photograph of a vast field of green grain, likely sorghum, under a sunset sky. The sun is a bright, glowing orb on the horizon, casting a warm orange and yellow light across the scene. The grain in the foreground is in sharp focus, showing its green leaves and developing seed heads. The word "Breaking" is written in a white, elegant script font in the upper right corner of the image.

Breaking

THE UNITED STATES STRIDES THE WORLD TODAY AS A FINANCIAL COLOSSUS, BUT THIS PERCEPTION IS A FAR CRY FROM THE US OF THE 19TH CENTURY. IN THOSE DAYS, BANK FAILURES WERE COMMON, FOREIGNERS WERE FICKLE IN THEIR TASTE FOR AMERICAN SECURITIES AND THE US GOLD STOCK WAS ALL TOO OFTEN ON THE VERGE OF POURING OUT TO OTHER NATIONS.

SIXTEEN

The Golden Fetters

The stability of the currency and the financial bases of growth were constantly vulnerable to attack from the outside.

The United States waited until 1900 before legally committing itself to a gold standard. Until then, Americans remained officially on Hamilton's bimetallic standard, in which silver had equal standing with gold. These arrangements were supported by the powerful and boisterous silver interests, and not just because silver mining was an important economic activity in the western states. Farmers and others who were characteristically debtors favored the largest possible money supply and saw inflation as a blessing rather than a curse. Silver plus gold equaled more money in the system than gold alone. But silver became something bigger than just a viable candidate to serve with gold as a monetary standard. Silver was a potent symbol for the struggle of the Little Man against the Established Powers. Propelled by universal suffrage, the rallying cries of equality and democracy rang out louder there than in the Old World.

No wonder foreigners were so skeptical of US intentions with regard to gold.

In 1861, early in the Civil War, a shrinking gold stock led Congress to suspend convertibility of dollars into gold, a customary move under the circumstances. In another customary wartime move, the Treasury financed part of the wartime expenditures by issuing a new paper currency. This paper money, convertible into nothing at all, carried the official imprint of 'US Notes'. They were more commonly known as greenbacks, a nickname that is still in use as a generic identifier of today's US dollar – which is also convertible into nothing at all. Like today's paper money, the greenbacks were legal tender, which means that a creditor could not legally refuse to receive them in payment of debts due; as a result, the greenbacks were also referred to on occasion as 'legal tenders'.

The harvest of 1979. The American harvest was good compared to poor yields in Europe

A dollar in greenbacks never commanded a dollar's worth of gold, because there was no certainty that the day would ever return when people could take their greenbacks to the bank and exchange them for gold coins. This condition was the same as in Britain during the Napoleonic Wars when sterling convertibility was suspended: if paper currency and bank deposits are not convertible into gold, gold ceases to serve as money and becomes a commodity whose price in terms of money is destined to fluctuate.

During most of the Civil War, a dollar's worth of gold cost about \$1.30 in greenbacks. Matters improved after the Union victory in 1865, and by 1869, the greenback notes were being valued at better than 90¢ on the dollar. In January 1875, Congress passed legislation that pledged full resumption of convertibility by 1 January 1879, and authorized the Treasury to borrow the money to acquire a gold reserve, if such a step should prove necessary.

By 1873, pressure was building to stop pussyfooting around about the promises for resumption. The time had come to tidy up the currency and finish the job. Congress held an extended series of hearings on the subject and then passed the Coinage Act of 1873, which listed the various denominations of gold and silver coins to be minted by the Treasury from that time forward. The legislation passed through Congress by an overwhelming margin.

The Act of 1873 contained a startling omission. There was not a single word that referred to the historical standard silver dollar of 371.25 grains that dated all the way back to Alexander Hamilton's establishment of the coinage of the United States. The only mention of silver is for subsidiary small-denomination silver coins. The consequences were momentous: the Coinage Act of 1873 finished off the legal status of bimetallism in the United States.

When the silver advocates woke up to what had happened, they were in a state of shock over what had taken place under their very noses. Later on, after much of the dust had settled, Senator William Stewart referred to the demonetization of silver as the 'crime of the 19th century'.

It is difficult to find a satisfactory explanation for why so few people were aware of what lay ahead. There had been no effort to keep the contents of the Coinage Act a secret. If, as Lincoln had asserted, you can fool all of the people some of the time, this was clearly a case in point.

The main drama of the 1870s was to occur in 1879, following the official resumption of convertibility of paper currency into gold that took place as scheduled on 1 January. Some of the gold on hand to meet demands for redemptions had been accumulated from a surplus of tax revenue (largely tariff receipts) over expenditure, with the remainder purchased abroad with funds that the Treasury borrowed in the capital markets.

Yet conditions were ominous and the viability of the resumption process remained uncertain. Despite passage of the Resumption Law, the credibility of the nation's

devotion to gold appeared to be every bit as questionable as it had been in the past. US export trade was slipping and the prospect of a significant loss of gold to Europe became a general expectation. How certain could anyone be that the Treasury would be either able or willing to live up to the obligation that the Resumption Act had imposed on it?



US oil drill

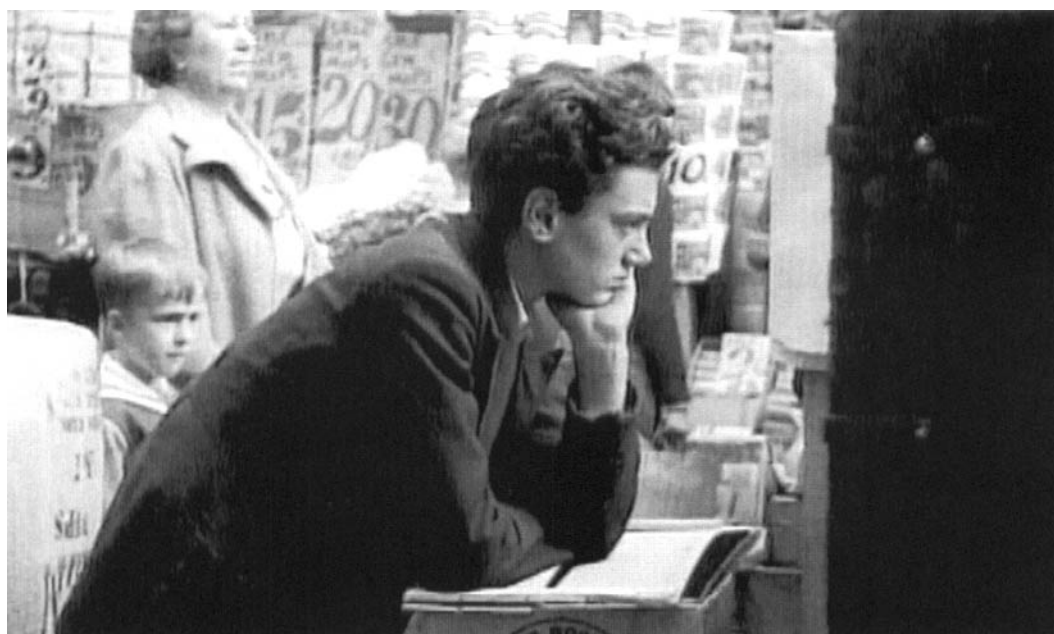
The day was saved for the US gold stock by a remarkable act of nature. In May, snow fell in France and Britain suffered a destructive frost, followed by rains that persisted until late summer. The disaster to British and French crops was almost unprecedented. Bad weather also hit Austria, Germany, and Russia, where the wheat crops were the poorest and smallest in ten years. Just to help matters along, the Indian cotton crop failed.

While the catastrophic European weather set wheat prices soaring, the weather in the United States was perfect. The US harvest was turning out to be enormous. In addition, an event with great future significance occurred to provide further help to the American trade balance: the tidewater pipeline from the newly discovered Pennsylvania oil wells was completed, swelling American exports of oil to the whole world.

Gold soon began to travel back from Europe toward the US. The government's reserve of gold, already rising, climbed from \$120 million at the close of June to \$157 million in early November. The Treasury was now in a position to pay out gold not just for notes presented for redemption but for the government's ordinary disbursements for goods and services. And for at least another three years, the extraordinary balance in the United States' favor in agriculture persisted and attracted even more gold from across the oceans.

We now fast-forward to 1890, the year the Barings crisis hit in London. Under these circumstances Europeans not only liquidated their holdings of American securities; they chose to convert the proceeds into their home currencies, which ultimately meant shipping their money back in the form of gold. The outflow made an alarming dent in the reserve the US Treasury held against redemption of the legal tender paper currency into gold. Now everything pointed to a run on the US gold stock, in which the only objective of each holder was to cash out his dollars into gold before every other holder could beat him to the windows of the Treasury.

Nature came to the rescue in 1891, just as it had saved the day in 1879. The South Russian wheat crop, the second most important European source of supply, was a total failure. The French harvest was the worst since the disasters of 1879. The farms of the United States, meanwhile, produced the largest grain crop in history, 65 per cent above the previous record. Once again, the outflow of gold was stemmed.



The Great Depression

This time the respite was brief. No sooner had 1892 rolled around than the golden hemorrhage resumed. By the end of May, the Treasury's gold reserve had fallen to \$114 million, just barely above the \$100 million minimum that Congress had intended the Treasury to maintain. The Treasury decided that there was no choice except to halt all government payments in gold; everything was now to be paid for in the form of legal tender paper currency. This unfortunate if unavoidable decision served only to increase the demand for gold among the public and foreigners and to raise even further doubts about the nation's commitment to the gold standard. Meanwhile, imports of merchandise had risen so rapidly in the first nine months of 1893 that they exceeded exports by the staggering sum of \$447 million.

The situation had by no means reached its climax. In May, National Cordage, the most widely held and actively traded industrial stock on the New York Stock Exchange, went bankrupt, pulling the whole stock market down with it in a panic collapse. Now, the solvency of the banks appeared to be in jeopardy, provoking the public to rush to withdraw their bank deposits – even paper currency seemed preferable. The run on the banks added to the hysteria and drove the interest rate on the shortest-term loans in the New York money market up to 74 per cent; time loans were unobtainable. Some 15,000 companies and 500 banks failed. In the face of these catastrophes, the only step the government took to relieve the situation was to repeal the Sherman Silver Purchase Act of 1890.

The aftermath of all these terrible events was extremely unpleasant. Unemployment in the United States exceeded 10 per cent of the labor force during every year from 1893 to 1898, by far the worst experience of any era in American history aside from the Depression of the 1930s.



*The New York Stock
Exchange*

In April 1894, a ragtag group of several thousand unemployed men organized themselves into what came to be known as Coxey's Army. Starting eastward from the Mississippi, they overran towns and seized railroad trains before appearing at the Capitol in Washington to demand relief. Troops had to be called out to disperse them. In the following month, downward pressure on wages provoked a strike at the Pullman Company that lasted for over two months before being terminated by a government injunction, the first use of the anti-trust laws against labor unions; the socialist leader Eugene Debs directed the strike and spent six months in jail for his efforts. In July, labor groups took possession of the railway system, converging on Chicago while their leaders opened formal headquarters there. The infantry had to be called out on that occasion. Nature decided to reverse her blessings and stunt crops in the United States in 1894 while providing bumper supplies to Europe.

Meanwhile, back in Washington, nothing hopeful was happening to the Treasury's financial position. By January 1895, its gold reserve was approaching \$40 million and sinking at a rate of \$2 million a day.

The five years from the passage of the Sherman Silver Act and the Barings crisis in early 1890 to the climax of January 1895 bracketed a series of gold-related disasters that slammed into the American economy with unparalleled magnitude and duration. Europe watched as the US suffered in isolation. Admittedly, the early 1890s were rough in Europe as well, but it was clear that the Bank of England, the Bank of France and their counterparts across the Continent were not about to soil their coveted gold bricks by offering them in the form of credits to the untrustworthy Treasury of the United States.

Yet the United States did not go over the brink in January 1895. Indeed, on the very last day of the month, the stock market leaped upward, the dollar suddenly started to

strengthen in the foreign exchange markets, and orders to export gold were abruptly canceled. Nine million dollars of gold on ships in the harbor were unloaded overnight.

What had happened? Lacking succor from the central banks of Europe and without any international organizations such as the International Monetary Fund, Americans had managed to concoct their own version of what would turn out to be the bail-outs of the late 20th century. They carried out the rescue operation with great skill by combining their genius for improvisation in the face of danger with their unabashed willingness to display raw power when required.

The power was deployed by none other than J Pierpont Morgan. By lucky coincidence, President Grover Cleveland was a friend of the House of Morgan, having worked at a law firm right next door to the bank during the four years between his two presidential terms. The two men were also neighbors in country homes in Princeton, New Jersey. Cleveland was the only Democrat for whom Pierpont Morgan had ever voted.

There was nothing easy or simple in what took place. In light of the distressed economic situation, popular hatred for New York bankers was more intense and widespread than ever, leaving Cleveland unable to turn to them as Secretary Carlisle had done only a year earlier. Even if the bankers had been more cooperative, there just was not enough gold in the United States to restore solvency to the Treasury. There was no choice but to turn to European financiers, and here the leadership continued to be with the Rothschilds. The Secretary and the Rothschilds agreed to attempt a European bond issue, and their first step was to approach the European branch of the Morgan bank in New York – J S Morgan & Co. J S Morgan, however, demurred, stipulating that they would participate only if Pierpont Morgan himself would handle the American side of the arrangements together with the Rothschild representative in New York, August Belmont, Jr.

The Cabinet in Washington was adamant in their opposition to any suggestion of a bond issue that would put the government of the United States in debt to a bunch of foreign bankers. Pierpont Morgan was apoplectic. He cabled his London partners that the United States was on 'the brink of the abyss of financial chaos' and set off at once in a private railroad car for Washington, taking Belmont along with him. When he was informed that there was no point in his seeing Cleveland, Morgan declared, 'I have come down to see the President and I am going to stay here until I see him.' He won his point. He soon joined a meeting with Cleveland, Carlisle, and the attorney general, during the course of which a clerk came in to inform the secretary of the Treasury that only \$9 million in gold coin remained in the government's vaults. Laconic as usual, Morgan stated, 'It will be all over before three o'clock.' Cleveland now realized he had no choices left. 'What suggestions have you to make, Mr Morgan?' he asked.

Morgan presented an audacious scheme. He proposed to sell a Treasury bond issue of approximately \$65 million to a European syndicate that Morgan and Rothschild would organize, payment to be made in some 3.5 million ounces of gold coin (about 100 tons), at least half of which would be obtained in Europe.

Morgan's plan contained three critical elements. The first was in the text of the contract between the Morgan–Rothschild syndicate and the Treasury, which said, in effect, that the syndicate was going to rig the gold market. The second element was that they would use their own supplies of European currencies to lend to Americans who owed money to Europeans on trade or financial transactions, thereby stanching the demand for conversion of dollars into gold. Finally, the syndicate bound together in this undertaking every banking house in New York City with important European connections, cutting them in on the bond issue as part of the deal.

When news of this unorthodox transaction broke, the public clamor was deafening against what appeared to be a sellout to the foreign bankers. *The New York World* described the syndicate as 'bloodsucking Jews and aliens'. In Congress, William Jennings Bryan asked the clerk to read Shylock's bond from *The Merchant of Venice*. The president was unmoved. In his annual message of 2 December 1895, Cleveland observed that he had 'never had the slightest misgiving concerning the wisdom of this arrangement'.

The actual daily execution of the plan was watched with skepticism in both London and New York, but it worked. It worked in part because of the mechanics of the plan, but the market's understanding that Europe was providing that kind of support was sufficient to soothe the bankers and investors. Soon gold was arriving at the Treasury from Europe at \$5 million a month; on 8 July, the Treasury reserve was back up to \$108 million. A virtual buying panic in American securities broke loose on all the European markets. During the spring, every outbound steamer carried piles of American stocks and bonds consigned to European houses.

Although the syndicate did not succeed in holding the pieces together indefinitely, and further weakness appeared later in 1895, the worst was over: 1896 would turn out to be the low point for business activity after so many years of recession.

Despite these dramatic victories, the silver enthusiasts in 1896 mounted the most powerful of all their attacks on the gold standard. In a convention held in tents in open fields in Chicago, they persuaded the Democratic Party to nominate William Jennings Bryan of Nebraska, only 36 years old, to oppose the 53-year-old Republican William McKinley in the presidential election of 1896.

This was the only election in American history in which the nature of the nation's monetary system came to occupy the central focus: McKinley was convinced that he could win the day by concentrating on the virtues of high-tariff legislation. He soon found out he would attract no votes from the Democrats on that. He knew, however, that many Democrats were vacillating over Bryan's unqualified support for the demand in their party's platform for 'free and unlimited coinage of both silver and gold'. On 30 July, Bryan came out flatly in favor of the mono-metallic gold standard.

Despite his youth, Bryan was a formidable opponent who was known as 'the silver-tongued orator' and 'the Great Commoner'. He was a man who always knew where he



William Jennings Bryan
(1908)



World War I

stood, with no words wasted. His view of economic matters was equally direct: 'Money is to be the servant of man, and I protest all theories that enthrone money and debase mankind.'

Bryan's famous speech at the Democratic Party convention on 9 July 1896, was in defense of the declaration for bimetallism in the party's platform. The simplicity of the language, the eloquence of

the phrasing, the indisputable confidence in the theme, the powerful organization of the arguments, and the alternation between a soaring sense of idealism and hard-headed political analysis are rare achievements.

Bryan began by claiming to speak 'in defense of a cause as holy as the cause of liberty – the cause of humanity'. Then he took up the cudgels against the gold standard, ending with:

If they dare come out in the open field and defend the gold standard as a good thing, we will fight them to the uttermost. Having behind us the producing masses of this nation and the world, supported by the commercial interests, the laboring interests and the toilers everywhere, we will answer their demand for a gold standard by saying to them: You shall not press down upon the brow of labor this crown of thorns, you shall not crucify mankind upon a cross of gold.

Bryan lost the election, and by a substantial margin. The impact of this on the financial markets was extraordinary. Bryan's noisy campaign had provoked a high degree of nervousness and uncertainty during the summer. Within a week after the election, stock prices soared, money rates were down to 4 per cent, and gold coin began to flow back into the Treasury for conversion back into paper currency.

Explanations for McKinley's great victory vary. Although bimetallism had been the traditional form of monetary standard for centuries, many voters in the 1890s were so unfamiliar with its operation that they perceived it as a newfangled idea that was not to be trusted. Despite the many hair-raising vicissitudes of recent history, gold had effectively been the single standard in the United States for over 20 years. Few people had any clear memory of any other set of arrangements.

Nature, however, was also at work once again. In October, news arrived of a failure of the Indian wheat crop serious enough to convert India from an exporter to an importer of wheat. Although Bryan claimed that the 'money power' was manipulating the market, the raw facts demonstrated that wheat prices could rise even under a full gold standard.

When the slaughter of World War I finally came to an end in November 1918, over eight million men had perished in combat, with somewhat more than half from the Allied side, including 1.8 million Russians and 1.4 million Frenchmen. Over a third of the German male population aged 19 to 22 years were gone; one in ten of British soldiers had fallen. The United States, by comparison, lost 114,000 men, less than half the losses of Romania. A total of 15 million men on both sides were wounded, many so badly hurt that they would be dependent for support on society and on their families for the rest of their lives, beyond hope of returning to a productive existence.

The financial consequences were just as appalling. National debts had swelled by many multiples of their 1914 values. Difficult as those burdens were to manage, each country owed most of that money to its own citizens. But the Allies also ended up in debt to the United States to the tune of nearly \$2 billion, while France, Italy, and Russia each owed Britain some \$500 million. Total holdings of gold by Britain, France and Germany at the end of the war amounted to no more than about \$2 billion. These debts amounted to substantial sums.

Rancor, bitterness, selfishness and envy stained almost every aspect of international relationships, corroding the congenial cooperative spirit of the earlier years. The French, having sustained most of the physical damage of the war, insisted on payment of the outlandishly harsh German reparations; until Germany came through, the French refused to repay their debts to the British, whose land was unharmed. The British, however, had also shed copious amounts of the blood of their youth and, in the process, had liquidated a substantial volume of their overseas wealth in order to pay for the war. Without payments from the French, the British were unwilling to pay down their debts to the Americans, who had come out of the war with only minor casualties and an economy that was both wealthier and as strong and vigorous as ever.

In addition, relations between the British and the French often led to serious problems that might otherwise have been avoided. Despite the high level of casualties, the British emerged from the war convinced that the Channel still kept them separate from 'Europe' and from the massive disorders on the Continent – France's war damage, Germany's revolution and impoverishment, the breakup of the Hapsburg Empire, the Russian Revolution, and upheaval from Poland on down through the Balkans. At the Versailles Peace Conference, this attitude kept Prime Minister Lloyd George from making much effort to block the French premier, Georges Clemenceau, from imposing a peace treaty on the Germans so brutal and impossible to fulfill that it would lead to the rise of Hitler and a war even more terrible in its toll than the war of 1914–18. This attitude also interfered with British–French cooperation on the matter of gold at crucial steps along the way.

Meanwhile, the Americans took the position: 'They hired the money, didn't they?' Standing on the indisputable evidence that the doughboys had come to the rescue of the



*The Capitol Building,
Washington*

Allies in the nick of time, the United States made concessions only when crisis conditions pushed matters in that direction or when others made proportionate sacrifices. Many Americans disapproved of the peace treaty and Wilson's helplessness to influence the outcome in the more noble directions he had promised. Consequently, there was persistent pressure to cling to traditional prejudices against 'foreign entanglements'. That the United States sat in comfort on by far the largest pile of gold reserves of any country only fortified the reluctance of Americans to become

involved in the economic turmoil in Europe.

A small number of officials recognized how essential American cooperation was to the recovery of Europe, but they were only putting their fingers in a dike that the flood would overwhelm in spite of their efforts. Before the structure collapsed, even they managed to shoot everybody else in the foot as they were taking aim at their own.

In 1920, when the British pound was trading in the foreign exchange markets at around \$4.00 – it would touch \$3.40 at its lowest – Parliament mandated a full return to the gold standard in Britain by the end of 1925. If Britain were to revert to the gold standard at Isaac Newton's longstanding metric, which had worked out to \$4.86 per pound sterling for Americans in the prewar era, that \$4.86 was not going to buy anywhere near as much in the Britain of the 1920s. Thus, it was clear that the stubborn character of the adverse trends in the balance of trade might well create unmanageable pressures on the gold stock. If, however, some lesser value were chosen for the pound as a means of helping the trade position, Britain's coveted credibility would be compromised and doubt would persist as to any kind of stability to British money for the indefinite future. Foreigners long accustomed to using the City of London as a banking center would flee elsewhere – New York or Paris, for example – and the pound would never look sterling again.

Although the debate swung back and forth all the way up to the 1925 deadline, informed opinion remained in firm support of \$4.86. The obstacles appeared small compared to the magnitude of the victory that appeared to lie ahead. 'Sacrifice' became a buzzword.

Presiding over most of the major economic policy decisions and much of the debate over the pound was a mysterious and powerful man named Montagu Norman. Norman was governor of the Bank of England for a record 24 years, serving 12 consecutive two-year terms from 1920 to 1944. According to his biographer Andrew Boyle, Norman's 'reputation for godlike aloofness and for tantalizing omniscience' enabled him to exert with gusto every bit of the power that 'the tremendous mystique surrounding the high office he held' placed in his hands.

Gaunt, elegant, with a small beard and a grayish visage, Norman had descended from a family that had long been a part of the upper-class establishment of the City. He was

alternately severe, austere, demanding, charming and seductive, as well as given to periodic nervous breakdowns that kept him out of the office for extended periods of time. Brilliant in finance but unshakable in his conservative ideologies, he was, in Boyle's words, 'the high priest of the City's dogma that the power of Britain had been founded on gold'.

Despite the disparities of price levels between Britain and its major trading partners, Norman was convinced that he could manage to get the pound back to \$4.86 without 'quasi-catastrophic effects' but that Britain must look forward to 'a long period of dear money'. 'Dear money' means high interest rates; high interest rates in turn tend to produce subdued business activity, which in turn means higher unemployment. The high unemployment keeps wages in check, which keeps price increases in check, which tends to strengthen the foreign exchange value of the currency and bring gold to the nation's shores. The human consequences of this inevitable sequence of events did not disturb Norman. When he jacked up interest rates in 1920 in order to squelch a budding boom, nearly a million men were laid off within 12 months, an outcome that was fully to be expected. According to Norman, the human pain was a matter for the government to worry about, not the Bank of England, whose primary responsibility was to add to the precious collection of gold bricks accumulated in its vaults below the ground.

To initiate the process up to the April 1925 deadline for restoration of the gold standard, Norman crossed the Atlantic in late December 1924 to consult with his close friend Benjamin Strong, president of the Federal Reserve Bank of New York, who, although in theory subject to the control of the Federal Reserve Board in Washington, was prepared to act independently when the circumstances suited him. Strong shared all of Norman's values and prejudices. Norman also took the occasion to sound out J P Morgan, Secretary of the Treasury Mellon, as well as other Federal Reserve officials. Norman would report back that they all agreed that the moment for resumption in Britain had arrived. He added that Strong in particular had assured him that there would be no Federal Reserve policy aimed at a 'deliberate policy of deflation' and that the Americans would attempt to lean on the expansionist side, to the extent that they could influence prices. Strong's own report of the meetings concluded that failure to resume the gold standard in Britain would lead to 'a long period of unsettled conditions too serious really to contemplate. . . It would prove an incentive to all those who were advancing novel ideas for nostrums and expedients other than the gold standard to sell their wares.'

Strong did warn Norman that his wishes would not necessarily prevail against a national mood opposed to international financial commitments and, in particular, the provincial attitude of most members of the Board in Washington. The most ominous possibility suggested by Strong was that the Federal Reserve might at some point have no choice but to raise interest rates in the United States to discourage stock market speculation, at which point domestic considerations would have to come first.



LEFT: Winston Churchill
RIGHT: John Maynard
Keynes

Encouraged by Norman's report, the chancellor of the Exchequer, Winston Churchill, notified Prime Minister Stanley Baldwin that 'It will be easy to attain the gold standard, and indeed almost impossible to avoid making the decision.' But it was not so easy to make the decision. Between the end of December and the deadline at the end of April, Churchill would spend a miserable four months trying to come to grips with the matter. 'When I held other offices under the Crown,' he complained to a friend, 'I could always find out where I was. Here I'm lost and reduced to groping.' He also grumbled that 'The governor [Norman] shows himself perfectly happy in the spectacle of Britain possessing the finest credit in the world simultaneously with a million and a quarter unemployed.' A senior advisor, Otto Niemeyer of the Treasury, observed that 'None of the witch-doctors see eye to eye and Winston cannot make up his mind from day to day whether he is a gold bug or a pure inflationist.'

The decision was reached on 20 March and announced in Parliament on 25 April; on 14 May, the king's signature made it official. Churchill went before Parliament on 4 May to defend the decision. In a fine Churchillian flourish, he finished with these ringing words: 'If the English pound is not to be the standard which everyone knows and can trust . . . the business not only of the British Empire but of Europe as well might have to be transacted in dollars instead of pounds sterling. I think that would be a great misfortune.'

The Gold Standard Act of 1925 did not completely restore the old arrangements. Bank notes remained legal tender but were no longer convertible into gold coin at the Bank. In other words, the ancient right to bring gold to the bank for minting into coin was abolished. Nevertheless, the Bank would continue to sell gold on demand in the form of 400-ounce bars – heavy things weighing 33.33 lb – at the traditional price of £3 17s 10 ½ d, or about £1700 a bar. The economist John Maynard Keynes drew a sad moral from this step. Looking back from the vantage point of 1930, he eloquently pointed out that

[Gold] no longer passes from hand to hand, and the touch of the metal has been taken away from men's greedy palms. The little household gods, who dwelt in purses and stockings and tin boxes, have been swallowed by a single golden image in each country, which lives underground and is not seen. Gold is out of sight – gone back again into the soil. But when gods are no longer seen in a yellow panoply walking the earth, we begin to rationalize them; and it is not long before there is nothing left.

Keynes pointed out that a return to gold would be 'a dangerous proceeding,' because it would put postwar Britain at the mercy of the Federal Reserve authorities in the United States. The US gold stock was six times the size of Britain's, he said, which meant that



The General Strike of 1926

Americans could absorb swings up and down in their gold stock that would ricochet back onto Britain with six times the impact. In addition, Britain was now in debt to the United States instead of being one of the Americans' largest creditors.

The grimmest problem was the failure of the vaunted revival of international trade to make its appearance on schedule. The pressures on the British domestic economy were intense. At the end of July, the coal-mining industry, with costs too high to be competitive in export markets, demanded that the miners either take a cut in wages or face losing their jobs. There had already been a sharp drop in coal exports, and the industry was losing £1 million a month. Meanwhile the miners, still smarting from bitter struggles against the employers in 1921 and 1922, flatly refused to accept the pay cut. Protracted negotiations continued into spring 1926, including threats of strike and a lockout, a temporary government subsidy, and a court of inquiry. One well-known member of this tribunal saw little alternative to a firm stand by the industry; the strike threat, he said, could be explained 'only by the immediate and necessary effects of the return to gold'.

Prime Minister Baldwin agreed. It was precisely the kind of situation that William Jennings Bryan had warned against when he cried, 'You shall not crucify mankind upon a cross of gold!' Baldwin summed up the position of the employers by demanding that 'All



Welsh pithead

workers of this country have got to take a reduction in wages to get this country on its feet.' On 1 May, when the leader of the miners, A J Cook, declared, 'Not a penny off the pay, not a minute on the day,' the owners proceeded to lock out a million men. The central organization of British unions, the Trades Union Congress, declared a national strike in support of the mine workers. The General Strike of 1926, as it came to be called, paralyzed large parts of the country, but it also brought out countless numbers of volunteers who kept supplies moving and essential services operating. The General Strike petered out, but the unpleasantness at the coal mines dragged on until November. At that point, the miners were on the verge of starvation. They folded up their protests and accepted the lower wage rates. The more liberal and leftist members of British society were convinced that the miners had been betrayed by stupid and venal leaders.

Blame for all the bad news soon landed on Churchill. The situation at the mines was dreadful enough, but there were more profound and distressing developments. Quite contrary to widespread hopes and expectations, world prices failed to move upward and, in fact, were drifting downward in most countries, including the United States, on which so much depended.

The French return to gold made a striking contrast with what had transpired in Britain. The French lurched from one crisis to another in the search for enough breathing space to make any kind of resolution at all – although there was never any doubt that France would rejoin the gold standard club at some point. In fact, when all the shouting was over, the French turned out to have made a wiser and more realistic decision than the British and were a great deal happier with where they ended up. Once the franc was stabilized, a flood of foreign gold and capital rushed toward Paris. That these repercussions would cause serious problems for London was more of a source of satisfaction than of concern.

The capital inflow had permitted the Bank of France to accumulate large sums to its credit in the City of London, from £5 million in November 1926 to £160 million by the end of May 1927. This was money that the Bank of France could withdraw on demand and represented a heavy potential claim against Britain's stock of gold. The relative positions were in any case shifting rapidly. From a position of approximate equality with the gold reserve of the Bank of England in 1926, the Bank of France's gold holdings were double Britain's by 1929; two years later, the French hoard would be approaching five times the size of Britain's.

Britain was under constant pressure to hold on to its gold stock in the face of the gains being made by France and Germany (the latter because of its high interest rates). In February 1931, Norman would describe his situation as having been continuously 'under the harrow' – a farmer's cultivating tool set with spikes for pulverizing the soil.

The growing financial tension between London and Paris turned into personal aggravation and frustration when Emile Moreau, former head of the Bank of Algiers, took



over in June 1926 as the President of the Bank of France. Strongly patriotic, canny, and sophisticated in financial matters, Moreau was well aware of the power that the ownership of gold conveyed in the postwar monetary system. In terms of personality, however, he was a maverick, for he was a laconic provincial from Auvergne who lacked the traditional polish of the central banker. He had little interest in international concerns, disliked travel, and spoke no foreign languages.

It would be hard to imagine anyone less to the taste of the suave, aristocratic, and worldly Montagu Norman. Norman patronized Moreau without mercy. Although Norman spoke fluent French, he insisted on speaking English at his meetings with Moreau; this meant that Moreau always had to have an interpreter present. Norman, who had spent one period of his youth in Germany, was always partial to Germans and antagonistic toward the French. The friction between him and Moreau was compounded by his warm friendship with the Reichsbank president, Hjalmar Horace Greeley Schacht, a powerful and brilliant financier who had been primarily responsible for ending the wild German hyperinflation of the early 1920s.

The fiercest scuffle between Norman and Moreau developed in the course of 1927, as France's balances in London were climbing at a rapid rate. Moreau made noises about wanting to convert these balances into gold by drawing from the Bank of England's gold

*A Wall Street trader:
Fortunes were lost in
minutes when the stock
market crashed in 1929*



stock; he suggested that Norman could avoid that unpleasant outcome by raising interest rates to persuade Frenchmen to stop converting their pounds. Norman could do no such thing in view of the high unemployment plaguing Britain. Instead, he insisted that France should legally restore the fixed relationship between francs and gold to halt the speculation that the franc might become even more expensive in terms of pounds. Benjamin Strong intervened in the conflict at this point, agreeing to provide the French with US gold in exchange for their sterling balances, thereby taking the pressure off London.

The arguments persisted, however. In July, Ogden Mills, the US secretary of the Treasury, organized a peace conference at his Long Island home, to which he invited Strong, Moreau, Norman and Schacht. As usual, Moreau declined to travel and sent a high-level representative in his place. Strong took the opportunity once again to step forward to help, agreeing that the Federal Reserve should lower interest rates in the US to take pressure off the pound while also making additional amounts of gold available to the French in exchange for French sales of sterling.

Despite Strong's earnest interventions, Norman and Moreau remained at loggerheads, with Moreau increasingly indignant about Norman's 'imperialism'. In February 1928, he complained to French premier Raymond Poincaré that Britain had been 'the first European country to re-establish a stable and secure money [and] had used that advantage to establish a basis for putting Europe under veritable financial domination. . . . Are we to let this continue?' Moreau was charged up enough to actually cross the Channel 'to offer Norman war or peace'. On his arrival at the Bank, he was politely informed that Norman was indisposed. That did not stop Moreau, who proceeded to negotiate a set of congenial agreements with Norman's staff. But that did not stop Norman. He accomplished a miraculous recovery the instant Moreau had departed and lost no time in repudiating the agreements.

Norman's mental and physical troubles were by no means over, however. Moreau noted (with some pleasure?) in his diary in April that, 'M. Norman [is] in a state of sickly neurosis as a result of the incidents in recent months.' Norman would receive another personal blow when his beloved friend Benjamin Strong succumbed to tuberculosis in October 1928. Norman would miss Strong, but the atmosphere at the Federal Reserve was already in the process of shifting even before Strong's final illness.

Indeed, in the United States, domestic considerations were now taking priority over international issues, as the authorities watched with increasing concern the thundering momentum of the bull market on Wall Street. Each breathtaking rise in stock prices served only to whet investors' appetite for more. In late 1928, when John J. Raskob, director of General Motors, friend of the DuPonts, and chairman of the Democratic Committee, wrote in the *Ladies' Home Journal* that 'Everybody Ought to Be Rich,' he evidently had plenty of believers.

*President Herbert Hoover*

It is fair to ask why the unfolding miracles in the stock market were of any concern to the Federal Reserve, which had been established in 1913 to supervise commercial banks and to provide liquidity for the economy as needed. The concern was not misplaced. Much of the boiling stock market was being financed by people who borrowed money to buy their stocks, often at interest rates well over 10 per cent. Banks began to lose their taste for financing anything except this speculative binge. Even worse, a flood of capital from other parts of the economy began to surge toward Wall Street.

George Harrison, Strong's successor as president of the New York Federal Reserve, argued in favor of an earlier suggestion of Strong's to take 'sharp, incisive action' to raise interest rates high enough to kill off the speculation, to be followed immediately by a decline in rates in order to avoid killing off the business prosperity. Strong had even persuaded Norman to support this notion, although Norman realized that any increase in rates in the United States would put further pressure on the Bank of England's gold stock. Central banks throughout Europe had been raising their interest rates all during the year.

The Federal Reserve rate was so far below what could be earned by lending to speculators that it failed in its purpose of serving as a serious deterrent. Its success, instead, was in rising high enough to contribute to weakness in the general economy; industrial production had already been falling for a few months before the big crash hit the stock market in October. Nor were the consequences of these moves limited to the United States. The vital flow of American lending to Europe, and especially to Germany, was unable to survive the rapid rise in domestic interest rates. By the time the stock market touched its peak, American lending abroad had essentially dried up. In fact, a rising stream of European capital had for some time been flowing to New York to join in the fun.

The outflow was no fun in the countries from which the capital moved, where interest rates shot upward in an effort to keep capital from leaving. Britain, Germany, Italy and Austria were already on their way into depression at the moment the stock market crashed in October; German unemployment alone had quadrupled between the summer of 1928 and the end of 1929.



Immigrants had rushed to the US for a bright new future but few found it

In the early days after the crash of 1929, Andrew Mellon – secretary of the Treasury and one of the wealthiest men in the United States – gave the following advice to President Hoover: ‘Liquidate labor, liquidate stocks, liquidate the farmers, liquidate real estate . . . purge the rottenness out of the system.’ At about the same time, Russell Leffingwell of Morgan offered his prescription for how to get the economy out of the depression: ‘The remedy is for people to stop watching the ticker, listening to the radio, drinking bootleg gin, and dancing to jazz . . . and return to the old economics and prosperity based upon saving and working.’

There were a few brave souls who were convinced that it was the entire economic system that suffered from looming maladjustments, not the morals of the players in the stock market. These men argued that the way out of the Depression was to attempt to short-circuit the misery, not intensify it. They sought means to relive the tottering banking system and somehow put money into people’s pockets so that they would be both willing and able to go out and spend it. They found themselves either talking to a brick wall or facing such vigorous opposition that in time they gave up and retired from the fray.

The gold standard dominated the mindsets of policymakers far into the Depression. Politicians, monetary authorities, business leaders and bankers and even most academics continued to genuflect before the gold bricks as though these gleaming hoards were all that mattered. They forgot that this was also the stuff that the Parthians had poured down the throat of Crassus after he had been defeated in battle.

The resulting infectious epidemic of disasters only strengthened the obstinate faith in the traditional approach. Indeed, as export trade shriveled in country after country, there seemed to be no choice but to restrain demand in order to hold down imports. Failure to act along those lines would be certain to lead to the most dreaded result: losses of gold to other nations. Although deflationary pressures were coming down so hard on Germany that Adolf Hitler would one day come roaring into power, and although

mounting unemployment and spreading bank and business failures throughout the world were tearing into the deepest roots of the capitalist system, these horrors served only to strengthen the determination to preserve gold reserves above all else.

Events during the first year after the Great Crash were disturbing, but 1930 in retrospect looks in many ways like the calm before the storm. On 7 March, US President Herbert Hoover reported that 'All the evidence indicates that the worst effects of the Crash upon unemployment will have passed during the next 60 days.' This was not a bad forecast under the circumstances. The domino effect of a major crisis would not make itself felt until the end of 1930. In Britain, the primary consequence was a series of significant losses of gold. By 13 January, Norman had already warned the chancellor of the Exchequer that he could no longer postpone the unpleasant task of raising interest rates.

The fuses were in place. They were lit on 11 May when the explosive news of the failure of the Creditanstalt Bank in Vienna stunned the world. This was Austria's largest commercial bank by far, holding more than half of all Austrian bank deposits. Its failure, in the words of the British Treasury official Ralph Hawtrey, 'sent a terrible spasm of panic through the financial centers of the world.' The spasm first ignited a run on the other Austrian banks and then panic hit the schilling in the foreign exchange markets. In desperation, the Austrian National Bank tried to borrow foreign exchange from other central banks. The Bank of France insisted that no loans would be possible unless the Austrians renounced their intention to form a customs union with Germany, which the Austrian government refused to do. Meanwhile, the Bank of France, busy substituting gold for deposits in foreign central banks, acquired \$539 million of gold – about 25 per cent more than the entire year's production of gold at the mines. Nevertheless, Montagu Norman, still fighting his personal war against the French, succeeded in infuriating his enemy once more by going ahead with a loan to the Austrians from the Bank of England.

It was too late. Everything was now focused on protecting the golden stockpiles from further damage. Norman's modest efforts would prove to be nothing more than a transitory stopgap. In short order, the run on Austria provoked runs on Hungary, Czechoslovakia, Romania and Poland. These panics were sufficient to disseminate alarm to still more countries, with the most serious repercussions coming down on Germany.

The knee-jerk response by the German chancellor, Heinrich Brüning, was to slash government spending. Brüning tried to offset the fury unleashed by the additional unemployment and deflationary pressure he was creating by proclaiming that Germany had reached the limit of its ability to pay reparations. His statement may have been welcome at home, but it led the flight from the Reichsmark to begin in earnest. The crisis became so threatening that, on 19 June, President Hoover, at Leffingwell's suggestion, proposed a one-year moratorium on both German reparation payments and payments by the Allies to the United States for war debts. The French were furious at this concession

to the Germans and refused at first to take part in any discussions. Although temporary credits were arranged for Germany, the haggling went on for so long that the panic started up all over again and the German hemorrhage of gold and foreign exchange accelerated. In an effort to contain the damage, the Germans began the imposition of a system of controls on foreign exchange transactions that became so tight and so complex that by 1932 Germany effectively ceased to be on the gold standard.

The crisis in central Europe rapidly communicated itself to the pound, in large part due to the actions of the French. Britain now began to suffer sharp losses of gold and heavy withdrawals of foreign-owned sterling balances. The sterling crisis was all the more remarkable because prices in Britain had already fallen 38 per cent from the level in 1925 when the gold standard had been re-established. When Harrison at the New York Federal Reserve Bank cabled Norman on 15 July that 'we are concerned and surprised at sudden drop in sterling today,' Norman's response was: 'I cannot explain this drop. it was sudden and unexpected.'

One can only wonder at the way Norman ducked Harrison's query. Two days earlier, the publication of the final report of the findings of a special governmental committee had disclosed the alarming deterioration in the condition of Britain's foreign trade position, with imports exceeding exports by an ever-widening margin. An even more immediate source of trouble was also coming to a head. London financiers had been borrowing at low interest rates in Paris and lending the proceeds to the Germans at much higher rates of interest, but now French financiers, uneasy about the outlook for the pound, were demanding repayment of their loans to London. The shocking sum of some £750 million was involved.

A week later, the situation had reached a point where Norman made the extraordinary decision to dispatch one of the Bank directors to ask the Bank of France for an immediate loan. The French were willing if the British government – in effect, the British taxpayers – would guarantee the loan. The Cabinet refused, the negotiations broke down and the pound resumed its slide.

This turn of events was too much for Norman. On 28 July, exhausted by the sequence of defeats to all his hopes and dreams, he went home from the Bank 'feeling queer'. After a week in bed, he sailed to Canada for a vacation of total rest. The last act of the drama that he had largely written and directed was about to play itself out without him. We will never know whether he was too sick to deal with the crisis or just unable to face the total failure of his efforts that loomed just ahead.

As the exodus of Britain's stock of gold persisted, sentiment was building for the government to take drastic action to put its financial house in order. The state of the economy made the task extremely disagreeable: by August, nearly one in four workers was unemployed, up from one in six a year earlier, while prices and wages continued to fall. With the Labour Party in power, headed by Ramsay MacDonald, the decision to cut the dole was turning out to be even more agonizing than if the Tories had been leading the government.

Yet something had to be done. On 24 August, King George invited MacDonald to form a National Government, a coalition of Labour, Liberal and Conservative parties, to make the pain of the budget cuts more politically palatable.

The National Government's Budget and Economy Bill provided for a £70 million reduction in government spending and a tax increase of £86 million. As a further lure to foreigners to hold sterling, the Bank Rate – the rate charged by the Bank of England to banks needing immediate credit – had been raised in a giant leap to 6 per cent from 2.5 per cent in June. The combination of higher taxes, reduced spending, and higher interest rates was lethal. The economy sank even lower and unemployment rose even higher, pulling tax revenues down along with the shrinking payrolls and profits. In the end, the government was left with a much larger deficit than the experts had predicted.

Before the Economy Bill had had much opportunity to rescue the pound, a bizarre but shocking event occurred while the debate on the legislation was under way. A small contingent of sailors at the British navy station of Invergordon went on strike against the pay cuts that were part of the proposed legislation. The press, both at home and abroad, gave the pocket mutiny huge black headlines. Foreigners received the news in a state of high alarm: if such a thing could occur in the British navy, of all places, the whole country must be on the verge of revolution. Nearly £40 million of gold was swept out of the Bank's vaults in a single week; £200 million had been lost since the middle of July.

By the third week of September, the jig was up. The Bank of England asked the government to relieve it at once of the obligation to provide gold bullion on demand, an obligation that had been in place a mere six years. The requisite legislation passed through parliament on 21 September. *The Economist* announced 'The End of an Epoch.' On 27 September, Keynes looked forward rather than backward. 'There are few Englishmen,' he wrote, 'who do not rejoice at the breaking of the golden fetters.'

A few days before these events, feeling much improved and ready to go back to work, Norman had sailed from Canada on a ship bound for home; he was totally unaware of the grand denouement of all his endeavors that lay just ahead. His colleagues at the Bank felt obliged to let him know about the momentous decision that was about to go into effect in his absence, but they did not want the news to leak out ahead of time. On the Saturday before the Monday on which the official announcement was to be made, and referring to the Old Lady of Threadneedle Street – the nickname of the Bank of England – they cabled Norman: 'old lady goes off on monday.' Poor Norman was so spaced out that he thought they were referring to his mother's plans for a vacation. Did he get a shock when the ship landed at Southampton!

The pound, no longer convertible into gold, took a steep tumble in the foreign exchange markets during the next three months. Central bankers around the world rapidly lost their appetite for holding their reserves in the currencies of other countries, even the almighty dollar itself.

President Roosevelt

Twenty-four out of the 47 nations on the gold standard immediately raced down the path that Britain had just blazed through the thickets of economic chaos: they suspended convertibility into gold within days of the British action. A year later, only the United States, France, Switzerland, Holland and Belgium remained on the gold standard; six years later, not a single country permitted their citizens to convert their currency or bank deposits into gold. The golden hoards were to be defended by rendering them inactive!

The dash toward gold hit the United States hard. This urgency to get out of dollars was a surprise, for the official US gold stock at that moment amounted to \$4.5 billion – over 40 per cent of the gold reserves of all central banks and treasuries around the world and 65 per cent larger than France's gold holdings. Nevertheless, on 22 September 1931, the Belgian national bank pulled \$106 million in gold from New York in one fell swoop; France took \$50 million on that day and another \$70 million a few weeks later. From the end of September to the end of October, a total of \$755 million in gold flowed out of the United States. About one in every seven gold bricks in the vaults of the Federal Reserve banks had departed. The panic induced by this news led Americans to follow suit by making massive withdrawals from commercial banks in the form of currency and gold coin, leading almost at once to another 800 bank failures.

At this point, hoarding of currency and coin by the public was restricting even further the ability of the banks to provide credit to their customers. President Hoover then invited Colonel Frank Knox of Chicago to conduct an educational campaign to discourage the hoarding; Knox must have been quite a salesman, for his pitch served at least to slow the rate of decline in the ratio for a while.

The savage deterioration in economic conditions forced the Federal Reserve authorities to conclude that they had pressed the policy of deflation far enough. Positive action could no longer be postponed. In the spring of 1932, the Reserve banks bought \$1 billion of US government securities. A fresh wave of anxiety and uncertainty broke out, prompted by concerns that the Federal Reserve was acting to promote inflation. The heroic step simply led to a renewed outflow of gold.

Now began a chilling replay of the events that had led to the final crisis of the British government in 1931: frantic efforts to extinguish a swelling budget deficit as tax revenues shriveled while the demand for government assistance to the unemployed increased in urgency. Following dutifully in the footsteps of Brüning and MacDonald, Hoover pressed Congress to act to reduce this forbidding deficit by cutting expenditures and raising taxes.

Hoover's determination in this matter was inexhaustible, but Congress was less motivated than the President for such noble objectives. Their spending cuts and tax increases were well below what Hoover sought. Nevertheless, the economic impact of these measures added to the prevailing deflationary pressures and led to an even greater budget deficit in 1933.

And so matters wobbled along until the election of November 1932, when the Democratic candidate, Franklin D Roosevelt, defeated Herbert Hoover in a landslide. It was a rough four months up to the inauguration on 4 March, with uncertainty growing as to what actually would lie ahead after the new president was installed. Roosevelt's campaign promise that the government would provide jobs for all the unemployed had the perverse effect of creating a new wave of unemployment by businessmen frightened by fears of socialism and reckless government spending.

Frightened Americans now joined agitated foreigners in seeking safety by moving their capital abroad or into gold. The mounting panic included withdrawals of gold coin from the commercial banks, with over \$80 million going out in the last ten days of February and over \$200 million during the first four days of March.

Over 10,000 banks disappeared from the scene during 1933, by which time the total number of banks in the United States had fallen to fewer than 10,000 from 30,000 in 1925. The frantic liquidation of bank accounts had driven the ratio of bank deposits to currency in circulation into a precipitous drop from \$6 of deposits for every dollar of currency in circulation at the end of 1932 to only \$4 for every dollar of currency in March 1933 – just about one-third of what this ratio had been before 1929.

Where was the Federal Reserve while all of this was going on? The eminent economists Milton Friedman and Anna Schwartz provide an answer to this question in their monumental monetary history of the United States: "The leadership which an independent central banking system was supposed to give to the market . . . [was] conspicuous by [its] absence."

On 9 March, five days after the inauguration, Roosevelt pushed the Emergency Banking Act of 1933 through both houses of Congress, authorizing him to regulate or prohibit the export or hoarding of gold or silver and empowering the secretary of the Treasury to require the surrender of all gold coin, bullion and certificates (paper notes fully secured by gold) held by the public. The public perceived the 9 March legislation as temporary! Anxiety eased, and gold and paper currency started to return to the banking system.

Everything seemed just fine for a few weeks, until the market began to suspect that Roosevelt really did plan to cut the gold value of the dollar. He had made no secret of his

intention to stem the deflationary cycle and start prices back upward, because he was confident that bold steps in that direction would encourage business to start rehiring and increase the depressed level of production. As the rumors about gold proliferated, the run on the dollar resumed.

Roosevelt was determined to proceed. On 5 April, only a month after his inauguration and acting under the authority of the Trading with the Enemy Act of 1917 and the Emergency Banking Act, he issued an executive order requiring all persons to deliver all gold coin, gold certificates and bullion to the banks in exchange for paper currency or bank deposits, and for the banks to deliver the gold to the Federal Reserve. According to Hoover, only about \$400 million in gold came into the banks, adding less than 10 per cent to the existing gold stock.

Further enabling legislation in April gave the president the authority, among other things, to fix the weight of gold in the dollar at not less than 50 per cent or more than 60 per cent below the weight that had established an ounce of gold at \$20.67 in 1837, nearly 100 years earlier. This act prompted Lewis Douglas, director of the Budget, to predict that 'This is the end of Western civilization.' There was more to come. Further congressional action on 5 June provided that any clause in any contract that provided for payment in gold was now abrogated – even including obligations of the US government.

The cancellation of the gold clause in US government obligations resulted in a rash of lawsuits that ended up in the Supreme Court. The Court agreed that Congress had no right to cancel the promise to redeem its debts in gold at the option of the holder. But then the justices went on to declare that, since the private ownership of gold coin was no longer legal, the plaintiff's demand for damages equal to the change in the gold value of the dollar from \$20.67 to \$35.00 was without merit!

On 3 July, Roosevelt issued a statement – which came to be known as the 'bombshell message' – declaring that efforts to stabilize exchange rates by going back to rigid relationships to gold were 'old fetishes of so-called international bankers' and that exchange-rate stability was 'a specious fallacy'. The statement scandalized most conservative politicians, financiers, and academic experts, but the president did find one supporter when John Maynard Keynes proclaimed Roosevelt 'magnificently right!' Raymond Moley, one of the president's chief advisors, subsequently quipped, 'Magnificently left, Keynes means.'

The game came to an end on 30 January 1934, when an executive order fixed the price of gold at \$35.00 an ounce, an increase of 69 per cent from the old value. That price would prevail without interruption for 37 years, which was three years longer than the *de jure* value of \$20.67 set in the Gold Standard Act of 1900. Although many other factors were at work, it is worth noting that from 1933 to 1937 industrial production jumped by 60 per cent and wholesale prices climbed 31 per cent while unemployment fell from 25 per cent to 14 per cent. By early 1937, the Dow Jones Industrial Average stood at 200, a

mighty surge from its nadir at 40 touched during the darkest days of 1932. Happy days were here again!

Except for the French, the Swiss, the Dutch and the Belgians, who remained nailed to the cross well into the 1930s. The Dutch and the Belgians clung to their 1913 parities right up into World War II. The French caught a bad case of the deflationary disease from which the rupture from gold had liberated Britain and the United States. Prices in France fell by nearly 25 per cent between 1931 and 1935, while French national income dropped by a third.

By 1936, the horrors had reached the stage where France was besieged by angry sit-down strikes. The political chaos led to the formation of a Popular Front government that included Communists and Socialists – not the most appropriate combination to restore international confidence in the franc. The break from gold and the devaluation of the franc occurred in September under cover of a Tripartite Agreement with Britain and the United States. This agreement added up to little in the way of action but did at least restore international cooperation to something better than the endless friction and backbiting that had characterized international financial relations since the end of World War I.

Meanwhile, something strange was happening to gold. As each currency broke away from gold and was devalued, a unit of each currency bought less gold than before – that is, the price of gold went up. In contrast, the prices of goods and services in all countries had fallen by substantial amounts. The result was that an ounce of gold in the mid-1930s could buy twice as many goods and services as that same ounce could have bought in 1929.

In a world in which production was deeply depressed for just about everything that people desperately wanted but could not afford – food, clothing, housing – this leap in the price of gold was a bonanza for gold miners and the equivalent of a whole new gold rush for the world economy. Gold production soared. This new gold came primarily from South Africa, although, as in the past, Russia remained an important producer. In addition, Asia – beset by the Great Depression like everyone else – for the first time in history dishoarded gold and shipped about \$100 million westward. In 1932, the two million tons of gold coming out of the world's gold mines amounted to nearly half of all monetary gold accumulated from the beginning of time to the middle of the 19th century. The growth in monetary gold around the world was so vast that by 1939 there was enough gold in the monetary reserves of the world to replace all ordinary currency 100 per cent with gold coin.

This was one rare moment when there seemed to be so much gold in existence that nobody knew what to do about it. Only one solution seemed acceptable: most of the new gold crossed the oceans to New York, along with a lot of old gold. About 20 per cent of this inflow came from France, but, in the pain of the Depression, even India was now exporting gold to the United States. When World War II broke out, some \$20 billion, or 60 per cent of the world's monetary gold, was lodged in the United States. This massive

hoard weighed more than 15,000 tons and was equal to 12 years' worldwide gold production at the time. What a pile it must have been! Atahualpa's chamber, when filled with gold, had contained only six tons – and even that was greater than the total annual output of gold in Europe in the early 1500s.

What could explain such a phenomenal migration of gold from the whole world to the United States? The most significant motivation was political: these were frightening days in Europe. Hitler was on the march, with Mussolini and the emperor of Japan rattling their swords by his side. While most other countries were still floundering in the mess of the Depression, Hitler's heavy spending on arms succeeded in pulling Germany out of the Depression as rapidly as Roosevelt's new policies had promoted recovery in the United States. Italy invaded Abyssinia (now Ethiopia) in 1935. The Nazis invaded Austria in 1938 and Czechoslovakia in early 1939; Hitler made off with their gold reserves as soon as his troops entered Vienna and Prague. Meanwhile, the Communist system was thriving in the Soviet Union, where output and employment kept rising all during the 1930s. In this alarming environment, shipping capital across the Atlantic to the United States seemed to make good sense, especially in the form of gold.

But the strangest thing of all was that the United States absorbed those billions of dollars' worth of gold without any sign of some natural force to throw the process in reverse. Money continued to heap up in the US far beyond its proper level just to sit quietly until the storms had blown over. The process came to be known as a 'liquidity trap'. The accumulations of cash would be put to use only later when the pressures of wartime spending demanded it.

When the bombings, the bloodshed, and the persecutions of World War II came to an end in the summer of 1945, Europe and Asia were a shambles. Industries, cities and transportation systems were little more than piles of rubble; millions of people were homeless or jobless, including many who had been forcibly transported far from their own countries; currencies in most countries were worthless – in Germany, cigarettes or nylon stockings were the preferred form of payment. In addition to the United States, only Britain, the Soviet Union and China had emerged from the war with surviving forms of government, and only Josef Stalin and Chiang Kai-shek were still in office among the wartime leaders.

Yet within less than 20 years, every vestige of the ravages of war had vanished. By the early 1960s, even western Germany, the Soviet Union and Japan, which had suffered the worst damage, were vibrant economies enjoying high rates of economic growth and broadly competitive with the one major nation that had emerged unscathed from the hostilities – the United States.

Despite this imposing progress, all the best-laid plans of both leaders and followers ended up far away from their announced destinations. The villain of the piece turned out to be gold.



Tearing Loose



THE LEADER MOST CONCENTRATED ON GOLD IN THE 1960S WAS GENERAL CHARLES DE GAULLE, WHO HAD COMMANDED THE FREE FRENCH FORCES IN WORLD WAR II AND WAS PRESIDENT OF FRANCE FOR MOST OF THE 1960S.

SEVENTEEN

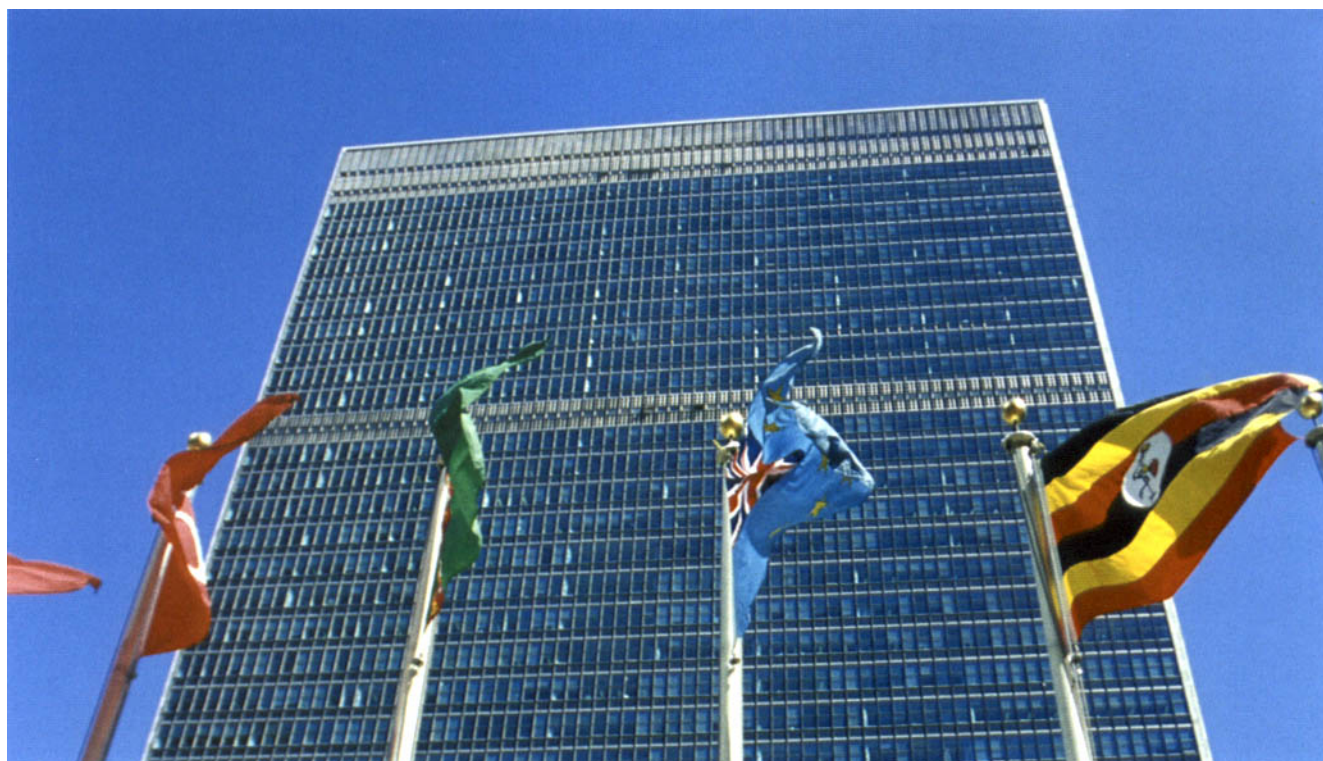
The Golden Anchor

Even 20 years after the fact, he was still smarting from what he considered the inexcusable insults to him and to France when Roosevelt and Churchill had excluded him from their most important consultations and their wartime meetings with Stalin. Nothing would have pleased de Gaulle more than to see the Anglo-Saxons on their knees before him. He was convinced he could accomplish that objective if the world were to follow his advice.

On 4 February 1965, he assembled 1000 journalists at the Elysées Palace, the official home of the president of France, and seated them in the gilded Salle des Fêtes. The general had called the press conference to demand nothing less than a return to the 19th-century gold standard.

He reminded his audience that 'The kind of transcending value attributed to the dollar has lost its initial foundation, which was possession by America of the greatest part of the world's gold.' His facts were indisputable. By early 1965, the US stock of monetary gold had fallen to its lowest level since March 1937. The stock was down to \$15 billion compared with its all-time peak of \$25 billion in 1949; the US share of the world's total monetary gold stock had shrunk from a high of 75 per cent to less than 50 per cent. By the end of the 1960s, it would be under 30 per cent.

Six days later, the French minister of finance, with the courtly name of Valéry Giscard d'Estaing, delivered a lecture at the University of Paris to a standing-room audience of 3000 students, in which he called for 'a solemn and unequivocal declaration' by the major financial powers to settle all international payments deficits with gold. This action, he contended, would 'stop the decay of the world money system'. He went on to announce that France was taking the lead by starting to convert all new accumulations of dollars into gold.



*The United Nations
Headquarters*

The French would proceed to make their point in no uncertain terms. Most of the time, other nations cashing in dollars for gold left the new gold for safekeeping in the vaults of the Federal Reserve Bank of New York. The French authorities made front-page news by announcing that all increases in their gold stock would travel across the Atlantic on French ocean liners to the secure shores of *la patrie*.

What had gone wrong? The press conference in the Salle des Fêtes was just one of a sequence of events in the 1960s that seemed destined to put the dollar in the shade and to restore gold to its former grandeur.

The story begins in the immediate aftermath of World War II, one of those rare moments in history when people have made all the right decisions. Most of the world outside the United States was in such a chaotic mess that the framers of the postwar era had a unique opportunity. The broad outlines of what was necessary were easy to define: the cascade of tragic errors in the 1920s and 1930s provided the leaders of the second postwar era with a perfect blueprint of what not to do.

Instead of exacting reparations, the Allies carried on vigorous financial and political efforts to bring Germany, Italy and Japan into the mainstream of democratic society. Instead of insisting on repayments for the huge amounts of military aid that the United States had provided during the war, Americans gave only lip service to demanding payment. Instead, they converted most of their contribution to the Allied war effort into gifts – and then added the prodigious assistance of the Marshall Plan and other substantial aid programs on top of that. Instead of a world where each nation stubbornly pursued its own self-interest, the United Nations was created to manage a world of international cooperation and harmony; unlike the League of Nations, the plans for the United Nations featured the enthusiastic participation of the United States.

The plans for a new international economic system were worked out by 730 delegates from 44 countries who gathered in the White Mountain resort of Bretton Woods, New Hampshire, in 1944. Most of the final design came from John Maynard Keynes, representing the British Treasury, and his counterpart, Harry White of the US Treasury Department.

Instead of an international economy where each nation was at the mercy of its stock of gold, the new system made the US dollar the centerpiece of the structure. The almighty dollar was then supported by 75 per cent of the world's stock of monetary gold. The huge US economy had sustained no war damage and was more productive than ever before in its history.

Under the new arrangements, the United States would be the only nation with a currency freely convertible into gold at a fixed rate. The conversion rate between dollars and gold was set at \$35 per ounce, and then the rest of the world defined their currencies in terms of the dollar rather than gold.

If that had been all there was to it, the new system would have left individual countries as much a prisoner of the dollar as once upon a time they had been prisoners of their gold stock. The Allied nations had few dollars left after paying for the monumental costs of the war, even with American aid; Britain owed massive debts to its imperial possessions; the enemy countries were flat broke. As a result, only the relationship between gold and the dollar at \$35 an ounce was expected to be permanent, but different arrangements were established for the relationship between the dollar and all the other currencies. These relationships, known as 'legal par values', were also expected to be fixed under the normal course of events, but provisions were established to provide for changes in the legal par values when a country's efforts to meet its international payment obligations ran into problems that were deemed more likely to be permanent than temporary. Setting these par values at a level that each country could manage to sustain was far from easy. Germany and Japan did not set theirs until 1953; Italy waited until 1960. Most of the original settings were changed more than once before the final breakdown of the system after 1971. Indeed, until well into the 1960s, Canada, Switzerland and the United States were the only advanced countries that allowed uncontrolled outflows of capital to other countries.

Arrangements for changes in legal par values was not the only step that the framers of the system took to avoid the suffocating deflations and unemployment that blind adherence to the old gold standard had mandated in the past. In the more frequent instances when trouble was expected to be temporary, the supply of foreign exchange available to be borrowed to tide a nation over such periods was to be augmented by the creation of a brand-new institution called the International Monetary Fund. The IMF was to function as a lender to governments and central banks requiring short-term accommodation that was not available from conventional sources.

Nevertheless, gold was still very much in the picture. The resources that the IMF had available to lend out came from contributions by each of its member countries, in two forms. Seventy-five per cent of the contributions consisted of the members' own currencies – the French paid in francs, the Dutch paid in guilders, the United States paid in dollars and so on. The other 25 per cent, however, had to be in gold, although a lesser percentage was permitted for countries that could not afford the payment. Over time, many transactions in gold took place between the IMF and its member countries.

The rest of the story is downhill all the way.

A grand underlying assumption supported this entire framework: that the dollar would continue to appear invincible – as good as gold – and therefore always acceptable everywhere and under all circumstances. This was an assumption that few people had stopped to question. Yet there was no reason to expect that state of affairs to last for ever.

On the contrary. The whole scheme was contrived with the primary purpose of putting Europe and Japan back on their feet. Once that goal was achieved, the dynamic would undergo fundamental change. The spanking new European and Japanese industrial facilities would be competition for American industry. They would also be attractive areas for American investment.

As the economies of Europe and Asia revived according to plan and in response to American aid, the lure of the profit motive drove massive amounts of private capital from the United States into thousands of enterprises across the seas and attracted rising imports of foreign merchandise to US shores. The United States undertook an unprecedented level of international economic assistance, launched large-scale social programs at home, deployed American military might around the world and fought two hot wars in Asia as well as the cold war against the Soviets.

The impact of the process on the United States' international financial position was already visible by the end of the 1950s. The US monetary gold stock, which had held steady at around \$22 billion from 1950 to 1958, shrank by over \$3 billion between the end of 1958 and the end of 1960. Meanwhile, as Americans spent their dollars abroad, foreign ownership of bank deposits in American banks and of short-term US Treasury obligations increased from only \$8 billion in 1950 to over \$20 billion in 1960. If foreigners at that moment had decided to convert all their liquid dollar assets into gold, the US gold stock would have been exhausted.

On 27 October 1960, provoked by news that the liberal John F Kennedy was probably going to defeat Richard Nixon in the presidential election in the following week, the price of gold jumped as high as \$40 an ounce in the London gold market. Here gold was freely bought and sold in the form of bullion, not coins, and where most of the new supply came from South Africa. Right after his election, Kennedy told his advisors that he ranked the dollar problem among his highest-priority concerns.

Inflation was the only force that should have threatened the viability of the dollar. And here is where questions began to be raised. True, the US economy had been free of inflationary pressures, despite high rates of economic growth, during the first half of the 1950s. Signs of trouble first appeared in 1956–7, as the business cycle approached its peak and inflation bubbled up to around 3 per cent. Even that rate of inflation might have been tolerated in the late phases of an economic expansion, but the increase of 2.7 per cent in consumer prices from the end of 1957 to the end of 1958 came as a shock. By then the boom was over, and a serious decline in business activity had begun. Unemployment had risen by over a million, to 7 per cent of the labor force.

Such a thing had never happened before. Prices had *always* fallen in recessions and depressions. The Federal Reserve was so alarmed that it jacked up the discount rate in short order from 1.5 to 4 per cent, a move that aborted the recovery from this recession just two years after it began. The chairman of the Federal Reserve, William McChesney Martin, announced that he considered inflation as serious a threat as communism to the safety and future of the United States.

Martin's determined stand against his hated enemy seems to have turned the trick. Once the bad news of 1958 was out of the way, the increase in consumer prices during the recovery years of 1959–60 was even smaller than in 1958. The Federal Reserve's aggressive policy even persuaded Westinghouse and General Electric employees to abandon the cost-of-living clauses in their contracts and the militant West Coast longshoremen to give their employers free rein in introducing labor-saving machinery.

Perhaps these inflation numbers would have sufficed to calm the worries over the dollar, but other significant US economic statistics spurred an accelerating drawdown of the gold stock. Just during the five years 1960–64, imports totaled more than \$80 billion, \$11 billion was spent by the government sustaining the armed forces outside the United States, American business spent \$29 billion investing in companies in Europe and Asia or building up financial assets in other countries, while foreign travel, other services and foreign aid consumed another \$18 billion. On the one hand, the American capital and the crowds of open-handed American tourists were welcome; on the other hand, resentment over the conspicuous display of American wealth festered throughout the 1960s. The French were most vocal in their concerns and were determined to clip the wings of the American juggernaut.

During the second half of the 1960s, the US inflation problem assumed more serious proportions. The intensification of the US commitment to the war in Vietnam was the primary source of inflationary pressure. Defense spending jumped by about 60 per cent during the second half of the 1960s. In a rare exhibit of almost complete unanimity, professional economists of all stripes urged President Johnson to finance the swelling



John F Kennedy 50 cent piece



President Nixon

volume of defense expenditures by an increase in taxes. Johnson refused to follow their advice. To make matters even more difficult, most of the proceeds of rising military expenditures in Vietnam were ending up in French banks.

The US gold stock continued to sink despite the establishment of a variety of elaborate arrangements for short-term credits so that the United States would have other currencies available to meet its obligations abroad. The Federal Reserve attempted to maintain short-term interest rates high enough to make dollars more attractive than Deutschmarks or Swiss francs and surely preferable to gold that paid no interest. The government pushed business groups and labor unions into 'voluntary' agreements to check domestic inflation by limiting price and wage increases.

There was more. In September 1964, a special tax called the Interest Equalization Tax was enacted to discourage American investment abroad by taxing interest earned on deposits and bonds held in foreign currencies. American investment in continental European companies was also curbed. Corporations were required to repatriate part of their profits earned overseas. Bank lending abroad was limited. American travelers were urged to stay home and explore the beauties of their own country instead of spending precious dollars touring in other peoples' countries. After Richard Nixon's victory in the presidential election of 1968, a 10 per cent tax surcharge was enacted to narrow the budget deficit – a belated reversal of Lyndon Johnson's adamant opposition to such a move.

Nothing sufficed. The US gold stock declined every year from 1958 to 1971, falling from \$19 billion to \$10 billion, while the country's liquid liabilities to foreigners rose every year over the same period of time until they had expanded to over \$60 billion.

The United States seemed unable to restore any equilibrium to its international position. The gold bricks continued to move from American ownership at Fort Knox to foreign ownership stored for safety in the vaults of the Federal Reserve Bank of New York (or, for the French, at the Bank of France). The Europeans gained gold, but the inflow failed to stimulate their demand for imports from the United States by enough to reverse the eastward movement of gold.

What appeared to be an unending sequence of bad news for everyone else was great news to the pessimists who were convinced that gold was the only hedge in a hopeless situation. These gold bugs, as they came to be known, had been accumulating gold in the London market since the late 1950s. The major financial powers were concerned that this buying could push gold above the official price of \$35 an ounce, provoking an uncontrollable rush into gold that could topple the entire system. Starting in 1961, the United States, Britain, Germany, France, Italy, Switzerland, Holland and Belgium had joined together to pool their resources in an effort to beat down these speculators. The pool would sell gold in the market whenever the price showed signs of moving more than a few cents above \$35.

At the beginning, with new production coming in large quantities into London, the pool was able to buy more gold at \$35 than it had to sell. The struggle with the speculators became increasingly one-sided, however, after General de Gaulle started beating the drums for an increase in the price of gold. In fact, de Gaulle had pulled France out of the gold pool during 1967. In early March 1968, the United States had to transfer \$950 million in gold to

London to keep the price close to \$35; nearly \$2.5 billion had been paid out from Fort Knox since 18 November 1967, when the British had devalued sterling for the second time since the end of World War II.

On 17 March 1968, after seven fruitless years of pouring their treasure into the maws of the speculators, the members of the gold pool decided the game was up. They announced that henceforth they would no longer supply gold to the London market, or any gold market, nor did they intend to buy any gold from those sources in the future. Official transactions in gold between central banks would continue at \$35 an ounce; central banks would still be free to buy gold from the United States for dollars at that price. From that point forward, the price of gold in the free market would be left to private parties to determine.

General de Gaulle now proceeded to call the whole gold-pool strategy a sham, merely a stopgap on the way to the ultimate devaluation of the dollar. His blunt statements might have been considered just another of his long series of efforts to damn the dollar if William McChesney Martin, the chairman of the Federal Reserve, had not been so indiscreet as to echo de Gaulle's prediction. At this point, another disturbing possibility came into view. What would prevent one of the central banks that had been a member of the pool from cashing in their dollars for gold at \$35 in New York and quietly feeding gold into the London market at a handsome profit?

The danger was clear, and something had to be done. The European members of the pool agreed among themselves that they would refrain from converting any dollar balances into gold at least for the duration of the Vietnam War, after which the Americans were expected at long last to come to grips with the gaping imbalance between the receipts from other countries and their payments abroad.

The willingness of the Europeans to play ball did not come for free. In return, the Americans were obliged to swallow the old-fashioned medicine of budget discipline and higher interest rates that had been routine in the 1920s and early 1930s. President Nixon was prevailed upon to support the 10 per cent surcharge on all income taxes, which went into effect in 1969, his first year in office. But the rate of price increases showed no sign of subsiding.



European Community flag

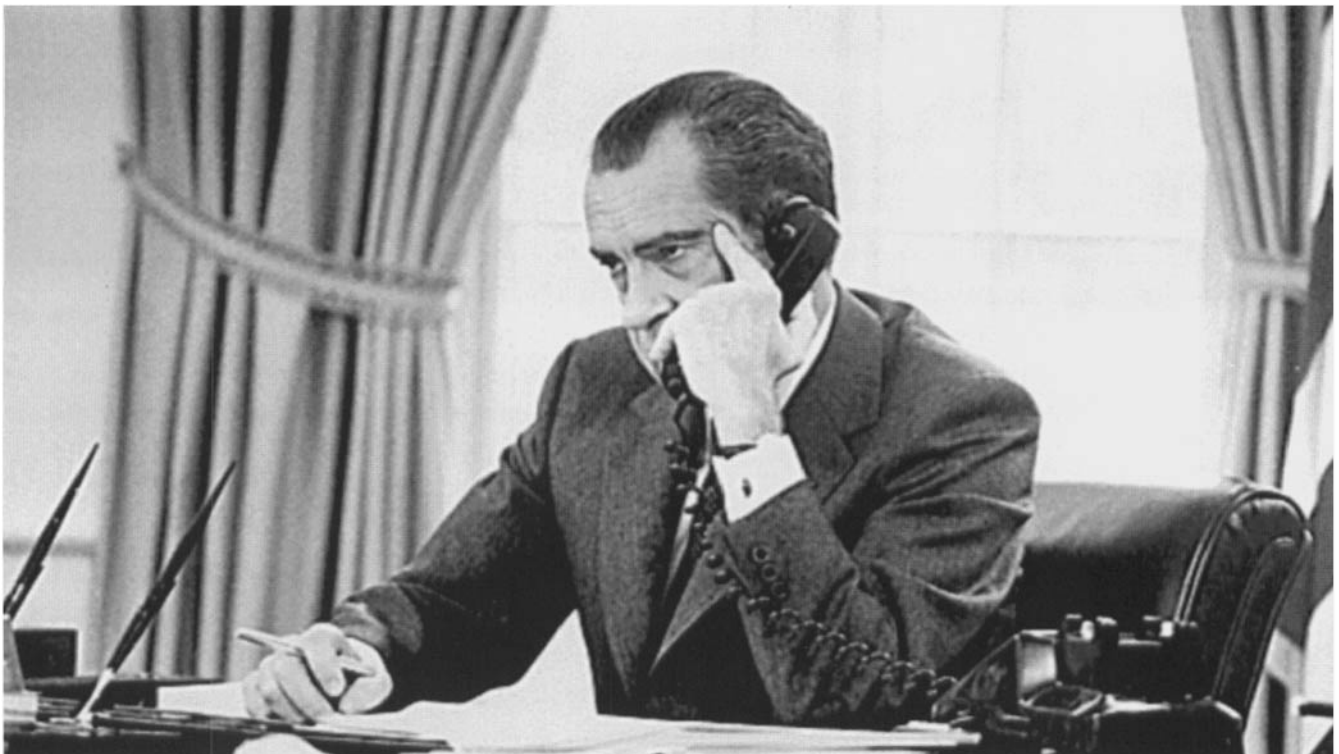
While all this was going on, General de Gaulle was busy. He and his associates called for an increase in the price of gold to \$70, which would have doubled the dollar value of all the monetary gold in the world. This step would have been a juicy benefit for the French, who held the largest share of monetary gold outside the United States.

De Gaulle's ultimate objective was to bring an end to the special position of the United States, in which Americans could offset the deficit in their international transactions simply by paying over dollars – their own currency – to foreigners, while everyone else had to settle up with some other country's currency or in gold. De Gaulle was far from alone in resenting a process in which only the United States could finance its spending abroad by what amounted essentially to printing dollars, while the rest of the world had to 'earn' their foreign currency or gold by running a surplus in their transactions with foreign nations.

De Gaulle's recommended solution drew no support. Quite aside from the inflationary potential in a doubling of the world's monetary base, this gain created by a stroke of the pen would have been a bonanza for the world's two largest gold producers, South Africa and the Soviet Union, neither of which would have won any popularity contests among the Western nations at that time.

The final blow to de Gaulle's hopes came in May 1968, when a wild swirl of rioting and strikes broke loose in France. The tumultuous uproar was similar in intensity to the

President Nixon



passionate uprisings that occurred in the United States at the same time, but the French were motivated by demands for higher wages rather than by protests over the Vietnam War. The strikes delivered a major blow to the French economy. The shocking violence of the uprising provoked a flight of capital reminiscent of the panics of the 1920s, accompanied by strong upward pressure on the Deutschmark and the Swiss franc as the most appropriate safe havens of the moment. By the time the dust had settled, de Gaulle had been forced to resign – and that decision ignited an even greater panic in the markets for the franc.

As inflation gathered steam in the United States during the course of 1968, the gold-based Bretton Woods system of fixed exchange rates loomed as an intolerable restraint on politicians struggling to finance rising costs of government. The result was renewed interest in gold among the public as a safe haven destined to fulfill the words that Herbert Hoover had thrown at President-elect Roosevelt in 1933: 'We have gold because we cannot trust governments.'

After 1968, inflation became a self-fulfilling prophecy that added momentum to the fundamental inflationary forces at work in the system. The unions were convinced they had to keep wages climbing faster than inflation, while business managements were convinced they had to keep prices rising in order to cover the increased labor costs.

The Nixon administration could see just two ways out of these dilemmas. The first alternative, the traditional one, was etched in agony by the British in 1931: raise taxes and interest rates so high that the economy would be pushed into a serious recession, not just a pause as in 1970. That step might kill off the inflation mentality and rescue the dollar, but at an unacceptable human cost, to say nothing of the consequences for an elected politician who selected such a strategy.

In August 1970, Congress gave President Nixon discretionary authority to impose comprehensive wage and price controls. He had no enthusiasm for the idea, but recognized that conditions were backing him into a corner. In December, he took a major step toward a more active policy when he appointed John Connally secretary of the Treasury. Connally had gained national attention when he was wounded while riding in the car when President Kennedy was assassinated in Dallas. He was just the man to overcome Nixon's reluctance to make spectacular moves in economic policy. And the United States could not postpone action much longer.

Nixon and Connally decided to consider a two-pronged move. Its two elements appeared at first glance to have no clear relationship to each other. As matters worked out, they neatly combined into an integrated program.

The first move was designed to solve the gold problem for the United States once and for all. The Treasury would simply shut down the gold window, which would mean refusing to sell gold at \$35 an ounce to governments or central banks coming to the

Treasury to exchange their dollars. The dollar would finally be liberated from the golden fetters, like all the other currencies of the world. Without any fixed anchor, the dollar would be free to 'float' in the foreign exchange markets.

Nixon and Connally would force the foreign central banks to face a nasty set of alternatives. The banks could continue to buy all dollars offered for sale, but that would only swell their already substantial dollar positions. If they held back from buying, however, the dollar would decline in price in terms of other currencies. That would mean heavy losses to their citizens who had acquired dollars or dollar-denominated assets in the past and were still holding them. Furthermore, if French francs now cost Americans more dollars than before, Americans would tend to buy less French perfume and wine; at the same time, the French would now have to pay fewer francs to buy \$1000 worth of American goods and services, which would give a boost to imports into France from the United States.

This was precisely the outcome that Nixon and Connally hoped to achieve. All of that would be good for American business and jobs. Furthermore, by making American exports more attractive and foreign imports less attractive, the devaluation of the dollar might tend to be self-correcting as American demands for foreign currencies diminished and foreign demands for dollars picked up.

The second part of the Connally–Nixon strategy was designed to suppress the potentially inflationary consequences of this stimulus to business. They recognized they would have to package the shattering of the link to gold with price and wage controls that would confirm their dedication to fighting inflation.

Nixon, convinced that the time for half measures was past, anticipated little difficulty. The combination of abandoning the gold standard and adopting mandatory price and wage controls made a perfect fit. The economic attractions were strong, but both Nixon and Connally appreciated even more the political appeal of this new policy: conservatives liked the free market implications of the break with gold while liberals liked the activist policy of wage and price controls.

The last straw came during the week of 9 August 1971, in a note of extraordinary irony, when the British economic representative came in person to the Treasury and asked for \$3 billion in gold. On the following Friday, the 13th, Nixon abruptly ordered the 16 major economic policymakers of the administration to accompany him by helicopter to Camp David. The president made certain that no leaks about the proceedings would occur by cutting the group off from all communication with the outside world.

The group brought forth what they dubbed a 'New Economic Policy', which combined the closing of the gold window with a mandatory and comprehensive freeze on prices and wages. The freeze was for 90 days, but the expectation was that this initial step would be followed by voluntary restraints by business and labor. In addition to the decisions on controls and gold, the new policy included recommendations for tax cuts for

business, reductions in government spending, and a 10 per cent surcharge on about half of all US imports. The import surcharge was the equivalent of a devaluation of the dollar because it automatically made those imports more expensive for Americans.

All the participants agreed that the announcement of the New Economic Policy should be designed to have major impact around the world. The president was urged to make a public statement on prime-time television that Sunday evening. By making his decisions known before the markets opened on Monday morning, no leaks or rumors would be able to dilute the force of his words. Nixon demurred: he hesitated to make his speech on Sunday evening, for fear of irritating the public by pre-empting *Bonanza*, one of the most popular programs of the era.

Under pressure from his advisors, Nixon managed to put the national interest ahead of the well-being of *Bonanza* fans. The news of his prime-time address broke with banner headlines on Monday morning's newspapers. On Wall Street, one salesman exulted, 'All brokers are the happiest people in the world today.'

With the golden anchor torn loose, once and for all, the New Economic Policy created instant pandemonium abroad. In contrast to Wall Street, foreign stock markets plummeted. In Tokyo, the market suffered a genuine panic, as the *New York Times* described it, 'with sell-at-any-price orders sending prices down sharply'.

The flood of dollar selling provoked foreign governments to demand a prompt return to some kind of system of fixed exchange rates. Even the Americans had to admit that such volatile exchange rates created uncertainty for all kinds of business decisions. A long series of negotiations culminated in a meeting in December 1971 at the Smithsonian Institution in Washington to re-establish order in the foreign exchange markets. A new official dollar price of gold was set at \$38 – the equivalent of a formal 7.9 per cent devaluation of the dollar – although by that time gold in the London market was trading at between \$43 and \$44, about \$5 higher than on 15 August.

To make matters worse, OPEC (Organization of Petroleum Exporting Countries), a consortium of major oil-producing countries, joined together in October 1973 to restrict their production until the price of oil had jumped from \$2.11 a barrel to over \$10, igniting additional powerful and irrepressible inflationary impulses throughout the world economy.

All of this was too much for the US system of price and wage controls. The administration had no choice but to abandon these arrangements in the face of the tremendous leap in the price of oil combined with the devaluation of the dollar, which automatically raised the price to Americans of most foreign goods and services.

Then, in November 1973, only a month after the OPEC countries had roiled the world economy, the central banks threw in the sponge on their 1968 decision to refrain from trading in gold except among themselves. Now they could buy and sell in the London market at prices far above the official price of \$42.22.



President Carter

Beginning in 1975, tentative steps were taken to liberate the monetary system even further from gold. Then in August, an International Monetary Fund committee reached two momentous decisions: they agreed to abolish the official price for gold, and they also decided to auction a portion of the IMF's gold holdings. The proceeds of the auction were to be used for the benefit of developing countries and also to return to member countries some of the money they had originally contributed to the IMF.

None of the international agreements to manage exchange-rate volatility seemed to hold up. Inflation in all countries was eating away at the values of stocks, bonds and cash. Inflation was most intense in Britain, France, and the United States among the developed countries, but even Germany's inflation averaged 5 per cent from 1974 to 1981 and included episodes above 7 per cent. Consequently, many speculators were only too happy to buy gold from the central banks.

The soaring demand for gold as a safe haven for wealth and as a hedge against inflation drove the price in the London market from \$46 an ounce at the beginning of 1972 to \$64 an ounce at the end of the year. The price broke through \$100 during 1973; from 1974 to 1977, gold fluctuated between \$130 and \$180. A second OPEC oil price increase in 1978 created a frenzy that ignited a new and precipitous climb in the gold markets: the price of gold hit \$244 an ounce before the year was out and then doubled to \$500 in 1979. In the spirit of the times, the famous comedienne Bette Midler, about to depart on a European tour, demanded on 3 July 1978, that her \$600,000 fee be paid in South African gold coins instead of in US dollars.

Political conditions were perhaps even more frightening. Iranian radicals in November 1979 took over the US embassy in Tehran and held the entire staff as hostages, initiating a crisis that would endure for more than 400 days. At the same time, the Soviets were building up their strength in southern Yemen near Saudi Arabia, near Afghanistan's border with Iran, and near Bulgaria's border with Yugoslavia – at a moment when Yugoslavia's 87-year-old Marshal Tito was in poor health.

January 1980 in the gold market turned out to be one of the wildest months in the history of any market, anywhere, any time. The price of gold jumped by \$110 an ounce to \$634 in just the first two business days of the month, while the value of the dollar in terms of German marks fell to a record low.

Then Treasury Secretary G. William Miller held a news conference at which he announced that the Treasury would hold no further gold auctions. 'At the moment,' he told the press, 'it doesn't seem an appropriate time to sell our gold.' With 220 million ounces (about 7000 tons) of gold stored away at Fort Knox, gold hoarding was regaining some traditional respectability.

Within 30 minutes of Miller's remarks, the gold price shot up \$30 an ounce to \$715. The next day it was up to \$760. The day after, gold hit \$820. The price touched its record high of \$850 on 21 January 1980. James Sinclair, a commodities broker, summed it up when he commented that 'We're in World War VIII, if you believe the market.' Late that afternoon, President Carter announced that the United States would have to 'pay whatever price is required to remain the strongest nation in the world.' His comment seemed to cool tempers in the gold and foreign exchange markets – the price of gold was down \$50 by the close of trading.

Indeed, the temper of the marketplace did a 180-degree turn with extraordinary abruptness. On 22 January, the price plunged by \$145. The high in 1981 was \$599 an ounce. By 1985, gold was down to around \$300. The subsequent high, touched only briefly, was \$486, in the wake of the stock market crash of 1987. At the end of 1997, gold broke below \$300. It had fallen by more than 60 per cent in the course of less than eight years.

In 1981, the US monetary gold stock amounted to approximately 8000 tons, a little more than a third of the 1949 peak, only 50 per cent more than in 1933, and equal to about a quarter of world monetary gold stocks. At the official price of \$42.22, the stock was carried at a mere \$11 billion, although the stock was worth \$120 billion at the 1981 average market price of \$460 an ounce. Liabilities to foreigners, however, were now over \$300 billion, an astonishing increase of nearly ten-fold over the level that had so agitated General de Gaulle 13 years earlier.

Fundamental economic trends in the early 1980s were finally shoving gold away from center stage. And as gold pays no income and incurs storage costs, owning gold was expensive indeed compared to alternative investment opportunities.

Gold would still have made sense in spite of these hurdles if people had expected inflation to remain out of control. The whole story of the 1980s, however, was the growing recognition, around the world, that the virulent inflation of the 1970s had been beaten back at long last and that the prices of goods and services for the foreseeable future would rise at a more moderate and manageable pace. It is indeed remarkable that US



The NY Mercantile Exchange: after Nixon abandoned the gold standard, gold could be bought and sold just like any other commodity



inflation fell from such precipitous heights at the end of the 1970s to as low as 3 per cent by 1985, but similar trends were at work in most countries, even in such areas as Italy, Latin America and the Middle East, where inflation had been a chronic problem. Holding gold can make little sense if inflation is dead or dying, because then there is little hope of recouping the storage costs and offsetting the lost income.

The cost of living doubled from 1980 to 1999 – an annual inflation rate of about 3.5 per cent – but the price of gold fell by some 60 per cent. In January 1980, one ounce of gold could buy a basket of goods and services worth \$850. In 1999, the same basket would cost five ounces of gold.

The lower the price of gold fell, the greater became the prospects of official sales, not just from the United States but from other countries and the International Monetary Fund itself. As the gold price rose from \$375 in 1982 to nearly \$500 after the stock-market crash of 1987, few central bank sales were executed. As the price then drifted downward toward \$350 in 1992, about 500 tons were disposed of. From 1992 to 1999, however, as the price sank below \$300, the central bankers sold off 3000 tons, or about 400 tons a year.

As the central banks were liquidating their gold, over 2000 tons a year of additional gold came into the markets from new mine production, about double the level of production before the price of gold broke free from the old \$35 price. Sales of 400 tons from the central banks sound small relative to the 2000 tons of supply forthcoming from the mines. Nevertheless, the volume of total central bank and official holdings was still so large – over 30,000 tons – that the overhang loomed like a black cloud over the markets. Who could say how much of that hoarded treasure might come to market?

A loud thunderclap along these lines hit the markets in October 1997, when a team of Swiss experts issued a report recommending an amendment to the Swiss constitution that would result in a radical restructuring of the Swiss currency system: 'A return to gold standards today is impossible. . . . The proposed draft of the new constitutional article does not contain any connection of the [Swiss] franc to gold.' The entire spirit of the report rested in its unquestioning confidence that the forecasting and managerial skills of the directors and staff of the central bank would perform a better job than obeisance to the gold stock in 'the priority of maintaining price stability'.

This view was by no means revolutionary doctrine in 1997 – on the contrary, it represented mainstream thinking. Nevertheless, this was the Swiss, not the British or the Americans or some minor-league country. The Swiss were legendary in their attachment to gold and in their aversion to holding currencies of countries whose devotion to the constant struggle to keep inflation in check was less passionate than theirs. The 'gnomes of Zurich' had been famous for their speculative attacks on the dollar and sterling during the crises of the 1970s. Now all that was forgotten.

Two years later, the British took a similar step with their gold stock, once upon a time the pride of British power. In May 1999, the British Treasury announced its intention of selling 415 tons out of its 715-ton stockpile. The price of gold promptly lost 4 per cent of its value.

The central banks would soon catch on: their enemy was themselves. The overhang of gold held by central banks meant that every time an official sale hit the headlines, the gold price would fall and the proceeds of the sale would be diminished.

How times had changed! In the 1960s, the major central banks had organized the gold pool to sell whatever amount of gold was necessary to keep the speculators from driving the price upward. In September 1999, owning nearly half of all gold held by central banks and other official institutions, they were in a position identical to Ruskin's man who strapped his golden wealth to himself as his ship was sinking and promptly sank to the bottom of the ocean. If the central banks all moved at the same time to sell off their hoards of gold, the price would run away from them on the downside and their sales would be a disaster.

They therefore agreed to limit their annual sales to 400 tons of gold over the next five years – about the same as the annual average liquidated over the previous eight years. The IMF announced that it would 'abide' by the spirit of the agreement. Australia and South Africa joined in an informal affiliation, bringing the amount of official gold covered up to 85 per cent of the total. The central banks also resolved to limit their lending transactions with the mining companies. The agreement covered the 1300 tons in the pipeline for the Swiss to sell and 365 tons for Britain, leaving only 335 tons for any other country that wanted to liquidate gold over the five-year period covered by the arrangements.

The central banks were not the only important sellers in the market, but they had been eager cooperators with the other major group: the mining companies. We would expect the mining companies to be sellers, because that is what they are in business for. During the 1990s, concerned like so many others about the future outlook for gold, the mining companies began to sell more than their current production. In effect, they mortgaged their future output at the prevailing price in order to avoid having to sell at a lower price later on. Buyers, however, want delivery when they contract to buy. The mining companies enlisted the central banks for this purpose: the miners borrowed gold from the central banks at a nominal rate of interest, secured by the promise to pay off the loan from future production, and then delivered that gold to the buyers. The central banks were delighted to earn anything at all on what was once the glory of their economic power but that they now considered a barren asset. This arrangement worked well – as long as the price of gold was falling. On the occasions when the price of gold went up, however, the central banks became more reluctant lenders and the mining companies got squeezed. As their current production was smaller than the amounts they had borrowed,

they had to go into the market and join the other buyers there in order to make their promised deliveries to people who had bought gold from them. The result was an added impetus to the upward movement in the gold price.

While all of this was going on, the worldwide demand for gold remained vigorous. Gold consumption doubled in the course of the 1990s, and for good reason. The price of gold was falling while the price of everything else was rising. As a result, gold was perceived as relatively inexpensive. The quantity of gold consumed in the production of jewelry – by far the most important component of demand – and in the electronics industry at the end of the 1990s was more than 50 per cent higher than in 1980 and about a third higher than in 1994.

The Asians, as in the past, continued to absorb large quantities of gold. *The Economist* reported in January 1999 that 'The Indian lust for gold remains unabated. . . . Gold jewelry [sic] is the only form of wealth that many women can claim as their own.' At that moment, the total amount of gold in India, estimated at around 9000 tons, exceeded even the great hoard stowed away in Fort Knox, Kentucky.

Over the centuries, gold has stirred the passions for power and glory, for beauty, for security, and even for immortality. Gold has been an icon for greed, a vehicle for vanity, and a potent constraint as a monetary standard. No other object has commanded so much veneration over so long a period of time.

In the world of money, the last vestiges of the golden fetters were discarded by Richard Nixon in 1971. Dispossessed of its power, gold is no longer at the centre of the universe of central banks, fixed exchange rates, and financial dealings. Now greed and lust run down different channels, where paper has shown how well it can serve that purpose.

That does not mean that gold will never make front-page news. The whole story of this book demonstrates that no régime lasts forever – not the stater, nor the bezant, not the dinar, not even the pound sterling. Gold may once again serve as the ultimate hedge, if the day should come when the dollar or the euro fail in their function as acceptable means of payment across international borders. Even if gold may appear as a safe haven under such chaotic conditions, its return to its historic role as universal money is doubtful.

Those who believe that gold is a hedge against the uncertainties of life do not understand that the pursuit of eternity is not to be satisfied by gold, or by anything else we choose to replace gold. Gold as an end in itself is meaningless. Hoarding does not create wealth. Gold and its surrogates make sense only as a means to an end: to beautify, to adorn, to exchange for what we need and really want.

Perhaps the wisest heroes of our story were the Africans who silently swapped gold for the precious salt that would keep them alive.





Bibliography

- Andrews, Kenneth, 1978. *The Spanish Caribbean: Trade and Plunder, 1530–1630*. New Haven, CT: Yale University Press.
- Andrews, Kenneth, 1984. *Trade, Plunder and Settlement: Maritime Enterprise and the Genesis of the British Empire, 1480–1630*. Cambridge: Cambridge University Press.
- Arnon, Arie, 1991. *Thomas Tooke: Pioneer of Monetary Theory*. Ann Arbor: University of Michigan Press.
- Attman, Artur, 1962. *American Bullion in the European World Trade, 1600–1800*. Goteborg: Kungl. Vetenskaps-och Vitterhets-Samhället. (I drew on the information in this book as provided in Kindleberger, 1989.)
- Austen, Ralph A, 1990. 'Marginalization, Stagnation, and Growth: The Trans-Saharan Caravan Trade in the Era of European Expansion, 1500–1900.' In Tracy, 1990.
- Bainton, Roland, 1952. *The Reformation of the Sixteenth Century*. Boston: Beacon Press.
- Bank for International Settlements. Various annual reports.
- Barnes, James A, 1937. 'Myths of the Bryan Campaign.' *Mississippi Valley Historical Review*, 34 (Dec.), pp. 367–400.
- Barsky, Robert, and Bradford DeLong, 1991. 'Forecast Pre-World War I Inflation: The Fisher Effect and the Gold Standard.' *Quarterly Journal of Economics*, 106, pp. 815–836.
- Bayoumi, Tamim, Barry Eichengreen, and Mark Taylor, eds, 1996. *Modern Perspectives on the Gold Standard*. Cambridge: Cambridge University Press.
- Becker, Gary S, Edward Glaeser and Ken Murphy, 1999. 'Population and Economic Growth.' *American Economic Review*, 89, no. 2 (May), pp. 145–149.
- Bernstein, Peter L, 1968. *A Primer on Money, Banking and Gold*, 2nd edition. New York: Random House.
-

- Bernstein, Peter L, 1970. *Economist on Wall Street*. New York: Macmillan.
- Bernstein, Peter L, 1996. *Against the Gods: The Remarkable Story of Risk*. New York: John Wiley & Sons.
- Bloch, Mark, 1933. 'Le problème de l'or au moyen âge.' *Annales d'histoire économique et sociale*, no. 19, January 31, pp. 1–33.
- Bloomfield, Arthur, 1959. *Monetary Policy Under the Gold Standard, 1800–1914*. Federal Reserve Bank of New York.
- Bonar, J, 1923. 'Ricardo's Ingot Plan.' *Economic Journal*, XXXIII, pp. 281–304.
- Boorstin, Daniel, ed., 1966. *An American Primer*. New York: Penguin Books.
- Bordo, Michael, 1986. 'Money, Deflation and Seigniorage in the Fifteenth Century: A Review Essay.' *Journal of Monetary Economics*, 18, no. 3, pp. 337–346.
- Bordo, Michael, and Tamim Bayoumi, 1999. 'Getting Pegged: Comparing the 1879 and 1925 Gold Resumptions.' Cambridge, MA: National Bureau of Economic Research, Working Paper 5497.
- Bordo, Michael, and Forrest Capie, eds, 1993. *Monetary Regimes in Transition*. Cambridge: Cambridge University Press.
- Bordo, Michael, and Barry Eichengreen, 1998. 'The Rise and Fall of a Barbarous Relic: The Role of Gold in the International Monetary System.' Cambridge, MA: National Bureau of Economic Research, Working Paper 6436.
- Bordo, Michael, and Ronald MacDonald, 1997. 'Violations of "Rules of the Game" and the Credibility of the Classic Gold Standard, 1890–1914.' Cambridge, MA: National Bureau of Economic Research, Working Paper 6115.
- Bordo, Michael, and Hugh Rockoff, 1996. 'The Gold Standard as the Good Housekeeping Seal of Approval.' Cambridge, MA: National Bureau of Economic Research, Working Paper 5340.
- Bordo, Michael, and Anna Schwartz, 1984. *A Retrospective on the Classical Gold Standard, 1921–1931*. Chicago: University of Chicago Press.
- Bordo, Michael, and Eugene White, 1991. 'A Tale of Two Currencies: British and French Finance During the Napoleonic Wars.' *Journal of Economic History*, 51, no. 2 (June), pp. 303–316.
- Bordo, Michael, Ehsan Choudri, and Anna Schwartz, 1999. 'Was Expansionary Monetary Policy Feasible During the Great Contraction?' Cambridge, MA: National Bureau of Economic Research, Working Paper 7125.
- Bovill, E W, 1958. *The Golden Trade of the Moors*. London: Oxford University Press.
- Bowle, John, 1964. *Henry VIII: A Study of Power in Action*. New York: Dorset Press.
- Boyer-Xambeu, Marie-Thérèse, Ghislain Deleplace, and Lucien Gillard, 1994. *Private Money & Public Currencies: The 16th Century Challenge*. Translated by Azizeh Azodi. Armonk, NY: M E Sharpe.
- Boyle, Andrew, 1967. *Montagu Norman: A Biography*. London: Cassell.

- Brace, Harrison H, 1910. *Gold Production and Future Prices: An inquiry into the Increased Production of Gold and Other Causes of Price Changes with a View to Determining the Future of Prices*. New York: The Bankers Publishing Co.
- Braudel, F, and F Spooner, 1967. 'Prices in Europe from 1450 to 1570.' In Rich and Wilson, 1967.
- Brimelow, Peter, 1998. 'Human Capital.' *Forbes*, July 6, pp. 52–62.
- Bryan, William Jennings, 1896. *The First Battle*. Chicago: W. B. Conkey.
- Bryant, Arthur, 1962. *Makers of England*. Garden City: Doubleday & Co.
- Bryant, Arthur, 1964. *The Age of Chivalry*. Garden City: Doubleday & Co.
- Bureau of the Census, U.S. Department of Commerce, 1975. *Historical Statistics of the United States: Colonial Times to 1970*.
- Burns, Arthur R, 1927. *Money and Monetary Policy in Early Times*. New York: Alfred A. Knopf.
- Cannan, Edwin, ed., 1919. *The Paper Pound of 1797–1821*. Contains a reprint of the *Bullion Committee Report*. Reprinted in 1969 by Augustus Kelley, New York.
- Challis, Christopher, 1978. *The Tudor Coinage*. Manchester: Manchester University Press (reviewed in Gould, 1979).
- Challis, Christopher, ed., 1992. *A New History of the Royal Mint*. Cambridge: Cambridge University Press.
- Chernow, Ron, 1990. *The House of Morgan*. New York: Atlantic Monthly Press.
- Cherry, John, 1992. *Medieval Craftsmen: Goldsmiths*. Toronto: University of Toronto Press.
- Chevalier, Michel, 1836. 'Lettres sur l'Amérique du Nord.' See Kindleberger, 1978.
- Chevalier, Michel, 1859. *On the Probable Fall in the Value of Gold: The Commercial and Social Consequences Which May Ensur, and the Measures Which It Invites*. Trans. Richard Cobden. Reprinted by Greenwood Press, New York, 1968.
- Cipolla, Carlo, 1956. *Money, Prices and Civilization in the Mediterranean World: 5th to 17th Centuries*. Princeton: Princeton University Press.
- Cipolla, Carlo, 1993. *Before the Industrial Revolution: European Society and Economy, 1000–1700*, 3rd edition. London and New York: Routledge. Originally published 1976 by W W Norton in New York.
- Clay, Sir Henry, 1957. *Lord Norman*. London: Macmillan.
- Clowes, Mike, 2000. *The Money Flood: A History of Pension Fund Investing*. New York: John Wiley & Sons.
- Court, W H B. 1964. *A Concise Economic History of Great Britain: 1750 to Recent Times*. Cambridge: Cambridge University Press.
- Cribb, Joe, Barrie Cook, and Ian Carradice, 1990. *The Coin Atlas: The World of Coinage from its Origins to the Present Day*. New York: Facts on File.
- Crone, G R, 1937. *The Voyages of Cadamosto and Other Documents on Western Africa in the Second Half of the Fifteenth Century*. London: Hakluyt Society.
- Crosby, Alfred W, 1997. *The Measure of Reality: Quantification and Western Society, 1250–1600*. Cambridge: Cambridge University Press.

- Davies, Glyn, 1995. *A History of Money from Ancient Times to the Present Day*. Cardiff: University of Wales Press.
- Day, John, 1978. 'The Great Bullion Famine of the Fifteenth Century.' *Past and Present*, 79 (May), pp. 3–54.
- Day, John, 1987. *Medieval Market Economy*. Oxford: Blackwell.
- Debeir, Jean-Claude, 1978. 'La crise du franc de 1924: Un exemple de speculation "internationale".' *Relations internationales*, no. 13, pp. 29–49.
- de Gaulle, Charles, 1965. *Discours et messages, 1962–1965*. Paris: Plon.
- Despres, Emile, Charles P Kindleberger, and Walter S Salant, 1966. 'The Dollar and World Liquidity: A Minority View.' *The Economist*, 218, no. 6389 (February 5).
- Dobbs, Betty Jo Teeter, 1992. *The Janus Faces of Genius: The Role of Alchemy in Newton's Thought*. Cambridge: Cambridge University Press.
- Drake, Louis S 1985. 'Reconstruction of a Bimetallic Price Level.' *Explorations in Economic History*, 22, pp. 194–219.
- Eichengreen, Barry, 1986. 'The Bank of France and the Sterilization of Gold.' *Explorations in Economic History*, 23, pp. 56–84.
- Eichengreen, Barry, 1987. 'Conducting the International Orchestra: Bank of England Leadership Under the Classical Gold Standard, 1880–1913.' *Journal of International Money and Finance*, 6.
- Eichengreen, Barry, 1992. *Golden Fetters*. New York: Oxford University Press.
- Eichengreen, Barry, 1996. *Globalizing Capital: A History of the International Monetary System*. Princeton: Princeton University Press.
- Eichengreen, Barry, and Ian McLean, 1994. 'The Supply of Gold under the Pre-1914 Gold Standard.' *Economic History Review*, 47.
- Emmerich, André, 1965. *Sweat on the Sun and Tears of the Moon: Gold and Silver in Precolombian Art*. Seattle: University of Washington Press.
- Encyclopedia Britannica On-Line (www.eb.com).
- Fay, C R, 1951. *Huskisson and His Age*. London: Longmans Green & Co.
- Feaveryear, A, 1963. *The Pound Sterling*. Oxford: Clarendon Press.
- Federal Reserve Board, 1943. *Banking and Monetary Statistics*.
- Fischer, David Hackett, 1996. *The Great Wave: Price Revolutions and the Rhythm of History*. New York: Oxford University Press.
- Fivaz, C E, 1988. 'How the MacArthur–Forrest Cyanidation Process Ensured South Africa's Golden Future.' *Journal of the South African Institute of Mining and Metallurgy*, September, pp. 309–318.
- Flandreau, Marc, 1996. 'The French Crime of 1873: An Essay on the Emergence of the International Gold Standard, 1870–1880.' *Journal of Economic History*, 56, no. 4 (Dec.), pp. 862–897.
- Flandreau, Marc, 1996. 'As Good as Gold? Bimetallism in Equilibrium, 1848–1873.' In Lawrence Officer, ed., *Monetary Standards in History*. London: Routledge.
- Fraser, Russell, 1973. *The Dark Ages and the Age of Gold*. Princeton: Princeton University Press.
- Friedman, Milton, 1992. *Money Mischief: Episodes in Monetary History*. New York: Harcourt Brace.

- Friedman, Milton, and Anna Schwartz, 1971. *A Monetary History of the United States, 1867–1960*. Princeton: National Bureau of Economic Research and Princeton University Press.
- Furness, William Henry, 1910. *The Island of Stone Money: Uap and the Carolines*. Philadelphia: J B Lippincott.
- Galbraith, John Kenneth, 1954. *The Great Crash: 1929*. New York: Houghton Mifflin.
- Gallarotti, Giulio M, 1995. *The Anatomy of an International Monetary Regime: The Classical Gold Standard 1880–1914*. New York: Oxford University Press.
- Gibbon, Edward, 1804. *The History of the Decline and Fall of the Roman Empire*. Originally published 1782. The citations herein, kindness of Charles Kindleberger, are from an edition of eight volumes that was published in 1804 in Philadelphia by William Y Birch and Abraham Small.
- Goldsmith, Raymond W, 1985. *Comparative National Balance Sheets: A Study of Twenty Countries, 1688–1978*. Chicago: University of Chicago Press.
- Gould, John D, 1970. *The Great Debasement*. Oxford: Clarendon Press.
- Gould, John D, 1976. 'How It All Began.' *Economic History Review*, XXIX, no. 2 (May).
- Gould, John D, 1979. Review of C E Challis, *The Tudor Coinage*. *The Economic History Review*, XXXII, no. 2 (May), pp. 271–272.
- Graham, Frank D, and Charles R Whittlesey, 1939. *The Golden Avalanche*. Princeton: Princeton University Press.
- Green, Timothy, 1993. *The World of Gold: The Inside Story of Who Mines, Who Markets, Who Buys Gold*. London: Rosendale Press.
- Gudde, Erwin G, 1936. *Sutter's Own Story: The Life of General John Augustus Sutter and the History of New Helvetia in the Sacramento Valley*. New York: G P Putnam's Sons.
- Habashi, Fathi, 1987. 'One Hundred Years of Cyanidation.' *Historical Metallurgy Notes*, 80, no. 905, pp. 108–114.
- Hackett, Francis, 1929. *Henry VIII*. New York: Horace Liveright. Reprinted September 1945.
- Hamilton, Alexander, 1791. *Report on the Establishment of a Mint*. Originally published in Philadelphia. Reprinted in Harold Syrett, ed., *The Papers of Alexander Hamilton*. New York: Columbia University Press, 1965.
- Hamilton, Earl, 1934. *American Treasure and the Price Revolution in Spain, 1501–1650*. Cambridge, MA. Reprinted by Octagon Books, New York, 1970.
- Hanke, Louis, 1951. *Bartolome de las Casas: An Interpretation of His Life and Writings*. The Hague: Nijhoff.
- Harvey, William H, 1894. *Coin's Financial School*. Reprint edited by Richard Hofstadter. Cambridge, MA: Harvard University Press, 1963.
- Hawtrey, R G, 1947. *The Gold Standard in Theory and Practice*, 5th edition. London: Longmans, Green & Co. Originally published 1927.
- Head, Barclay Vincent. *The Coinage of Lydia*. San Diego: Pegasus Publishing (no date).

- Heilbroner, Robert L, 1953. *The Worldly Philosophers*. New York: Simon & Schuster.
- Heilbroner, Robert L, 1956. *The Quest for Wealth*. New York: Simon & Schuster.
- Helleiner, Karl F, 1967. *The Population of Europe from the Black Death to the Eve of the Vital Revolution*. In Rich and Wilson, 1967.
- Hellferich, Karl, 1927. *Money*. New York: The Adelphi Company.
- Hemming, John, 1970. *The Conquest of the Incas*. San Diego: Harcourt Brace & Co.
- Herodotus, 1992. *The Histories*. Translated by Walter Blanco. New York: W W Norton.
- Herrington, Richard, Chris Stanley, and Robert Symes, 1999. *Gold*. London: The Natural History Museum.
- Hicks, John, 1931. *The Populist Revolt*. Minneapolis: University of Minnesota Press.
- High Court of South Africa, 1986. *Official Reports: Hay v. African Gold Recovery Co.*, pp. 244–301.
- Hofstadter, Richard, 1948. *The American Political Tradition and the Men who made it*. New York: Knopf.
- Holtfrerich, Carl Ludwig, 1999. *Frankfurt as a Financial Centre: From Medieval Trade Fair to European Banking Centre*. Munich: Verlag C H Beck.
- Hoover, Herbert, 1952. *The Memoirs of Herbert Hoover: The Great Depression, 1929–41*. New York: Macmillan.
- Hughes, Robert, 1987. *The Fatal Shore: A History of the Transportation of Convicts to Australia, 1787–1868*. New York: Alfred A Knopf.
- Hume, David, 1752. 'Of the Balance of Trade.' In *Essays, Moral, Political, and Literary*, vol. I, 1898 edition. London: Longmans Green, pp. 330–345.
- Ibbotson, Roger, and Laurence Siegel, 1983. 'The World Market Wealth Portfolio.' *Journal of Portfolio Management*, 9, no. 2 (winter), pp. 5–17.
- Jacob, William, 1831. *An Historical Inquiry into the Production and Consumption of the Precious Metals*. London: John Murray.
- Jastram, Roy, 1977. *The Golden Constant: The English and American Experience, 1560–1976*. New York: John Wiley & Sons.
- Jevons, W Stanley, 1875. *Money and the Mechanism of Exchange*. New York: Appleton. Reprinted 1896.
- Kaye, Joel, 1998. *Economy and Nature in the Fourteenth Century: Money, Market Exchange, and the Emergence of Scientific Thought*. Cambridge: Cambridge University Press.
- Kemmerer, Edward, 1944. *Gold and the Gold Standard*. New York: McGraw-Hill.
- Keynes, John Maynard, 1923. *A Tract on Monetary Reform*. London: Macmillan.
- Keynes, John Maynard, 1931. *Essays in Persuasion*. London: Macmillan.
- Kindleberger, Charles, 1978. 'Chevalier, 1806–1870, *The Economic de Tocqueville*.' Unpublished manuscript reprinted as Chapter 2 in Kindleberger, 1985.
- Kindleberger, Charles, 1985. *Keynesianism versus Monetarism and Other Essays in Financial History*. London: George Allen & Unwin.
- Kindleberger, Charles, 1986. *The World in Depression*, 2nd edition. Berkeley: University of California Press. Original edition 1973.
-

- Kindleberger, Charles, 1989. 'Spenders and Hoarders: The World Distribution of Spanish American Silver, 1550-1750.' Singapore: Institute of Southeast Asian Studies. Reprinted in Kindleberger, 1990, pp. 35-85.
- Kindleberger, Charles, 1990. *Historical Economics: Art or Science?* Berkeley: University of California Press.
- Kindleberger, Charles, 1993. *A Financial History of Western Europe*. New York: Oxford University Press.
- Kindleberger, Charles, 1996a. *Manias, Panics, and Crashes: A History of Financial Crises*, 3rd edition. New York: John Wiley & Sons.
- Kindleberger, Charles, 1996b. 'The French Crime of 1871: A Comment.' Unpublished manuscript.
- Kindleberger, Charles, 1996c. *World Economic Primacy, 1500-1990*. New York: Oxford University Press.
- Kindleberger, Charles, 1998. *Economic and Financial Crises and Transformations in Sixteenth Century Europe*. Princeton: Department of Economics, Princeton University.
- King, Gail, 1997. 'A Mother Lode of Klondike Lore.' *Wall Street Journal*, July 17, Op-Ed page.
- Krooss, Herman E, 1969. *Documentary History of Banking and Currency in the U.S.* New York: Chelsea House Publishers. Contains Alexander Hamilton's report on the establishment of a mint.
- Lacey, Robert, and Danny Danziger, 1999. *The Year 1000: What Life Was Like at the Turn of the First Millennium*. Boston: Little, Brown.
- Lane, Frederic, and Reinhold Mueller, 1985. *Money and Banking in Medieval and Renaissance Venice, vol. I, Coins and Money of Account*. Baltimore: Johns Hopkins University Press.
- Langer, William, 1952. *An Encyclopedia of World History*. Boston: Houghton Mifflin. A revision and update of Ploetz, 1883.
- Leary, Francis, 1959. *The Golden Longing*. New York: Charles Scribner's Sons.
- Li, Ming-Hsun, 1963. *The Great Recoinage of 1696-1699*. London: Weidenfeld & Nicolson.
- Lopez, Robert S, 1951. 'The Dollar of the Middle Ages.' *Journal of Economic History*, II, pp. 209-234.
- Lopez, Robert S, 1956. 'Back to Gold, 1252.' *Economic History Review*, 2nd series, 9, pp. 219-240.
- Martin, David, 1977. 'The Impact of Mid-Nineteenth Century Gold Depreciation upon Western Monetary Standards.' *Journal of European Economic History*, 6, no. 3, pp. 641-658.
- Marx, Jenifer, 1978. *The Magic of Gold*. New York: Doubleday.
- Mauro, Frédéric, 1990. 'Merchant Communities, 1350-1750.' In Tracy, 1990.
- Mayhew, Nicholas J, 1992. 'From Regional to Central Mining, 1158-1464.' In Challis, 1992.
- Mayhew, Nicholas J, 1999. *Sterling: The Rise and Fall of a Currency*. London: Allen Lane/Penguin Press.
- McCulloch, John R, ed., 1856. *A Select Collection of Scarce and Valuable Tracts on Money*. London: Political Economy Club. Reprinted 1966 by August M Kelley, New York.

- Menias, G-P, 1969. *Napoleon et l'Argent*. Paris: Editions de l'Epargne.
- Metropolitan Museum, New York, 1985. *The Art of Precolumbian Gold*.
- Miskimin, Harry, 1963. *Money Prices and Foreign Exchange in Fourteenth Century France*. New Haven: Yale University Press.
- Miskimin, Harry A, 1977. *The Economy of Later Renaissance Europe: 1460–1600*. Cambridge: Cambridge University Press.
- Miskimin, Harry A, 1984. *Money and Power in Fifteenth Century France*. New Haven: Yale University Press.
- Miskimin, Harry A, 1989. *Cash, Credit, and Crisis in Europe, 1300–1600*. London: Variorum Reprints.
- Moggridge, Donald, 1969. *The Return to Gold 1925: The Foundation of Economic Policy and its Critics*. Cambridge: Cambridge University Press.
- Moggridge, Donald, 1972. *British Monetary Policy, 1924–31: The Norman Conquest of \$4.86*. Cambridge: Cambridge University Press.
- Moggridge, Donald, 1980. *The Collected Writings of John Maynard Keynes*, vol. XXVI, 'Activities.' London: Macmillan.
- Morineau, Michel, 1985. *Incroyables gazettes et fabuleux métaux: Les retours des trésors américains d'après les gazettes hollandaises (XVIe–XVIIIe siècles)*. Paris: Maison des sciences de l'homme.
- Mun, Thomas, c. 1620. 'England's Treasure by Foreign Trade.' Reproduced in *Early English Tracts on Commerce*, J R McCulluch, ed., Cambridge: Cambridge University Press, 1954.
- Neal, Larry, 1998. 'The Financial Crisis of 1825 and the Restructuring of the British Financial System.' *Review of the Federal Reserve Bank of St Louis*, 80, no. 3 (May/June), pp. 53–76.
- Niedringhaus, Lee I, 1998. 'The Panic of 1893.' *Financial History*, winter, pp. 16–23.
- North, Michael, 1990. *Geldumlauf und Wirtschaftskonjunktur im südlichen Ostseeraum an der Wende zur Neuzeit (1440–1570)*. Sigmaringen: Jan Thorbecke Verlag.
- Noyes, Alexander Dana, 1898. *Forty Years of American Financial History*. New York: Putnam.
- Officer, Lawrence, 1996a. *Between the Dollar-Sterling Gold Points*. Cambridge: Cambridge University Press.
- Officer, Lawrence, 1996b. *Monetary Standards in History*. London: Routledge.
- O'Leary, Paul M, 1960. 'The Scene of the Crime of 1873 Revisited: A Note.' *Journal of Political Economy*, 68, August, pp. 398–392.
- Parry, J H, 1967. 'Transport and Trade Routes.' In Rich and Wilson, 1967.
- Phillips, Carla Rahn, 1986. *Six Galleons for the King of Spain: Imperial Defense in the Early Seventeenth Century*. Baltimore: Johns Hopkins University Press.
- Pindar, 1927. *The Odes of Pindar Including the Principal Fragments*. Sir John Sandys.
- Ploetz, Carl, 1883. *Epitome of Ancient, Medieval, and Modern History*. Translated by William H. Tillinghast. Boston: Houghton, Mifflin & Co.
- Polo, Marco, 1289. *The Travels of Marco Polo*. Translated by Ronald Latham. London: Penguin Books, 1958.

- Postan, M M, and H J Habakkuk, 1952. *The Cambridge Economic History of Europe*, vol. II: *Trade and Industry in the Middle Ages*. Cambridge: Cambridge University Press.
- Prescott, William H, 1847. *History of the Conquest of Peru*. Boston: Harper & Brothers. Republished 1957 by The Heritage Press, New York, from which the citations herein are taken.
- Pressnell, L S, 1968. 'Gold Reserves, Banking Reserves, and the Barings Crisis of 1890,' in C R Whittlesey and J S G Wilson, eds., *Essays in Honour of R S Sayers*. Oxford: Clarendon Press, pp. 167–288.
- Quinn, Stephen, 1996. 'Gold, Silver, and the Glorious Revolution: Arbitrage Between Bills of Exchange and Bullion.' *Economic History Review*, XLIX, no. 3, pp. 473–490.
- Ramage, Andrew, and Paul Craddock, 2000. *King Croesus' Gold: Excavations at Sardis and the History of Gold Refining*. London: British Museum Publications.
- Redish, A, 1990. 'The Evolution of the Gold Standard in England.' *Journal of Economic History*, vol. L, pp. 789–806.
- Ricardo, David, 1809. *The Price of Gold*. Reprinted by The Johns Hopkins Press, Baltimore, 1903, Jacob Hollander, ed.
- Ricardo, David, 1811a. 'Reply to Mr. Bosanquet's Practical Observations on the Report of the Bullion Committee.' In *Economic Essays*, E C K Gonner, ed. London: G Bell & Sons, 1923, pp. 63–149.
- Ricardo, David, 1811b. 'The High Price of Bullion and the Depreciation of Bank Notes.' In *Economic Essays*, E C K Gonner, ed. London: G Bell & Sons, 1923, pp. 3–60.
- Ricardo, David, 1816. 'Proposals for an Economical and Secure Currency; with Observations on the Profit of the Bank of England, as They Regard the Public and the Proprietors of the Bank of England.' In *Economic Essays*, E C K Gonner, ed. London: G Bell & Sons, 1923, pp. 63–149.
- Rich, E E, and C H Wilson, eds., 1967. *The Cambridge Economic History of Europe*, vol. IV: *The Economy of Expanding Europe in the Sixteenth and Seventeenth centuries*. Cambridge: Cambridge University Press.
- Robertson, Dennis H, 1929. *Money*. New York: Harcourt, Brace & Co.
- Rockoff, Hugh, 1990. 'The Wizard of Oz as a Monetary Allegory.' *Journal of Political Economy*, 98 (August), pp. 739–760.
- Roll, Eric, 1995. *Where Did We Go Wrong: From the Gold Standard to Europe*. London: Faber & Faber.
- Rolnick, Arthur J, and Warren E Weber, 1986. 'Gresham's Law or Gresham's Fallacy.' *Journal of Political Economy*, 94 (February), pp. 185–199.
- Rolnick, A J, F R Velde, and W E Weber, 1996. 'The Debasement Puzzle: An Essay on Medieval Monetary History.' *Journal of Economic History*, December.
- Rosenthal, Eric, 1970. *Gold. Gold. Gold*. London and New York: Macmillan.
- Rousseau, Peter L, and Richard Sylla, 1999. 'Emerging Financial Markets and Early U.S. Growth.' Cambridge, MA: National Bureau of Economic Research, Working Paper 7448.

- Rueff, Jacques, 1972. *The Monetary Sin of the West*. New York: Macmillan.
- Ruskin, John, 1862 [1982]. 'Unto This Last': *Four Essays on the First Principles of Political Economy*. London: Smith, Elder & Co. These interesting essays also appear in *The Complete Works of John Ruskin*, vol. XVII, E. T. Cook and Alexander Wedderburn, eds. London: George Allen, 1905.
- Schubert, Aurel, 1991. *The Credit-Anstalt Crisis in 1931*. Cambridge: Cambridge University Press.
- Schuker, Stephen A, 1976. *The End of French Predominance in Europe: The Financial Crisis of 1924 and the Adoption of the Dawes Plan*. Chapel Hill: The University of North Carolina Press.
- Schwab, Gustav, 1946. *Gods and Heroes: Myths and Epics of Ancient Greece*. Translated from German by Olga Marx and Ernst Morwitz. New York: Pantheon Books.
- Schwartz, Anna, 1973. 'Secular Price Change in Historical Perspective.' *Journal of Money, Credit and Banking*, 5, pp. 243–269.
- Select Committee of the House of Commons on the High Price of Gold Bullion (*The Bullion Committee Report of 1810*). London: P S King, 1925.
- Shaw, Bernard, 1928. *The Intelligent Woman's Guide to Socialism and Capitalism*. Reprinted by New Transaction Books, New Brunswick, NJ, 1984.
- Skousen, Mark, 1977. *Economics of a Pure Gold Standard*. Irvington-on-Hudson: The Foundation for Economic Education.
- Smith, Adam, 1776. *The Wealth of Nations*. Amherst, NY: Prometheus Press, 1991.
- Smith, Martin, 1996. *Bookwatch: The General Strike*. International Socialism, March (from www.google.com).
- Sraffa, P, *The Works and Correspondence of David Ricardo*. Cambridge: Cambridge University Press.
- Stein, Herbert, 1984. *Presidential Economics: The Making of Economic Policy from Roosevelt to Reagan and Beyond*. New York: Simon & Schuster.
- Stigler, Stephen, 1977. 'Eight Centuries of Sampling Inspection: The Trial of the Pyx.' *Journal of the American Statistical Association*, 72, pp. 493–500.
- Supple, B E, 1959. *Commercial Crisis and Change in England, 1600–1642*. Cambridge: Cambridge University Press.
- Sutherland, C H V, 1959. *Gold: Its Beauty, Power, and Allure*. London: Thames & Hudson.
- Swiss Expert Group, 1997. 'Der Neue Geld-Und Währungsartikel in der Bundesverfassung.' Press release dated October 24.
- Tallman, Ellis, 1998. 'Gold Shocks, Liquidity, and the United States Economy During the National Banking Era.' *Explorations in Economic History*, 35, pp. 381–404.
- Tassel, Janet, 1998. 'The Search for Sardis.' *Harvard Magazine*, April, pp. 51–60, 95–96.
- Temin, Peter, 1989. *Lessons from the Great Depression*. Cambridge, MA: MIT Press.
- Tracy, James D, ed., 1990. *The Rise of Merchant Empires: Long-Distance Trade in the Early Modern World, 1350–1750*. Cambridge: Cambridge University Press.

- Triffin, Robert, 1960. *Gold and the Dollar Crisis*. New Haven: Yale University Press.
- Tuchman, Barbara W, 1978. *A Distant Mirror: The Calamitous 14th Century*. New York: Ballantine Books.
- US House Committee on Foreign Affairs, 1961. 'Gold and the US Balance of Payment Deficits.'
- US House of Representatives, Committee on Banking and Currency, Report No. 31, Forty-first Congress, Second Session. 1870. 'The New York Gold Conspiracy.'
- US Treasury Department, 1982. 'Report to Congress on the Role of Gold in the Domestic and International Monetary Systems.'
- Vilar, Pierre, 1976. *A History of Gold and Money, 1450-1920*. London: New Left Books.
- Watson, Andrew M, 1967. 'Back to Gold—and Silver.' *Economic History Review*, 2nd series, 20, pp. 1-34.
- Weil, Gordon, and Ian Davidson, 1970. *The Gold War: The Story of the Monetary Crisis*. New York: Holt Rinehart.
- Wheatcroft, Geoffrey, 1985. *The Randlords*. London: Atheneum.
- White, Michael, 1977. *Isaac Newton: The Last Sorcerer*. Reading, MA: Addison-Wesley.
- Wilkic, A D, 1994. 'The Risk Premium on Ordinary Shares.' A presentation to the Faculty of Actuaries and the Institute of Actuaries, London.
- Wimmer, Larry, 1975. 'The Gold Crisis of 1869: Stabilizing or Destabilizing Speculation Under Floating Exchange Rates.' *Explorations in Economic History*, 12, pp. 105-122.
- Wirth, Max, 1893. 'The Crisis of 1890.' *Journal of Political Economy*, 1, no. 2, pp. 214-235.
- Wright, Louis B, 1970. *Gold, Glory, and the Gospel: The Adventurous Lives and Times of the Renaissance Explorers*. New York: Atheneum.



Index

- Acts for Remediating the Ill State of the Coin,
 An 181, 182
Africa Company 175
African Gold Extracting Company 216
Age of Iconoclasm 66
Agnus Dei (Lamb of God) coin 92
Alboin, King of the Lombards 61
alchemy 56
Alexander the Great 46-7, 48-9, 52
Alyattes, King 34, 38
American Revolution 199
Aquinas, St Thomas 77
Arab dinars 66, 70
Ardys, King 34, 37, 38
Atahualpa 123, 125, 127-9, 130, 132, 160
augustalis coin 94
Australian gold rush 213-14
- Balboa, Vasco Núñez de 120-2, 133
 n accent
Baldwin, Stanley 246, 247
Bank notes see paper money
Bank of England 59, 165, 178, 180, 194-201,
 204-5, 219, 227-30, 239, 245, 248, 249, 255
Bank of France 165, 219, 230, 239, 248, 253,
 254, 269
Bank Restriction Act (1797) 201
Baring Brothers 153, 227-30, 237, 239
Baring, Alexander (Lord Ashburton) 229
bars, gold 30-1
Battle of Poitiers 104
Battle of Waterloo 202
Bernard, St 78, 79
bezant 57, 60, 63, 64, 66, 67, 190
Biblical references 14
bills of exchange 151, 152-3, 154, 198
bimetallic currency system 39, 222, 223, 241-2
- Black Death 99-101, 102, 103, 106, 146
Blondeau, Pierre 174-5, 176, 177
Bodin, Jean 148
Bohun, Edward 182
Bovill, E.W. 71, 75
Boyle, Andrew 244
Breasted, James Henry 16-18
Bretton Woods 265
Briot, Nicholas 174
Brüning, Heinrich 253, 257
Bryan, William Jennings 241-2, 247
Bulgaroctonus, Basil 7, 61
Bullion Committee 203-5, 209, 218, 222
Byzantine empire 59-67, 69-70
- Cadamosto, Alvise da 116
Californian gold rush 7, 13, 19-20, 20-1
camels 71-3, 114
Candaules, King 33
carats 29-30, 37
Catherine of Aragon 143, 145
Cawdor, Lord 193
Cellini, Benvenuto 142, 143
Charlemagne 63, 79, 80-1, 82
Charles II of England 173, 174-5, 177
Charles le Mauvais, King of Navarre 103-4
Charles V of Spain 83, 130, 131, 136, 140, 142,
 143, 145, 147
Chaucer, Geoffrey: 'Shipman's Tale' 152
Churchill, Winston 246, 248, 264
chrysography 79
Cicero 101
Clemenceau, George 243
Clement V, Pope 100
Clement VI, Pope 102
clipping of coins 173-4, 175-6
Coinage Act (1873) (US) 236
-

coinage

Byzantine 60, 66-7
of Charlemagne 81, 82
Chinese 163-4
debasement 53-6
English 96-7, 102
Genoese 95
Greek 37-8, 46
Japanese 167-8
Lydian 37-43, 45
Macedonian 48-9
of Offa 79-80, 82
Persian 46-7
Roman 49-52, 55-7, 59
Columbus, Christopher 7, 111, 115, 117-20, 162, 207
commercial bills under discount 200-1
Conduitt, John 189-90
Connally, John 272-3
Constantinople 60-2, 64
Cook, A.J. 248
Crassus 7, 51-2, 60, 252
Creditanstalt Bank 253
Croesus, King 7, 34-5, 37, 38-43, 45, 50, 60, 63, 160, 162, 170, 190, 207, 221
Cromwell, Oliver 174
Crusades 88-90, 93, 95
currency 25-6, 29
 see also coinage
Cyrus, King of Persia 40, 42, 46

da Gama, Vasco 117
Darius 46-7, 169
Dark Ages 59
Davies, Glyn 85, 147
Day, John 110, 111
de Soto, Hernando 125-6
debasement 53-6, 67, 82, 85, 147
Defoe, Daniel 178
Dias, Bartholomeu 116-17
Diocletian, Emperor 56-7
Diodorus 19

Dionysius of Syracuse 55
Disraeli, Benjamin 232
dollar, US 222, 235-6
Domitian, Emperor of Rome 50
Drake, Sir Francis 137-8
ducat 96, 97, 190
dumb barter 73, 74, 75
Dunstan, St 61

earmarking 28
East India Company 157, 158, 177, 189
Edward I 97
Edward II 101
Edward III 85, 97, 102, 103, 104-5, 106, 109, 145, 182
electrum (white gold) 33
Elizabeth I 142, 174, 179, 190
Eloi, St (St Eligius) 61
Emergency Banking Act (1933) (USA) 257, 258
environmental damage 19-20, 25

Federal Reserve Bank of New York 27-8, 169, 245, 250-1, 264, 269
fei monetary system (Yap island) 26-7, 28
Ferdinand of Aragon 20, 108, 109, 119, 120, 121, 136, 140
fiorino d'oro (florin) 96
Fischer, David Hackett 78
Fishguard "invasion" (1797) 193-5
florin 97, 190
Forbes, Sir William 197
Francis I of France 63, 83, 140, 142, 143-5, 146, 147, 159, 160
Francis, Father Alban 186
Franks 61-3
Frederick Barbarossa (Holy Roman Emperor) 87, 93
Frederick II (Holy Roman Emperor) 93-4, 95
Friedman, Milton 148, 203, 257
Fuggers family 140, 142, 153
Furness, William Henry III 26, 27

- Gallienus, Emperor 55-6
 Gaulle, Charles de 7, 264, 270-2, 276
 Genoa/Genoese 94-6, 99
 genovino (genoin) 95-6, 97, 190
 Gibbon, Edward 50, 56, 133
 gilding 8
 Giscard d'Estaing, Valéry 264
 Godescalc Gospels 79
 gold bugs 269
 gold : silver ratio 30, 48, 55, 224
 gold standard 93-4, 205, 221-2, 224-8, 231-2, 244-5, 252, 256
 Gold Standard Act (1900) (USA) 258
 Gold Standard Act (1925) (UK) 246
 golden calf 14, 25
 Golden Fleece, legend of 7, 21-3, 25, 66
 goldsmiths 35-7, 61, 78, 109
 Goldsmiths Company of London 83
 Great Debasing, The 147, 180, 182
 Great Depression 53, 238-9, 252, 253
 Great Famine 99
 Great Rebellion 180
 Great Recoinage 180-4, 188, 190
 Great Strike 248
 Green, Timothy 20
 greenbacks 235-6
 Gregory IX, Pope 93
 guinea 74, 175, 177, 178-9, 188-90, 198-9
 Gyges, King 33-4, 35, 37, 38, 42
- Hamilton, Alexander 223, 224, 235, 236
 Hargraves, Edward Hammond 213, 214, 215
 Harrison, George 215, 251, 254
 Hatshepsut 18, 160
 Haynes, Hopton 176
 Henry Bolingbroke 101, 109
 Henry I of England 85
 Henry II of England 87
 Henry III of England 96
 Henry VII of England 119
 Henry VIII of England 142-3, 145-7, 180, 182
 Henry the Navigator (Portugal) 111, 113-14, 115-16
- Herodotus 33, 35, 38, 42, 71, 73
 Heyn, Piet 137
 Hien Tsung 163, 200
 Hitler, Adolf 130, 243, 252, 260
 hoarding 59, 61, 169, 219, 225, 256-7
 Hoover, Herbert 252, 253, 256, 257, 258, 272
 Horner, Francis 202, 203, 204, 205
 Hughes, Richard 213
 Hume, David 158, 159-60, 168, 203
 Hundred Years' War 97, 101, 182
 Huskisson, William 203
 hydraulicking 19-20
- inflation 55-6, 146-8, 200-1, 219, 267
 Innocent IV, Pope 90-2, 93, 94, 95
 International Monetary Fund (IMF) 265-6, 275, 279
 Irene, Emperor of Byzantium 80
 Isabella of Castile 108, 109, 119, 130, 136
- Jacob, William 19
 James II 177, 186
 Jean II (Jean Le Bon) of France 103-6, 132, 145
 Jevons, Stanley 8
 Joanna the Mad 109, 130
 John I of Portugal 109
 John II of Portugal 118-19
 John I Tzimiscēs, Emperor 67
 Johnson, Lyndon B. 267-9
 Jones, John Paul 113
 Justinian I 6, 60, 78, 132, 160
 Justinian II 66
- Kennedy, John F. 266
 Keynes, John Maynard 185, 246, 255, 258, 265
 Khan, Aly 10
 Kindleberger, Charles 109, 178, 230
 King, Peter 205
 Klondike gold rush 215
 Knox, Colonel Frank 256
 Kublai Khan 160-2, 164-5, 167, 169-70

- Langland, William 102
Lansdown, Lord 200
Leo III 66
Lidderdale, William 229-30
Lloyd George, David 243
Locke, John 180, 184, 187, 222, 223
Lombards 61-3
Lopez, Robert Sabatino 63, 66, 94, 95, 96
Louis, St 87, 92, 95, 96
Louis X of France 101
Louis XIV 18, 177, 198
Lowndes, William 179, 180, 184, 187
Luther, Martin 142
Lydia, Asia Minor 33-43, 45
- Ma Twan-lin 164, 170
MacArthur, John Stewart 216, 217
Macedon, Alexander of 33
Magellan, Ferdinand 111
Malthus, Thomas 201-2
Marco Polo 90, 119, 160-2, 164-5, 167, 169-70
Marshall, James 212, 215
Martel, Charles 69
Martin, William McChesney 267, 270
McKinley, William 241, 242
Meeting on the Field of Cloth of Gold 145
Mellon, Andrew 252
Mermnadae 33-5
Mestrell, Eloy 174
Midas, King of Phrygia 10, 31-3
Mill, John Stuart 7
milled coins 174
Mills, Ogden 250
mining
 Arab 69
 Californian 7, 13, 19-20
 Egyptian 19
 Frank 80
 Nubian 63
 Peruvian 133
 Roman 19, 51, 52
 Russian 211
 salt 74
 South Africa 10, 13, 20, 215-17
- Miskimin, Harry A. 102
Misselden, Edward 177
Montagu, Charles 180, 185, 187
Moreau, Emile 248-50
Morgan, J. Pierpoint 240-1
Morgan, J.S. & Co. 240
Morison, Samuel Eliot 117
Mun, Thomas 158, 159
- Napoleon 48, 130, 178, 193, 198, 200, 202, 209, 218, 224, 229
Neale 187, 188
Nero, Emperor 55
New York Stock Exchange 238
Newton, Sir Isaac 185-90, 201, 208, 223, 244
Nicephorus 63
Nicholas, Jemima 193
Niemeyer, Otto 246
Nixon, President Richard 167, 266, 269, 270, 272-4, 280
noble 97
Norman, Montagu 244-6, 248, 249-50, 253, 254
North, Sir Dudley 179
Nubia 18-19, 63
- Offa, King 79-80, 82, 184, 222, 223
Organization of Petroleum Exporting Countries (OPEC) 274
- Pactolus, River 31, 32, 35, 43, 170
panning 20-1
paper money 163-5, 169-70, 194-6, 198-201, 203-5, 209, 235-6
Peachell, John 186
pepper 110
Petrarch 100, 106
Petty, William 159
pharaohs 16-18
Phidias 46, 50
-

- Philip 142
 Philip 52
 Philip Augustus, King of France 87
 Philip II of Macedon 47-9
 Philip II of Spain 140
 Philip IV, King of France 100-1
 Pindar 7
 pirates 113, 135, 136, 137
 Pitt, William 195
 Pizarro, Francisco 7, 120, 122, 123-32, 133, 147, 207
 Pizarro, Hernando 125, 126, 133
 Pliny the Elder 50
 Polk, President 212
 Prescott, William H. 122, 125, 132
 Price Revolution of the 16th Century 146-7, 148, 149, 150, 207, 208
 Ptolemy, II, King of Egypt 8

 Qin Shu Huangdi 163
 Quinctius Flaminius 49

 ransoms 82-3, 90, 103-6, 132
 Raymond of Tripoli 90
 Reichsbank 228, 231
 Restriction Bill (1797) (UK) 195, 196-7, 200, 204
 Resumption Act (USA) 236-7
 Ricardo, David 201-2, 204, 222
 Richard I (Richard the Lionheart) 82-3
 Richard II of England 101, 109
 Richelieu, Cardinal 175
 Rienzi, Cola di 101
 Rogers, Will 13
 Roosevelt, Franklin D. 257-8, 264
 Rothschilds 153, 230, 240

 Sadyattes, King 34, 38
 Saint Sophia, church of, Constantinople 16, 60, 78, 132
 salt 73-4, 75
 San Vitales, Church of, Ravenna 61

 Schacht, Hjalmar Horace Greeley 249, 250
 Schwartz, Anna 149, 257
 seignorage 85
 Sheba, Queen of 14, 28
 Sherman Silver Purchase Act (1890) (US) 238, 239
 shilling 39, 179, 180
 silver 30, 31, 39, 78, 168, 176-7
 coins 39, 177, 178-9, 188-90
 standard 188, 222-3, 235-6
 Sinobaldo Fiesco (Pope Innocent IV)
 Smith, Adam 132, 147-8, 168, 169, 201, 203
 Solomon, King 14, 16, 28, 60
 Solon 35, 38, 42, 45
 South African gold rush 215-17
 sovereign 190, 205, 221
 Spanish Armada 142, 143, 196
 State Bank of Russia 229
 Statute of Labourers 102
 Statute of Purveyors 85
 Stewart, William 236
 St-Méthard Gospel Books 79
 Strong, Benjamin 245, 250, 251
 Suger, Abbé 78-9, 160
 Sutter, Johann 7, 212-13, 217
 Sutter's Mill 211-13, 225

 tabernacle 9-10, 14, 16
 Tacitus 147
 tari gold standard 93-4, 95
 i grave
 Theodora, Empress 61, 66-7
 Theophilus, Emperor 60, 69
 Thornton, Henry 203, 204
 Thutmose I 18
 Tmolus, Mount 32
 touchstone 37, 83
 trade fairs 150-1, 154
 Treaty of London 104-6
 Treaty of Utrecht 188
 Trials of the Pyx 83-5
 troy ounce 30, 39

Tuchman, Barbara 103, 105

Tyler, Watt 101, 102

United States of America

Constitution 37

devaluation of gold 257-8, 272-4

dollar:gold conversion rate 265

economic decline (1890s) 239-41

gold hoarding 256-7

gold standard, restoration of 245-8

inflation 267-9, 272

international financial position 266-9

migration of gold to 259-60

Wall Street crash 250-3

war debts owed to 243-4, 253-4

see also Federal Reserve Bank of New York

Valencia, Pedro de 139

Venice 61, 107-8

Versailles 18

Villani, Matteo 106

Visconti, Galeazzo 106

Webster, Daniel 229

Western Union Company 212

William III 178

William the Conqueror 82

Wilson, Woodrow 244

Wolsey, Cardinal 145

Yap, island of 26-7, 28, 30, 31, 167

Photo credits

Prologue

- 6 Detail of gold bars: PhotoDisc Inc.
- 7 John Ruskin: Corbis (Supplied by ZooID Pictures Ltd.)
- 8 Eiffel Tower: PhotoDisc, Inc.
- 9 Inca Gold: Tinopolis for S4C International
- 10 Gold foil: Tinopolis for S4C International
- 11 Buddha Statue: © PhotoDisc

Chapter 1

- 12 San Sophia, Turkey: Tinopolis for S4C International
- 13 Gold Ingots: Tinopolis for S4C International
- 15 Golden calf: akg-images (Supplied by ZooID Pictures Ltd.)
- 16 Wailing Wall: © PhotoDisc
- 17 Tutankhamen's mask: akg-images (Supplied by ZooID Pictures Ltd.)
- 18 Egyptian jewellery: Tinopolis for S4C International
- 18 Inca jewellery: Tinopolis for S4C International
- 19 Les Médulas mountains: Tinopolis for S4C International
- 20 Roman canal: Tinopolis for S4C International
- 20 Mountain: Tinopolis for S4C International
- 21 South African miners: Tinopolis for S4C International
- 22-3 South African mine: Tinopolis for S4C International

Chapter 2

- 24 Midas: akg-images (Supplied by ZooID Pictures Ltd.)
- 25 Crowns: PhotoDisc, Inc.
- 29 Weighing gold: Tinopolis for S4C International
- 30 Shekel: Edgar L. Owen www.edgarlowen.com
- 31 Lydian empire: Tinopolis for S4C International
- 32 Burial mound: Tinopolis for S4C International
- 34 Sardis: Tinopolis for S4C International
- 36 King Croesus: akg-images (Supplied by ZooID Pictures Ltd.)
- 38 Lydian coin: Tinopolis for S4C International
- 39 Sardis: Tinopolis for S4C International
- 40 Sardis: Tinopolis for S4C International
- 41 Delphi: Photo Researchers, Anita Duncan.
- 43 Ephesus: Photographer Adam Salkeld for S4C

Chapter 3

- 44 Alexander the Great: Tinopolis for S4C International
- 46 Daric: Edgar L. Owen www.edgarlowen.com
- 47 Alexander the Great: Tinopolis for S4C International
- 49 Hercules coin: Edgar L. Owen www.edgarlowen.com
- 49 Nike coin: Tinopolis for S4C International
- 52-3 Hadrian's Wall: PhotoDisc Inc.

- 54 Mosaic: Tinopolis for S4C International
 56 Bezan: Tinopolis for S4C International

Chapter 4

- 58 Egyptian temple: © PhotoDisc
 59 Interior of Blue Mosque in Istanbul, Turkey: © PhotoDisc
 60 San Sophia: Tinopolis for S4C International
 61 Constantinople images: Photographer Adam Salkeld for S4C
 62 St. Mark's: PhotoDisk, Inc.
 63 Statue: PhotoDisk, Inc.
 65 San Sofia: Tinopolis for S4C International
 66 Coins: Tinopolis for S4C International
 67 Constantinople: Tinopolis for S4C International
 67 San Sofia: Photographer Adam Salkeld for S4C

Chapter 5

- 68 Pyramid: Corbis Digital Stock
 70 Dinar: Edgar L. Owen
www.edgarlowen.com
 71 Camel train: Tinopolis for S4C International
 72 Camel rider: Corbis Digital Stock
 75 Salt: Photo Researchers, Anita Duncan

Chapter 6

- 76 Lion: Photos.com
 79 Illuminated manuscript: Tinopolis for S4C International
 81 Charlemagne: Photos.com
 81 Offa's coin: Tinopolis for S4C International
 82 Charles V: akg-images (Supplied by ZooID Pictures Ltd.)
 84 Trials of Pyx: Tinopolis for S4C International

Chapter 7

- 86 Gold jewellery: Tinopolis for S4C International
 88 The Duomo: © PhotoDisc
 89 Oxford University: Corbis Digital Stock
 91 Steps to the Dome of the Rock, Jerusalem: © PhotoDisc
 92 Notre Dame: Corbis Digital Stock
 94 Venice: Corbis Digital Stock
 97 Notre Dame: Corbis Digital Stock
 97 Ponte Vecchio, Florence: PhotoDisk, Inc.

Chapter 8

- 98 Wales, Europe: Tinopolis for S4C International
 100 Plague woodcut: Corbis
 105 Chartres Cathedral at Night: © PhotoDisc
 107 Venetian Masks: Tinopolis for S4C International
 108 Mosaic: Tinopolis for S4C International
 109 Ducat: Edgar L. Owen
www.edgarlowen.com
 111 Gold pot: Tinopolis for S4C International

Chapter 9

- 112 Inca Gold: Tinopolis for S4C International
 115 Camel Train: Tinopolis for S4C International
 117 Columbus: aaa images (Supplied by ZooID Pictures Ltd.)
 118 Columbus landing: aaa images (Supplied by ZooID Pictures Ltd.)
 120 The Andes: Photographer Adam Salkeld for S4C
 121 Atahualpa: Tinopolis for S4C International
 122 Machu Picchu: Photographer Adam Salkeld for S4C
 123 Machu Picchu: Photographer Adam Salkeld for S4C

- 124 Atahualpa: Tinopolis for S4C International
- 124 Atahualpa's prison room: Tinopolis for S4C International
- 125 Inca Gold: Tinopolis for S4C International
- 126 Machu Picchu: Photographer Adam Salkeld for S4C
- 127 Machu Picchu: Photographer Adam Salkeld for S4C
- 128 Inca Gold: Tinopolis for S4C International
- 129 Inca Gold: Tinopolis for S4C International
- 130 Pizarro's tomb: Photographer Adam Salkeld for S4C
- 131 Cuzco Cathedral, Peru from S4C International
- 133 Gold jewellery: Tinopolis for S4C International

Chapter 10

- 134 Spanish Galleon: Tinopolis for S4C International
- 136 Seville Cathedral: Tinopolis for S4C International
- 136 Seville Cathedral: Tinopolis for S4C International
- 137 Seville Cathedral: Tinopolis for S4C International
- 138 Tower of gold: Tinopolis for S4C International
- 141 Spanish Armada: Photos.com
- 144 St Peter's: Photo Researchers, Anita Duncan
- 151 Bills of Exchange: Tinopolis for S4C International
- 152 Helmet: Photos.com
- 155 Vatican © PhotoDisc

Chapter 11

- 156 Silks: Photos.com
- 158 Tea Harvest: PhotoDisk, Inc.
- 161 Map: Tinopolis for S4C International
- 162 Cash Coin: Edgar L. Owen
www.edgarlowen.com
- 164 River Li: Flat Earth
- 166 Tapestry: Photos.com
- 171 Vase: Photos.com

Chapter 12

- 172 Molten gold: Tinopolis for S4C International
- 175 Coin press: Hulton/Getty Archive (Supplied by ZooID Pictures Ltd.)
- 176 Royal Exchange: Getty Images
- 179 Clipped Coin: Tinopolis for S4C International
- 181 Globe: Corbis Digital Stock
- 183 Newton: National Portrait Gallery
- 184 Cambridge University: PhotoDisk, Inc.
- 186 The Tower of London: Corbis Digital Stock
- 188 Guinea: Edgar L. Owen
www.edgarlowen.com
- 191 Modern mint press: Tinopolis for S4C International

Chapter 13

- 192 French Frigate: Tinopolis for S4C International
- 194 Jemima Nichols: Tinopolis for S4C International
- 194-5 Tapestry images: Tinopolis for S4C International
- 196-7 The Royal Oak: Tinopolis for S4C International
- 199 Bank of England - is also a S4C International
- 202 Houses of Parliament: PhotoDisk, Inc.
- 203 Sovereign: Tinopolis for S4C International

Chapter 14

- 206-7 Molten gold: Tinopolis for S4C International
- 208 Prospectors from around the world: Tinopolis for S4C International
- 208 Prospectors crossing river: akg-images (Supplied by ZooID Pictures Ltd.)
- 209 Panning: PhotoDisk, Inc.
- 210 Sacramento River: Tinopolis for S4C International
- 210 Johann Sutter: akg-images (Supplied by ZooID Pictures Ltd.)
- 211 Panning: Tinopolis for S4C International
- 214 Monument: Tinopolis for S4C International
- 215 Cyanidation: Tinopolis for S4C International
- 217 Coins: Tinopolis for S4C International
- 218 Smelting gold: Tinopolis for S4C International

Chapter 15

- 220 Gold bars: PhotoDisk, Inc.
- 224 Coin bag: PhotoDisk, Inc.
- 226 Iguassu Falls: Corbis Digital Stock
- 229 Gold piggy bank: PhotoDisk, Inc.
- 233 US Dollars: PhotoDisk, Inc.

Chapter 16

- 234 Harvest: PhotoDisk, Inc.
- 237 Oil Drill: PhotoDisk, Inc.
- 238 Great Depression: Tinopolis for S4C International
- 239 Stock Exchange: Image State
- 241 Jennings Bryan: Library of Congress
- 242 World War II: Tinopolis for S4C International
- 244 Capitol Building: Corbis Digital Stock
- 246 Churchill: Tinopolis for S4C International
- 246 Keynes: Tinopolis for S4C International

- 247 General Strike Images: Tinopolis for S4C International
- 247 Pit Head: Tinopolis for S4C International
- 249 Trader: Tinopolis for S4C International
- 250 Wall Street sign: Tinopolis for S4C International
- 251 Hoover: Tinopolis for S4C International
- 252 Immigrants: Tinopolis for S4C International
- 256 Roosevelt: Tinopolis for S4C International
- 261 Bank of England: Getty Images

Chapter 17

- 262 Oil Pumps: PhotoDisk, Inc.
- 264 United Nations: Photos.com
- 267 50 cent piece: Photos.com
- 268 Nixon: Tinopolis for S4C International
- 270 Flag: PhotoDisk, Inc.
- 271 Nixon: Tinopolis for S4C International
- 275 President Carter: Library of Congress
- 277 Stock Exchange: Tinopolis for S4C International
- 281 Nuggets: PhotoDisk, Inc.

Bibliography

- 282 Gold Wire: Tinopolis for S4C International

Index

- 294 Gold Foil: PhotoDisk, Inc.
-